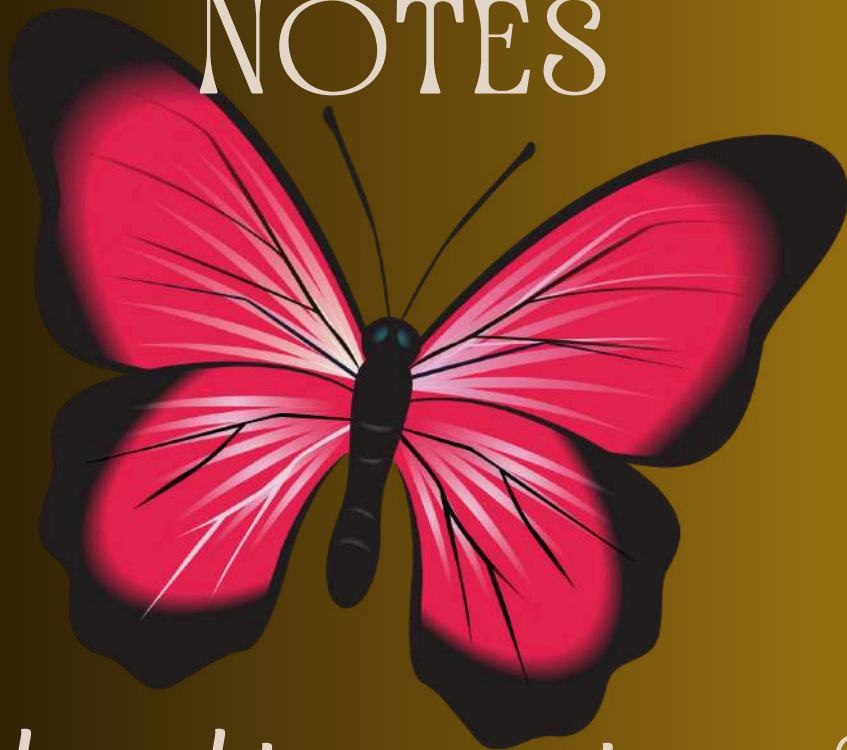


DIGITAL CIRCUITS NOTES



*Your thoughts are gates – AND
what matters, OR what inspires,
NOT what breaks you.”*

*I strongly believe that
the moment you decide
better at your chosen
men, you'll become me*

INDEX

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[illegible]



जय श्री कृष्ण

jaya prakash-

Number Systems

$10^3, 10^2, 10^1, 10^0 \rightarrow$ Base (or) radix

$$(N)_x = \overbrace{d_n d_{n-1} d_{n-2} \dots d_1 d_0}^{\text{Integer}} \cdot \underbrace{d_{-1} d_{-2} \dots d_{-m}}_{\text{Fraction}}$$

Radix point

Here,

$x^n =$ Weight

$d_n x^n \rightarrow$ Weighted value.

Base / radix:

The number of different symbols use to represent a number in Number System is called base / radix.

$\rightarrow$ A number is consisting of two parts integer and fraction and separated by radix point.

1. Decimal ($x=10$) (0, 1, 2, 3, 4, 5, 6, 7, 8, 9)
2. Binary ($x=2$) (0, 1)
3. Octal ($x=8$) (0, 1, 2, 3, 4, 5, 6, 7)
4. Hexa decimal ($x=16$) (0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F)

$x: 0 \rightarrow (x-1)$

$\hookrightarrow$ In any number system with radix (x) decimal equivalent of symbol ranging from 0 to $x-1$.

$\rightarrow$ In any number system with radix (x),

1. Decimal to any base

$$(319.6875)_{10} = ()_2 = ()_8 = ()_{16} = ()_4$$

$$\begin{array}{r} 2 \overline{) 319} \\ 2 \overline{) 159-1} \\ 2 \overline{) 79-1} \\ 2 \overline{) 39-1} \\ 2 \overline{) 19-1} \\ 2 \overline{) 9-1} \\ 2 \overline{) 4-1} \\ 2 \overline{) 2-0} \\ 1-0 \end{array}$$

$$\begin{aligned} 0.6875 \times 2 &= 1.375 \\ 0.375 \times 2 &= 0.75 \\ 0.75 \times 2 &= 1.5 \\ 0.5 \times 2 &= 1.0 \end{aligned}$$

$$(319.6875)_{10} = (100111111.1011)_2$$

$$\begin{array}{r} 8 \overline{) 319} \\ 8 \overline{) 39-7} \\ 4-5 \end{array}$$

$$\begin{aligned} 0.6875 \times 8 &= 5.5 \\ 0.5 \times 8 &= 4.0 \end{aligned}$$

$$(319.6875)_{10} = (457.540)_8$$

$$\begin{array}{r} 16 \overline{) 319} \\ 16 \overline{) 19-15(F)} \\ 1-3 \end{array}$$

$$0.6875 \times 16 = 11.0 (B)$$

$$(319.6875)_{10} = (B13F.B)_{16}$$

$$\begin{array}{r} 4 \overline{) 319} \\ 4 \overline{) 79-3} \\ 4 \overline{) 19-3} \\ 4 \overline{) 4-3} \\ 1-0 \end{array}$$

$$\begin{aligned} 0.6875 \times 4 &= 2.75 \\ 0.75 \times 4 &= 3.0 \end{aligned}$$

$$(319.6875)_{10} = (10333.23)_4$$

2. Any Base to decimal

$\hookrightarrow$ Summation of Weighted values

$$(i) (100111.101)_2 = ()_{10}$$

$$\begin{aligned} &\hookrightarrow (1 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}) \\ &= (32 + 0 + 0 + 4 + 2 + 1 + \frac{1}{2} + \frac{1}{8}) \\ &= 39 + 0.5 + 0.125 \end{aligned}$$

$$(100111.101)_2 = (39.625)_{10}$$

$$(ii) (A3F.C)_{16} = (10 \times 16^2 + 3 \times 16 + 15 \times 16^0 + 0.12 \times \frac{1}{16})$$

$$= 2560 + 48 + 15 + 0.75$$

$$= (2623.75)_{10}$$

$$(A3F.C)_{16} = (2623.75)_{10}$$

$$(iii) (237.24)_8 = ()_{10}$$

$$\hookrightarrow (2 \times 8^2 + 3 \times 8^1 + 7 \times 8^0 + 2 \times \frac{1}{8} + 4 \times \frac{1}{8^2})$$

$$= 128 + 24 + 7 + 0.25 + 0.0625$$

$$(237.24)_8 = (159.3125)_{10}$$

$$* (6082.135)_9 = ()_3$$

$$9 \rightarrow 3$$

$$3^2 \rightarrow 3$$

$$\begin{array}{r} 3 \overline{) 6} \\ 2-0 \end{array} \quad \begin{array}{r} 3 \overline{) 8} \\ 2-2 \end{array} \quad \begin{array}{r} 3 \overline{) 2} \\ 0-2 \end{array} \quad \begin{array}{r} 3 \overline{) 5} \\ 1-2 \end{array}$$

$$b^n \rightarrow b$$

$$= (20002202.011112)_3$$

$$* (8) \xrightarrow{2^3} 2$$

$$(16) \xrightarrow{2^4} 2 \rightarrow 4$$

$$\begin{array}{cc} 0 & 000 \\ 1 & 001 \\ 2 & 010 \end{array}$$

$$\begin{array}{cc} 0 & 0000 \\ 1 & 0001 \\ 2 & 0010 \end{array}$$

$$\begin{array}{cc} 3 & 011 \\ 4 & 100 \end{array}$$

$$\begin{array}{cc} 3 & 0011 \\ 4 & 0100 \end{array}$$

$$\begin{array}{cc} 5 & 101 \\ 6 & 110 \end{array}$$

$$\begin{array}{cc} 5 & 0101 \\ 6 & 0110 \end{array}$$

$$\begin{array}{cc} 7 & 111 \\ 8 & 1000 \end{array}$$

$$\begin{array}{cc} 7 & 0111 \\ 8 & 1000 \end{array}$$

$$\begin{array}{cc} 9 & 1001 \\ 10(A) & 1010 \end{array}$$

$$\begin{array}{cc} 9 & 1001 \\ 10(A) & 1010 \end{array}$$

$$\begin{array}{cc} 11 & 1011 \\ 12 & 1100 \end{array}$$

$$\begin{array}{cc} 11 & 1011 \\ 12 & 1100 \end{array}$$

$$\begin{array}{cc} 13 & 1101 \\ 14 & 1110 \end{array}$$

$$\begin{array}{cc} 13 & 1101 \\ 14 & 1110 \end{array}$$

$$\begin{array}{cc} 15 & 1111 \\ 16 & 10000 \end{array}$$

$$\begin{array}{cc} 15 & 1111 \\ 16 & 10000 \end{array}$$

$$\begin{array}{cc} 17 & 10001 \\ 18 & 10010 \end{array}$$

$$\begin{array}{cc} 17 & 10001 \\ 18 & 10010 \end{array}$$

$$\begin{array}{cc} 19 & 10011 \\ 20 & 10100 \end{array}$$

$$\begin{array}{cc} 19 & 10011 \\ 20 & 10100 \end{array}$$

$$\begin{array}{cc} 21 & 10101 \\ 22 & 10110 \end{array}$$

$$\begin{array}{cc} 21 & 10101 \\ 22 & 10110 \end{array}$$

$$\begin{array}{cc} 23 & 10111 \\ 24 & 11000 \end{array}$$

$$\begin{array}{cc} 23 & 10111 \\ 24 & 11000 \end{array}$$

$$\begin{array}{cc} 25 & 11001 \\ 26 & 11010 \end{array}$$

$$\begin{array}{cc} 25 & 11001 \\ 26 & 11010 \end{array}$$

$$\begin{array}{cc} 27 & 11011 \\ 28 & 11100 \end{array}$$

$$\begin{array}{cc} 27 & 11011 \\ 28 & 11100 \end{array}$$

$$\begin{array}{cc} 29 & 11101 \\ 30 & 11110 \end{array}$$

$$\begin{array}{cc} 29 & 11101 \\ 30 & 11110 \end{array}$$

$$\begin{array}{cc} 31 & 11111 \\ 32 & 100000 \end{array}$$

$$\begin{array}{cc} 31 & 11111 \\ 32 & 100000 \end{array}$$

$$\begin{array}{cc} 33 & 100001 \\ 34 & 100010 \end{array}$$

$$\begin{array}{cc} 33 & 11111 \\ 34 & 100000 \end{array}$$

Convert into Binary :-

$$(i) (601.72)_8$$

$$= (110000001.111010)_2$$

$$01(2F.6F0E) = 2F(2.96A)$$

$$(ii) (D09.AF)_{16}$$

$$= (110100001001.10101111)_2$$

$$(iii) (310021.2233)_4 \rightarrow \text{Hexadecimal.}$$

$$\hookrightarrow (D09.AF)_{16}$$

$$* 001011011.101010 \rightarrow ()_8$$

$$\hookrightarrow (133.52)_8 \rightarrow \text{For signed bit don't use this.}$$

$$* 01011011.10101000 \rightarrow ()_{16}$$

$$\hookrightarrow (5B.A8)_{16}$$

$$* (657.43)_8 \rightarrow ()_{16}$$

$$\hookrightarrow (000110101111.10001100)_2$$

$$\hookrightarrow (1AF.8C)_{16}$$

$$* (D74.6B)_{16} \rightarrow ()_8$$

$$\hookrightarrow (110101110100.011010110)_2$$

$$\hookrightarrow (6564.326)_8$$

If $23 + 22 = 100$ then what is base?

$$(2 \times x^1 + 3 \times x^0) + (2 \times x^1 + 2 \times x^0) = (1 \times x^2 + 0 \times x^1 + 0 \times x^0)$$

$$\Rightarrow 2x + 3 + 2x + 2 = x^2$$

$$x^2 - 4x - 5 = 0$$

$$(x+1)(x-5) = 0$$

$$x = 5$$

$$(2) \sqrt{41} = 5 \rightarrow \text{find Base.}$$

$$41x = 5^2(25)$$

$$\sqrt{4x+1} = 5$$

$$4x+1 = 25$$

$$4x+1 = 25$$

$$2x = 4$$

$$4x = 24$$

$$x = 2$$

$$x = 6$$

Here $5^2 = 25$ should not be considered because it is not mentioned as decimal.

$$3. \frac{66}{6} = 11$$

$$\frac{6x+6}{6} = x+1 \Rightarrow 6x+6 = 6x+1$$

x is valid for any base.

$$\begin{aligned} \hookrightarrow x &> 6 \\ \text{or } x &\geq 7 \end{aligned}$$

$$4. \frac{312}{20} = 13.1$$

$$\frac{3x^2+x+2}{2x} = x+3+\frac{1}{x} = \frac{x^2+3x+1}{x}$$

$$\Rightarrow 3x^2+x+2 = 2x^2+6x+2x$$

$$x^2-5x=0$$

$$x(x-5)=0 \Rightarrow \boxed{x=5}$$

Conclusion:

1. Minimum possible base value is 2.
2. The base value can't be -ve, 0, 1.

* find no of valid solution of x and y ,

which can satisfy $(43)_8 = (x0)_y$

$$\begin{aligned} (43)_8 &= (x0)_y \\ \Rightarrow 32+3 &= xy \\ xy &= 35 \end{aligned}$$

$$\begin{aligned} \boxed{y > x} \\ \boxed{y > 0} \end{aligned}$$

$$\begin{array}{r} x \ y \\ 35 \rightarrow 1 \ 35 \\ 35 \ 1 \times \\ 5 \ 7 \\ 7 \ 5 \times \end{array}$$

$\therefore$ 2 valid sol are possible.

$$(ii) (123)_5 = (x8)_y$$

$$\begin{aligned} \Rightarrow (25+10+3) &= xy+8 \\ 38 &= xy+8 \\ xy &= 30 \end{aligned}$$

$$\boxed{y > x}$$

$$\boxed{y > 8}$$

$$\begin{array}{r} x \ y \\ 30 \rightarrow 1 \ 30 \\ 30 \ 1 \times \\ 2 \ 15 \\ 15 \ 2 \times \\ 3 \ 10 \\ 10 \ 3 \times \\ 5 \ 7 \times 5 \ 6 \vee (\alpha) \\ 6 \ 5 \times \end{array}$$

$\hookrightarrow$ 3 valid solutions.

Complements

1. Radix Complements (r 's)
2. Diminished radix complements ($(r-1)$'s)

$$(N)_r = \underbrace{d_{n-1}d_{n-2}\dots d_1d_0}_{n} \dots \underbrace{d_{-1}d_{-2}\dots d_{-m}}_m$$

$$r's = r^n - N \quad (\forall N \neq 0)$$

$$= 0 \quad (\forall N = 0)$$

$$(r-1)'s = r^n - r^{m-N}$$

$$1. 06092.105200_{10}$$

$$9's \ 93907.804799$$

$$10's \ 93907.894800$$

$$2. 00450.10230_8$$

$$7's \ 77327.67547$$

$$8's \ 77327.67550$$

$$3. 0609A.F304A00_{16}$$

$$15's \ F9F65.0CFB600$$

$$\hookrightarrow F's \ F9F65.0CFB5FF$$

$$4. 01101.1011000$$

$$1's \ 10010.0100111$$

$$2's \ 10010.0101000$$

$$10 \xrightarrow{2's} 10$$

$$100 \xrightarrow{2's} 100$$

$$1000 \xrightarrow{2's} 1000$$

Q. 2's complement of a non-zero binary integer with minimum 3 bits is itself when except _____

(a) LSB are one $\rightarrow (1110)$

(b) MSB are one (0111)

(c) LSB are Zero (0001)

(d) MSB are Zero (1000)

$$\begin{aligned} 0 &\rightarrow 1111 \\ 1 &\rightarrow 0000 \end{aligned}$$

$$\begin{aligned} 1 &\rightarrow 0000 \\ 0 &\rightarrow 1111 \\ 1 &\rightarrow 0000 \\ 0 &\rightarrow 1111 \\ 1 &\rightarrow 0000 \\ 0 &\rightarrow 1111 \end{aligned}$$

Signed Numbers Representation

1. Signed Magnitude Representation
2. 1's Complement representation
3. 2's Complement representation

1. Signed Magnitude Representation

↳ Sign + Magnitude

MSB (n-1) bits

MSB $\rightarrow 0 \rightarrow +ve$ 1001 $\rightarrow -1$

1 $\rightarrow -ve$ 1100 $\rightarrow -4$

0100 $\rightarrow 4$

-7 to 7

$-(2^{n-1}-1)$ to $+(2^{n-1}-1)$

1000 $\rightarrow -0$

0000 $\rightarrow +0$

0001 $\rightarrow +1$

$n \rightarrow -(2^{n-1}-1)$ to $(2^{n-1}-1)$

0100 $\rightarrow +4$

Note:

In this representation MSB is used only for sign and all the remaining bits used for magnitude with binary weights

$\rightarrow$ If MSB = 0, number is +ve

MSB = 1 number is -ve

$\rightarrow$ The range of decimal integers using n bits is $-(2^{n-1}-1)$ to $(2^{n-1}-1)$

$\rightarrow$ In this representation, 0 is having 2 representation (+0) $\rightarrow$ 000

-0 $\rightarrow$ (1000)

2. 1's complement Representation

MSB

$-(2^{n-1}-1)$ 2^{n-2} 2^{n-3} ... 2^1 2^0

-7 +4 +2 +1

b3 b2 b1 b0

1 0 0 0 $\rightarrow -7$

1001 $\rightarrow -6$

1100 $\rightarrow -3$

1110 $\rightarrow -1$

1111 $\rightarrow -0$

0000 $\rightarrow +0$

0001 $\rightarrow +1$

0100 $\rightarrow +4$

0101 $\rightarrow +5$

In this representation, the only MSB has -ve weight and weight is $-(2^{n-1}-1)$ and all remaining weights are having +ve weights.

$\rightarrow$ The range of decimal integer can be represented using $-(2^{n-1}-1)$ to $2^{n-1}-1$.

$\rightarrow$ In this representation also 0 is having 2 representation +0 (0000) & -0 (1000)

3. 2's complement representation

MSB

-2^{n-1} 2^{n-2} 2^{n-3} ... 2^1 2^0

n=3 -8 +4 +2 +1

b3 b2 b1 b0

1000 $\rightarrow -8$

1001 $\rightarrow -7$

1100 $\rightarrow -4$

1110 $\rightarrow -2$

1111 $\rightarrow -1$

0000 $\rightarrow 0$

0001 $\rightarrow +1$

0100 $\rightarrow +4$

0101 $\rightarrow +5$

0111 $\rightarrow +7$

$\rightarrow$ In this representation the only bit MSB has -ve weight and weight is -2^{n-1} & all remaining bits are having binary +ve weights.

$\rightarrow$ The range of decimal integers can be represented using n bits is -2^{n-1} to $+(2^{n-1}-1)$

$\rightarrow$ In this representation, 0 is having only 1 representation 0(0000).

	5-bit	1's	2's
4-bit	-7 to 7	-7 to 7	-8 to 7
5-bit	-15 to 15	-15 to 15	-16 to 15
6-bit	-31 to 31	-31 to 31	-32 to 31

<u>1's</u>		<u>2's</u>	
1010 → -5		1010 → -6	
(1's) 0101 → +5		(2's) 0110 → +6	
0110 → +6		0111 → +7	
(1's) 1001 → -6		(2's) 1001 → -7	

overflow: The no of bits used is not sufficient to represent result.

<u>2's</u>		<u>1's</u>	
1010 → -6		1010 → -5	
11010 → -6		11010 → -5	
111010 → -6		111010 → -5	
11111010 → -6		1111010 → -5	
0111 → +7		0110 → +6	
00111 → +7		000110 → +6	
000111 → +7		000110 → +6	
000000111 → +7		000110 → +6	

↳ Extension of MSB won't change value

↳ Sign extension property.

S.M

1101 → 5
11101 → 13 } Not satisfied.

Note :-

In 2's complement and 1's complement representation of signed no's extension of sign bit is not changing decimal equivalent value & it is called as sign extension property.

* If x_3 is signed bit of 8 bits —

a. $0000 x_3 x_2 x_1 x_0$

b. $1111 x_3 x_2 x_1 x_0$

c. $x_3 x_3 x_3 x_3 x_3 x_2 x_1 x_0$

d. $\bar{x}_3 \bar{x}_3 \bar{x}_3 \bar{x}_3 x_3 x_2 x_1 x_0$

Find Signed Magnitude, 1's, 2's representation of $(-27)_{10}$ using 8 bits.

↳ Signed → 100 11011

0's → 000 11011 (+27) in all

(1's) → 111 00100 (-27) in 1's

(2's) → 111 00101 (-27) in 2's

(ii) $(-35)_{10}$ → 8 bits

32 16 8 4 2 1
1 0 0 0 1 1

↳ Signed → 10100011

0's → 00100011 (+35)

(1's) → 11011100 (-35) in 1's

(2's) → 11011101 (-35) in 2's

25-07-85

Find Signed Magnitude, 1's, 2's representation of $(+24)_{10}$ using 8 bits.

24 → 11000

↳ Signed → 00011000

Complement need to done when it is a negative number.

we is same in all 1's, 2's.

(ii) $(-127)_{10}$

127 → 1111111

Signed → 11111111 (-127)

0's → 01111111 (+127)

(1's) → 10000000 (-127) in 1's

(2's) → 10000001 (-127) in 2's.

2 | 127
2 | 63-1
2 | 31-1
2 | 15-1
2 | 7-1
2 | 3-1
1-1

Find 2's complement representation of $(-128)_{10}$ using 8 bits

$$\rightarrow 128 \rightarrow 100000000$$

$(+128)$ is not possible is using 8 bits.

$(-128) \rightarrow$ Weights & range.

$$\hookrightarrow -2^{n-1}, (n=8)$$

$$= -2^{8-1} = -128 (100000000)$$

2's-Range is from -128 to 127 .

Subtraction using complements:

$$(A)_x - (B)_x = (\text{difference})_x$$

I. x's complement method

1. Add A to x's complement of B

2. observe the obtained in step-1

(i) If $A > B$, Sum is producing final carry. The result is +ve and discard carry, sum is expected difference.

(ii) If $A < B$, Sum is not producing final carry, the result is -ve, take x's complement of sum & produce -ve sign to get expected difference.

II. $(x-1)$'s Complement method

1. Add A to $(x-1)$'s Complement of B.

2. observe the result obtained in step 1.

(i) If $A > B$, Sum is producing final carry the result is +ve & add carry to LSB (least significant digit) of sum to get expected difference.

(ii) If $A < B$, Sum is not producing final carry, the result is -ve, take $(x-1)$'s complement of sum and produce -ve sign in front to get expected difference.

$$(i) (11110)_2 - (1001)_2$$

$\hookrightarrow$ Make equal digits in both A & B then solve.

$$(11110)_2 - (01001)_2$$

2's

$$\begin{array}{r} 11110 \\ + 10111 \\ \hline 10101 \end{array}$$

$$\begin{array}{r} 11110 \\ + 10110 \\ \hline 10100 \end{array}$$

1's

$$\begin{array}{r} 11110 \\ - 10001 \\ \hline 01111 \end{array}$$

A > B

$$\begin{array}{r} 11110 \\ - 10110 \\ \hline 10100 \\ - 10101 \\ \hline \end{array}$$

$$(ii) (01001)_2 - (11110)_2$$

2's

$$\begin{array}{r} 01001 \\ + 00010 \\ \hline 01011 \end{array}$$

A < B

$$2's \rightarrow (10101)$$

1's

$$\begin{array}{r} 01001 \\ + 00001 \\ \hline 01010 \end{array}$$

01010 $\rightarrow$ A < B

$$-(10101)$$

(iii) $(6059.62)_{10} - (504.9)_{10}$ using 10's complement & 9's comp.

$$\hookrightarrow (6059.62)_{10} - (0504.90)_{10}$$

10's

$$\begin{array}{r} 6059.62 \\ + 9495.10 \\ \hline 15554.72 \end{array}$$

$$10's \text{ comp } (0504.90) = 9's + 1$$

$$\begin{array}{r} 9's \quad 6059.62 \\ 9495.09 \\ \hline \end{array}$$

$$\begin{array}{r} 15554.72 \\ + 1 \\ \hline 5554.72 \end{array}$$

$$(iv) (0504.90)_{10} - (6059.62)_{10}$$

10's

$$\begin{array}{r} 0504.90 \\ + 3940.38 \\ \hline 4445.28 \\ 10's \rightarrow 5554.71 \\ + 1 \\ \hline 5554.72 \end{array}$$

9's

$$\begin{array}{r} 0504.90 \\ + 3940.37 \\ \hline 4445.27 \end{array}$$

A < B

$\hookrightarrow$ 9's

$$\begin{array}{r} 4445.27 \\ - 5554.72 \\ \hline \end{array}$$

Boolean Algebra

↳ Based on switching theory,

1. logical OR (+)

$$0+0=0$$

$$0+1=1$$

$$1+0=1$$

$$1+1=1$$

2. logical AND (·)

$$0 \cdot 0 = 0$$

$$0 \cdot 1 = 0$$

$$1 \cdot 0 = 0$$

$$1 \cdot 1 = 1$$

3. logical Inverse (Not) (-)

$$\bar{0} = 1$$

$$\bar{1} = 0$$

Dual - To get dual function replace,

(i) $OR \leftrightarrow AND$

$0 \leftrightarrow 1$

Note:

If a function is true, then dual of that function is also true.

Boolean Rules:

$$1. A+0=A$$

$$2. A+1=1$$

$$3. A+A=A$$

$$4. A+\bar{A}=1$$

$$5. \bar{\bar{A}}=A$$

$$6. A \cdot 1=A$$

$$7. A \cdot 0=0$$

$$8. A \cdot A=A$$

$$9. A \cdot \bar{A}=0$$

Note:

If dual of a function is itself then the function is called self dual.

$$1. A+B=B+A$$

$$A \cdot B=B \cdot A$$

$$2. A+(B+C)=(A+B)+C$$

$$A \cdot (B \cdot C)=(A \cdot B) \cdot C$$

$$3. A \cdot (B+C)=A \cdot B+A \cdot C$$

$$A+B \cdot C=(A+B)(A+C)$$

$$1. A+AB=A$$

$$\hookrightarrow A(1+B)$$

$$A \cdot 1$$

$$A$$

$$\xrightarrow{\text{dual}} A(A+B)=A$$

$$\hookrightarrow A \cdot A+A \cdot B$$

$$A+AB$$

$$A$$

$$2. A+\bar{A}B=(A+B) \xrightarrow{\text{dual}} A \cdot (\bar{A}+B)=AB$$

{ Distributive law
apply LHS }

$$3. \overline{A+B}=\bar{A} \cdot \bar{B} \xrightarrow{\text{dual}} \overline{AB}=\bar{A}+\bar{B}$$

{ De-morgan
law }

$$4. \overline{A+B+C}=\bar{A} \cdot \bar{B} \cdot \bar{C} \xrightarrow{\text{dual}} \overline{ABC}=\bar{A}+\bar{B}+\bar{C}$$

$$5. AB+\bar{A}C+BC=AB+\bar{A}C$$

{ consensus law }

$$\hookrightarrow (A+B)(\bar{A}+C)(B+C)=(A+B) \cdot (\bar{A}+C)$$

$$6. AB+\bar{A}C+BC=AB+\bar{A}C$$

$$\hookrightarrow AB+\bar{A}C+BC(A+\bar{A})$$

$$=AB+\bar{A}C+ABC+\bar{A}BC$$

$$=AB(1+C)+\bar{A}C(1+B)$$

$$=AB+\bar{A}C$$

Eg: of consensus

$$\hookrightarrow (i) \check{A}B+\check{A}\check{C}+\overset{aa}{BC}=AB+\bar{A}C$$

$$(ii) \check{A}\check{B}+\check{B}\check{C}+\overset{aa}{AC}=AB+\bar{B}C$$

$$(iii) \overset{aa}{AB}+\overset{aa}{B}\check{C}+\overset{aa}{AC}=AB+\bar{B}C$$

$$(iv) Af_1+\bar{A}f_2+f_1f_2=Af_1+\bar{A}f_2$$

$$(v) \bar{f}g_1+\bar{f}g_2+g_1g_2=\bar{f}g_1+\bar{f}g_2$$

$$* f(A,B,C)=(\underbrace{A\bar{A}}_{\text{variables}}, \underbrace{B\bar{B}}_{\text{Literals}}, \underbrace{C\bar{C}}_{\text{Literals}})$$

variables

Literals

$$\begin{aligned}\overline{A + \overline{B}C} &= \overline{A} \cdot \overline{\overline{B}C} \\ &= \overline{A} \cdot (\overline{\overline{B}} + \overline{C}) \\ &= \overline{A} \cdot (B + \overline{C})\end{aligned}$$

For complex function or any function using de-morgan principle, we can find easy.

- Note
1. Write dual of given function
 2. complement each literal.

priority → 1st go with AND in case of (+, ·) are present. Next OR.

eg:- $A + \overline{B}C = A(\overline{B} + C)$
 $= \overline{A}(B + \overline{C})$

2. $\overline{A} + B(C + \overline{D})$
 $= \overline{A}(B + C \cdot \overline{D})$ → Here, priority is given to paranthesis.
 $= A(\overline{B} + \overline{C}D)$

3. $A + \overline{B}(C + \overline{D}E)$
 $= A[\overline{B} + C(\overline{D} + E)]$
 $= \overline{A}[B + \overline{C}(D + \overline{E})]$

4. $\overline{A}\overline{B} + \overline{C}D$
 $= (\overline{A} + \overline{B})(\overline{C} + D)$
 $= (\overline{A} + B)(C + \overline{D})$

* Truth Table for $f(A, B, C)$

A	B	C	$f(A, B, C)$
0	0	0	1
1	0	0	0
2	0	1	1
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

$$\begin{aligned}f(A, B, C) &= \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + \overline{A}B\overline{C} + \overline{A}BC + ABC \\ &\quad \downarrow \text{Min terms} \\ &= m_0 + m_2 + m_3 + m_5 + m_7 \\ &= \sum m(0, 2, 3, 5, 7)\end{aligned}$$

SOP Canonical SOP

$$\overline{f}(A, B, C) = \overline{A}\overline{B}C + A\overline{B}\overline{C} + AB\overline{C}$$

$$\begin{aligned}f(A, B, C) &= \overline{\overline{f}(A, B, C)} \\ &= \overline{\overline{A}\overline{B}C + A\overline{B}\overline{C} + AB\overline{C}} \\ &= \overline{\overline{A}\overline{B}C} \cdot \overline{A\overline{B}\overline{C}} \cdot \overline{AB\overline{C}}\end{aligned}$$

$$\begin{aligned}\overline{f}(A, B, C) &= (\overline{A} + \overline{B} + \overline{C})(\overline{A} + B + C)(\overline{A} + \overline{B} + C) \\ &\quad \downarrow \text{Max terms} \\ &= M_1 \cdot M_4 \cdot M_6 \\ &= \prod M(1, 4, 6)\end{aligned}$$

POS Canonical form.

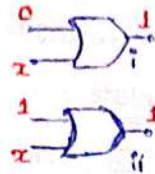
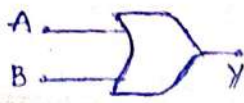
↳ SOP and POS are 2 equations of same function.

$$\sum m(0, 2, 3, 5, 7) = \prod M(1, 4, 6)$$

Logic Gates

1. OR

A	B	$Y = A + B$
0	0	0
0	1	1
1	0	1
1	1	1



(i) → Enable input = 0

(ii) → Disable input = 1

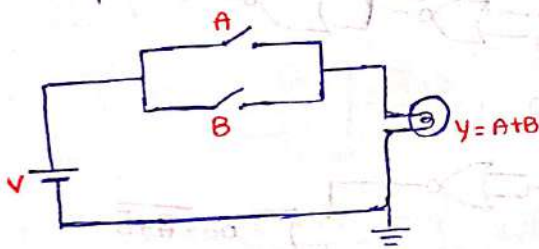
Enable input:

The input which allowing other inputs to pass through gate.

Disable input:

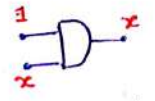
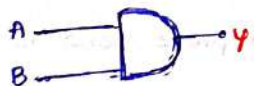
The input which is not allowing other inputs to pass through gate.

Circuit representation:



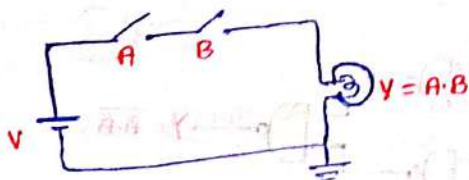
2. AND

A	B	$Y = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

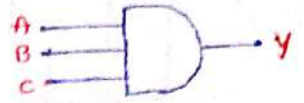


disable = 0
i/p

Enable = 1
i/p

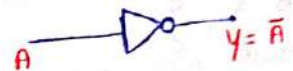


A	B	C	$Y = ABC$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

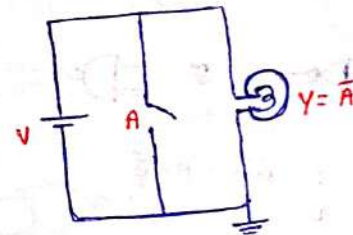


3. INVERTER (NOT)

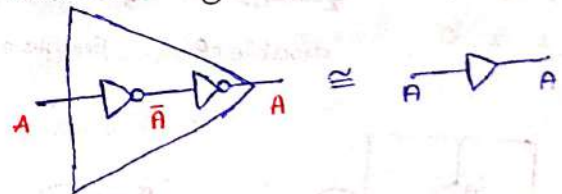
A	$Y = \bar{A}$
0	1
1	0



Here No enable or disable (single i/p)



Non-inverting Buffer:



4. NOR

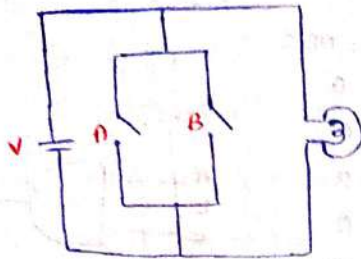
A	B	$Y = \overline{A + B}$
0	0	1
0	1	0
1	0	0
1	1	0



Enable = 0

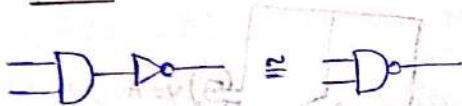
disable = 1



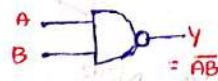
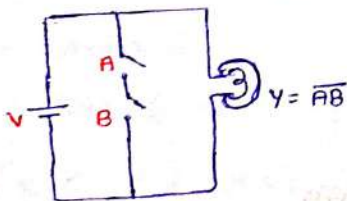
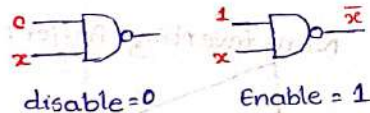


A	B	C	$Y = \overline{A+B+C} = \bar{A} \cdot \bar{B} \cdot \bar{C}$
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

5. NAND



A	B	$Y = \overline{A \cdot B} = \bar{A} + \bar{B}$
0	0	1
0	1	1
1	0	1
1	1	0



A	B	C	$Y = \overline{ABC} = \bar{A} + \bar{B} + \bar{C}$
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0



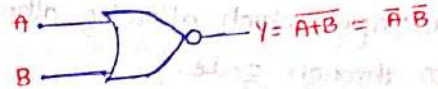
Universal Gate:

If it is possible to implement any logic function using only the gate then gate is called universe.

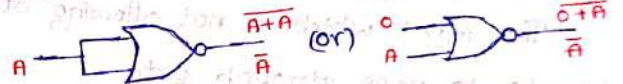
Note:

To prove the gate is universal, it must be possible to implement 3 basic gates (AND, OR, NOT) using only the gates.

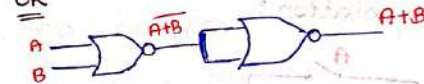
①



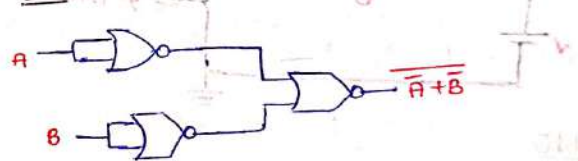
↳ NOT



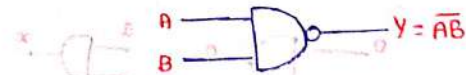
OR



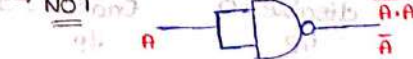
AND



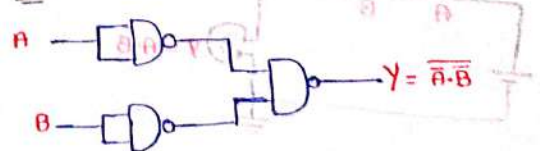
2. To prove NAND is universal.



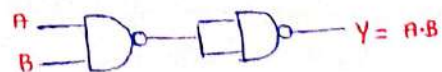
↳ NOT

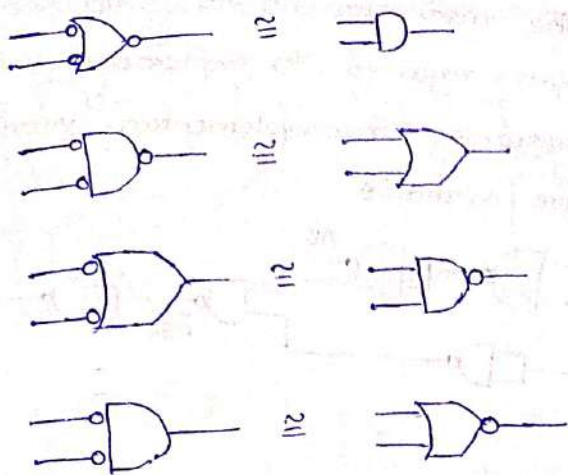


↳ OR

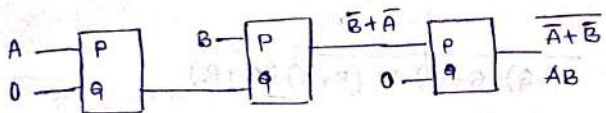
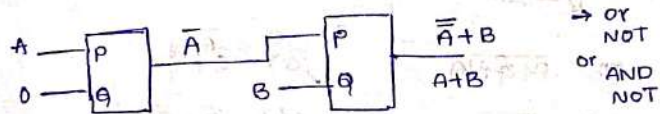
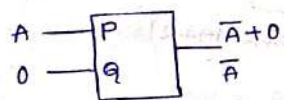
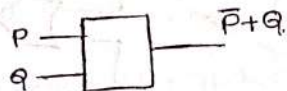


↳ AND

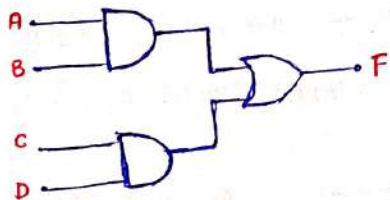




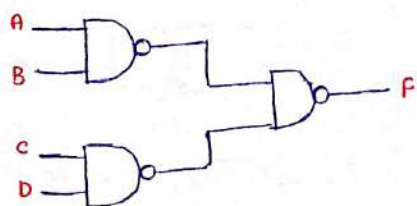
Q. In the given gate, which universal gate is possible?



Q. $F = AB + CD$

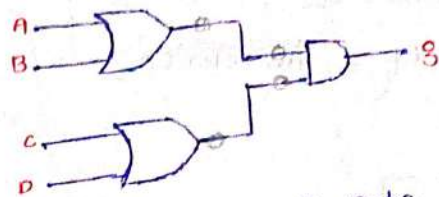


2 level $\rightarrow$ AND, OR

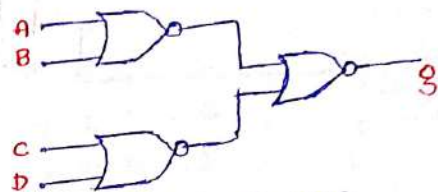


2-level NAND-NAND

Q. $g = (A+B)(C+D)$



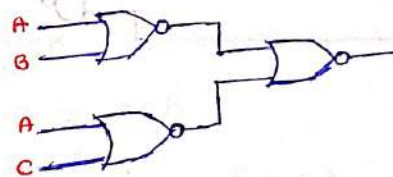
2 level NOR-AND Gate



2 level NOR-NOR

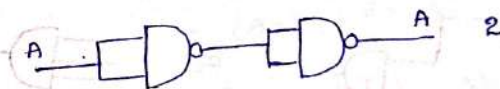
Q. The min. no of 2 i/p's NOR Gates required to implement the function $A+BC$ 3.

$$A+BC = (A+B)(A+C)$$



Q. The min no of 2 i/p NAND gates required to implement $A+A\bar{B}+ABC$

$$A+A\bar{B}+ABC = A$$



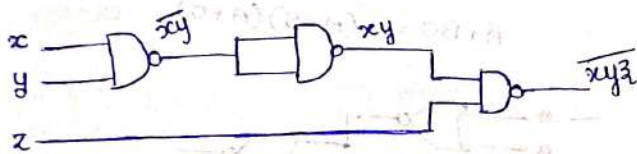
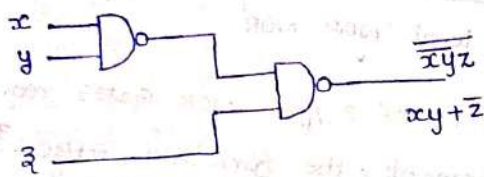
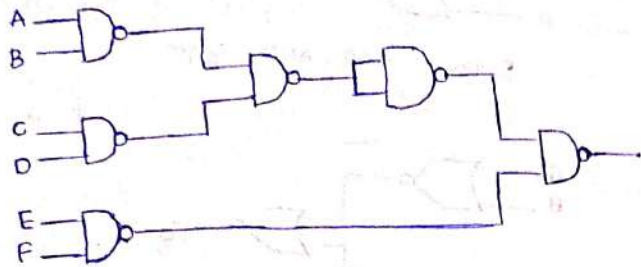
$$f = A(1+BC) + A\bar{B}$$

$$f = A + A\bar{B}$$

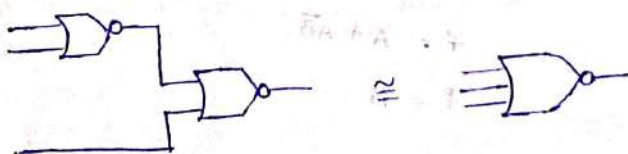
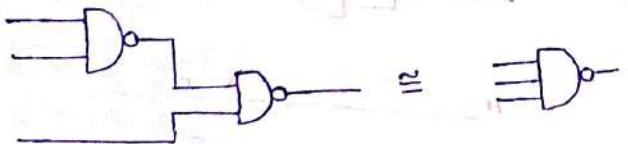
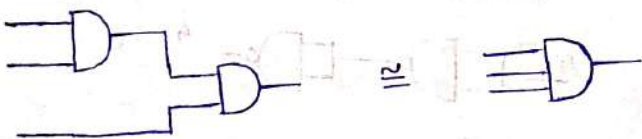
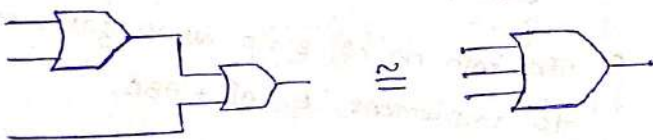
$$f = A$$

Q. The min No of NAND Gates required to $AB + CD + EF$.

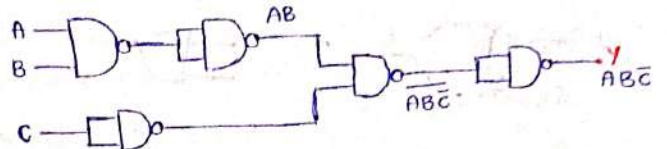
$$AB + CD + EF = \overline{\overline{AB + CD + EF}}$$



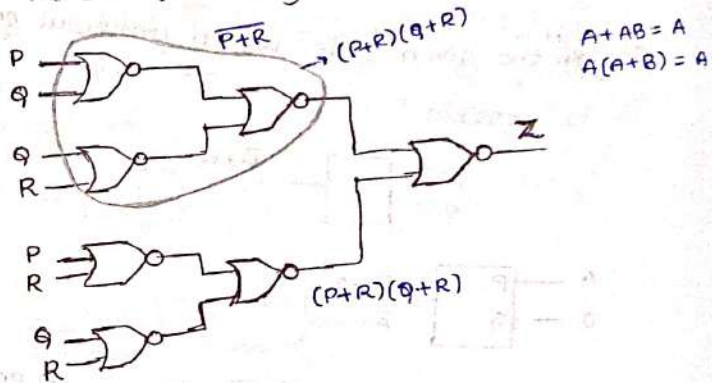
*



Q. The minimum no of 2 I/P NAND Gates required to implement $AB\bar{C}$, assuming uncomplementary variables are available.



Q. In the following circuit o/p Z is.



(a) $\overline{P+Q+R}$

(b) $\overline{P+R}$

(c) $\overline{Q+R}$

(d) $\overline{P+Q+R}$

$$Z = \overline{(P+Q)(Q+R) + (P+R)(Q+R)}$$

$$= \overline{(Q+R)\{P+(Q+R)\}}$$

$$= \overline{Q+R} + \overline{P+Q+R}$$

$$= \overline{Q} \cdot \overline{R} + \overline{P} \overline{Q} \overline{R}$$

$$= \overline{Q} \overline{R} (1 + \overline{P})$$

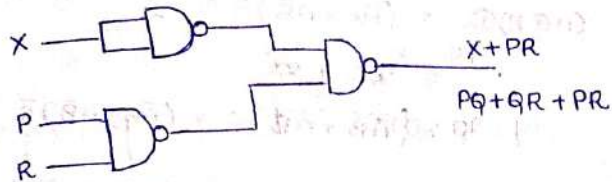
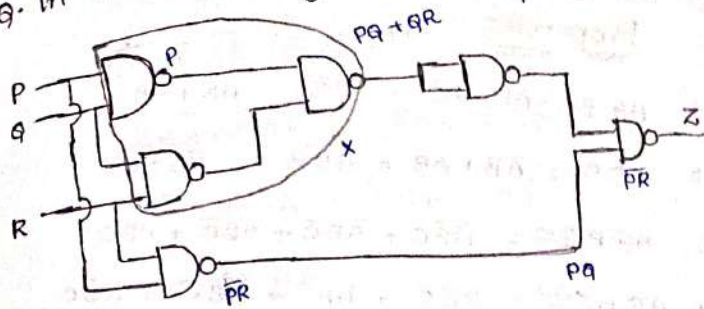
$$= \overline{Q} \overline{R} (1)$$

$$= \overline{Q+R}$$

$$[\because \overline{A \cdot B} = \overline{A} + \overline{B}]$$

$$[\overline{A \cdot B} = \overline{A+B}]$$

Q. In the following circuit o/p z is.



$$\overline{PQ+QR+PR} = \overline{PQ} \cdot \overline{QR} \cdot \overline{PR} = \overline{PQ} + \overline{QR} + \overline{PR}$$

$$f = (P+Q)(Q+R)(P+R)$$

$$= (PQ + Q + PR + QR)(P+R)$$

$$= (PQ + PQ + PR + PQR + PQR + QR + PR + QR)$$

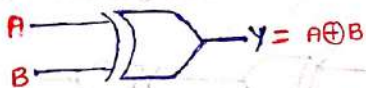
$$= PQ + QR + PR + PQR$$

$$= PQ + QR + PR$$

Ex-OR

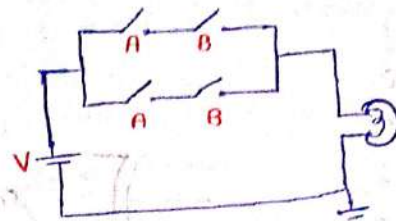
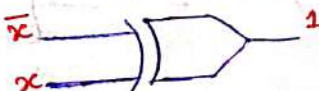
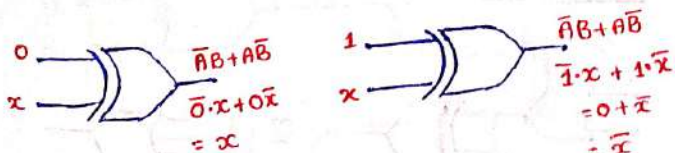
$$A \oplus B = A \oplus B$$

0	0	0
0	1	1
1	0	1
1	1	0



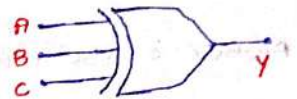
$$A \oplus B = \overline{A}B + A\overline{B} \rightarrow \text{sop using min terms}$$

$$= (A+B) \cdot (\overline{A}+\overline{B}) \rightarrow \text{pos using max}$$



$$A \oplus B \oplus C = A \oplus B \oplus C$$

0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1



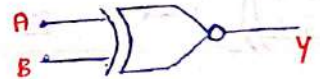
$$A \oplus B \oplus C = \overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\overline{C} + ABC$$

$$= (A+B+C)(A+\overline{B}+\overline{C})(\overline{A}+B+\overline{C})(\overline{A}+\overline{B}+C)$$

Ex-NOR

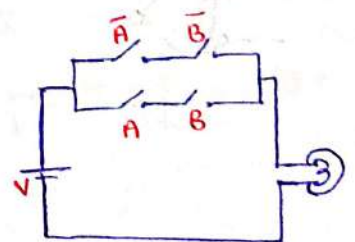
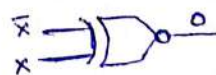
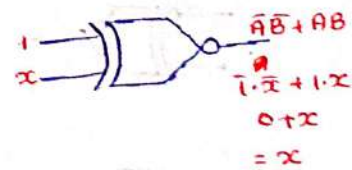
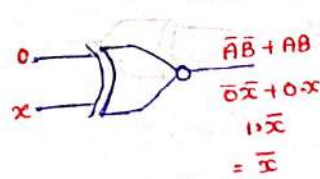
$$A \odot B = A \odot B$$

0	0	1
0	1	0
1	0	0
1	1	1



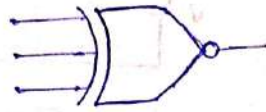
$$A \odot B = \overline{A}\overline{B} + AB$$

$$= (A+\overline{B})(\overline{A}+B)$$



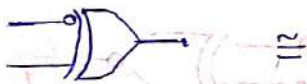
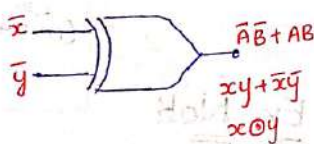
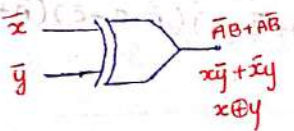
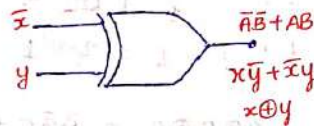
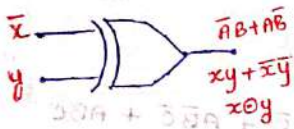
$$A \quad B \quad C \quad y = A \oplus B \oplus C$$

0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0



$$A \oplus B \oplus C = \bar{A}\bar{B}\bar{C} + \bar{A}BC + A\bar{B}\bar{C} + ABC$$

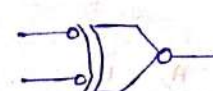
$$= (A+B+C)(A+\bar{B}+\bar{C})(\bar{A}+B+C)(\bar{A}+\bar{B}+\bar{C})$$



≡



≡



≡



26-07

Properties

$$1. A \oplus B = \bar{A}B + A\bar{B} = \overline{A\bar{B}} = \overline{\bar{A}B + AB}$$

$$2. A \oplus B = \bar{A}\bar{B} + AB = \overline{A\bar{B}} = \overline{\bar{A}B + AB}$$

$$3. A \oplus B \oplus C = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

$$4. A \oplus B \oplus C = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

$$5. (A \oplus B) \oplus C = (\bar{A}B + A\bar{B}) \oplus C$$

$$= \bar{x}C + x\bar{C}$$

$$= (\bar{A}B + A\bar{B})C + (\bar{A}B + A\bar{B})\bar{C}$$

$$= (\bar{A}\bar{B} + AB)C + \bar{A}B\bar{C} + A\bar{B}\bar{C}$$

$$= \bar{A}\bar{B}\bar{C} + ABC + \bar{A}B\bar{C} + A\bar{B}\bar{C}$$

$$= A \oplus B \oplus C$$

$$6. (A \oplus B) \oplus C = (\bar{A}B + A\bar{B}) \oplus C$$

$$= \bar{x}C + x\bar{C}$$

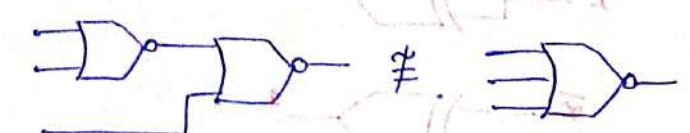
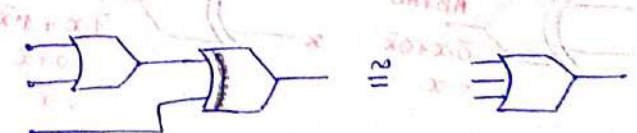
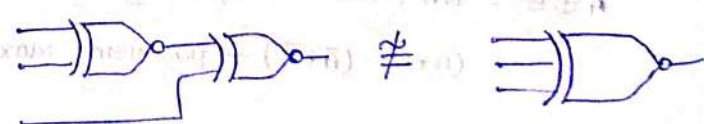
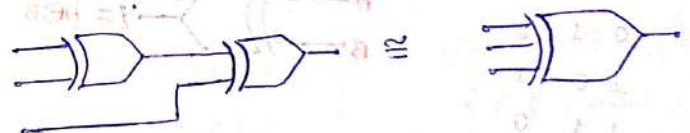
$$= (\bar{A}\bar{B} + AB)\bar{C} + (\bar{A}\bar{B} + AB)C$$

$$= (\bar{A}B + A\bar{B})\bar{C} + \bar{A}\bar{B}C + ABC$$

$$= \bar{A}B\bar{C} + A\bar{B}\bar{C} + \bar{A}\bar{B}C + ABC$$

$$\neq A \oplus B \oplus C$$

$$= A \oplus B \oplus C$$



Q. $(A \oplus B) \oplus A$

$$= (\bar{A}B + A\bar{B}) \oplus A$$

$$= (\bar{A}B + A\bar{B})A + (\bar{A}B + A\bar{B})\bar{A}$$

$$= (\bar{A}\bar{B} + AB)A + (\bar{A}B + 0)$$

$$= 0 + AB + \bar{A}B$$

$$= (A + \bar{A})B$$

$$= B$$

* $(A \oplus B) \oplus A = B$
 $(A \oplus B) \oplus B = A$
 $(A \odot B) \odot A = B$
 $(A \odot B) \odot B = A$

Q. $A \oplus B \oplus AB$

$$A \oplus B \oplus C = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

$$= \bar{A}\bar{B}(AB) + \bar{A}B(\bar{A}B) + A\bar{B}(\bar{A}B) + AB(AB)$$

$$= 0 + \bar{A}B(\bar{A} + \bar{B}) + A\bar{B}(\bar{A} + B) + AB$$

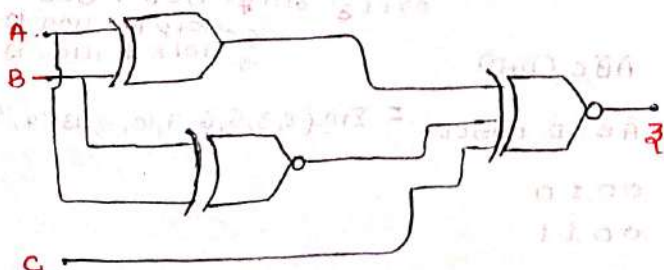
$$= \bar{A}B + 0 + 0 + A\bar{B} + AB$$

$$= \bar{A}B + A\bar{B} + AB$$

$$= B(\bar{A} + A) + A(\bar{B} + B)$$

$$= A + B$$

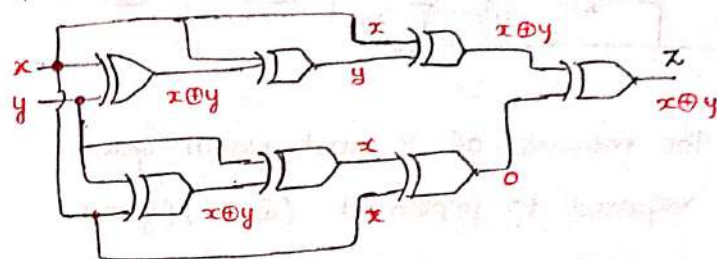
Q. Which of the following ABC combination can result o/p $z=1$ in the circuit shown.



- A B C
- a. 0 1 0
 - b. 1 0 0
 - c. 1 1 0
 - d. 0 0 1

L, at least 2 ones should be there to become $z=1$

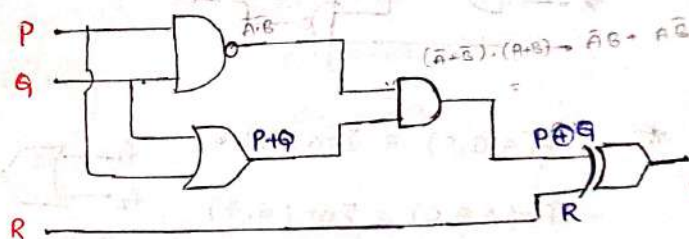
Q. In the following circuit output z is.



- a. 0 b. 1 c. $x \oplus y$ d. $x \odot y$



Q. In the following circuit output z is.



$$z = P \oplus Q \oplus R$$

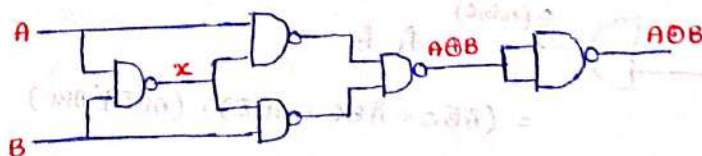
* $A \oplus B = \bar{A}B + A\bar{B}$

$$(A+B)(\bar{A} + \bar{B})$$

$$= (A+B)\bar{A}\bar{B}$$

$$= (A+B)x$$

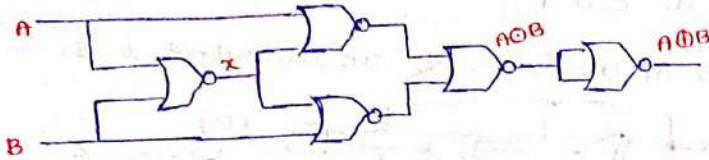
$$= Ax + Bx$$



$$A \odot B = \bar{A}\bar{B} + AB = (A+B)(\bar{A}+\bar{B})$$

$$= x + AB$$

$$= (x+A)(x+B)$$

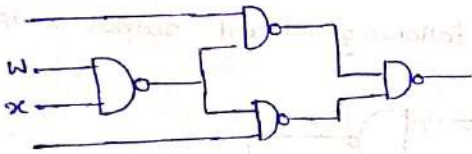


Q. The min. No of 2 input NAND gates required to implement $(\bar{w} + \bar{x})(y+z)$

$$(\bar{w} + \bar{x})(y+z) = \bar{w}\bar{x}(y+z)$$

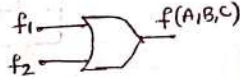
$$= P(y+z)$$

$$= Py + Pz$$



$$* f_1(A,B,C) = \sum m(1,3,6)$$

$$f_2(A,B,C) = \sum m(6,7)$$



$$(i) f_1(A,B,C) = \bar{A}\bar{B}C + \bar{A}BC + AB\bar{C}$$

$$f_2(A,B,C) = AB\bar{C} + ABC$$

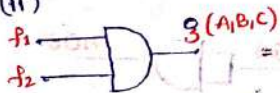
$$f(A,B,C) = f_1 + f_2$$

$$= \bar{A}\bar{B}C + \bar{A}BC + AB\bar{C} + ABC + AB\bar{C}$$

$$= \bar{A}\bar{B}C + \bar{A}BC + AB\bar{C} + ABC$$

$$= \sum m(1,3,6,7) \rightarrow \text{Union}$$

(ii)



$$= (\bar{A}\bar{B}C + \bar{A}BC + AB\bar{C}) \cdot (AB\bar{C} + ABC)$$

$$= 0 + 0 + AB\bar{C} + 0 + 0 + 0$$

$$= AB\bar{C}$$

$$= \sum m(6) \rightarrow \text{Intersection}$$

(iii)



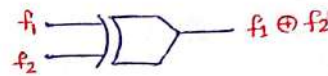
$$\bar{f}_2 = \overline{AB\bar{C} + ABC}$$

$$= \overline{AB\bar{C}} \cdot \overline{ABC}$$

$$= (\bar{A} + \bar{B} + C)(\bar{A} + \bar{B} + \bar{C})$$

$\sum m(6,7) = x$ $\pi m(6,7) = (\sum m(0,1,2,3,4,5))$
 $\pi m(0,1,2,3,4,5)$ These not equal built func.
 Same.

(iv)



$$f_1 = \sum m(1,3,6)$$

$$\bar{f}_1 = \sum m(0,2,4,5,7)$$

$$f_2 = \sum m(6,7)$$

$$\bar{f}_2 = \sum m(0,1,2,3,4,5)$$

$$f_1 \oplus f_2 = \bar{f}_1 f_2 + f_1 \bar{f}_2$$

$$= 7 + 1,3$$

$$= \sum m(1,3,7)$$

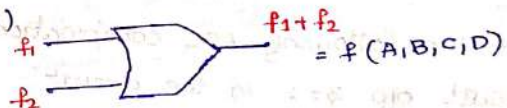
$$Q. f_1(A,B,C) = \sum m(1,3,6,7)$$

$$f_2(B,C,D) = \sum m(2,5,6)$$

$$f_1(A,B,C) = \bar{A}\bar{B}C + \bar{A}BC + AB\bar{C} + ABC$$

$$f_2(B,C,D) = \bar{B}C\bar{D} + B\bar{C}D + BCD$$

(i)



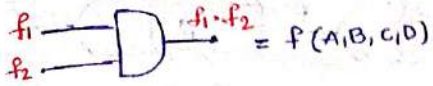
$$f(A,B,C,D) = \bar{A}\bar{B}C + \bar{A}BC + AB\bar{C} + ABC + \bar{B}C\bar{D} + B\bar{C}D + BCD$$

$$\bar{A}\bar{B}C(\bar{D}+D)$$

$$\bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}CD = \sum m(2,3,5,6,7,10,12,13,14,15)$$

$$0010$$

$$0011$$



$$f(A, B, C, D) = (\bar{A}\bar{B}C + \bar{A}BC + A\bar{B}C + ABC) \cdot (\bar{B}C\bar{D} + B\bar{C}D + BC\bar{D})$$

$$= (\bar{A}\bar{B}C\bar{D} + 0 + 0) + (0 + 0 + \bar{A}BC\bar{D}) + (0 + A\bar{B}C\bar{D} + 0) + (0 + 0 + ABC\bar{D})$$

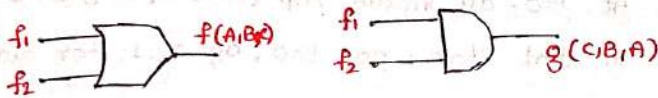
$$f(A, B, C, D) = \bar{A}\bar{B}C\bar{D} + \bar{A}BC\bar{D} + A\bar{B}C\bar{D} + ABC\bar{D}$$

0010 0110 1101 1110
 42 6 13 14

$$f(A, B, C, D) = \sum m(2, 6, 13, 14)$$

Q. $f_1(A, B, C) = \sum m(1, 3, 7)$

$$f_2(C, B, A) = \sum m(5, 6, 7)$$



$$f_1 = \bar{A}\bar{B}C + \bar{A}BC + ABC$$

$$f_2 = C\bar{B}A + CB\bar{A} + CBA$$

$$= \bar{A}BC + \bar{A}BC + ABC$$

$$f_2(A, B, C) = \sum m(3, 5, 7)$$

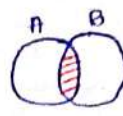
$$\rightarrow \therefore f(A, B, C) = \sum (1, 3, 5, 7)$$

$$f_1(C, B, A) = \sum m(4, 6, 7)$$

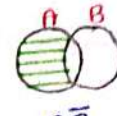
$$\rightarrow g(C, B, A) = \sum m(4, 6, 7) \cdot \sum m(5, 6, 7)$$

$$= \sum m(6, 7)$$

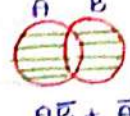
Venn Diagram



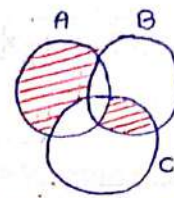
$$A \cdot B$$



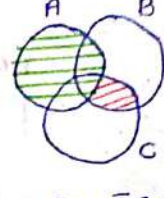
$$A\bar{B}$$



$$A\bar{B} + \bar{A}B$$



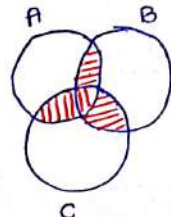
$$A\bar{B}\bar{C} + \bar{A}B\bar{C}$$



$$A + \bar{A}BC$$

$$\text{or } (A + \bar{A})(A + BC)$$

$$= A + BC$$

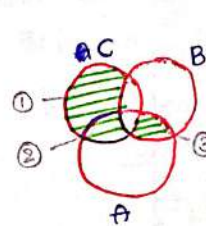


$$AB + BC + AC$$

$$\Rightarrow AB\bar{C} + \bar{A}BC + A\bar{B}C + ABC$$

$$= AB(C + \bar{C}) + BC(\bar{A} + A) + AC(\bar{B} + B)$$

$$= AB + BC + CA$$



$$\rightarrow \textcircled{1} \bar{A}\bar{B}C + \textcircled{2} \bar{A}BC + \textcircled{3} A\bar{B}C$$

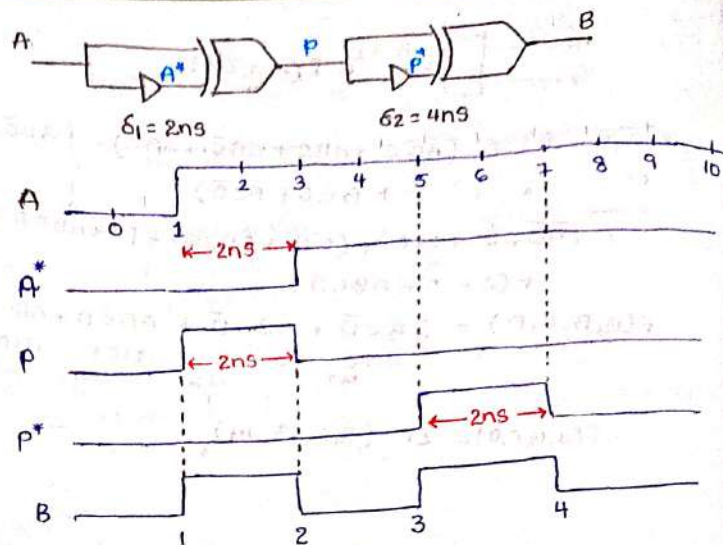
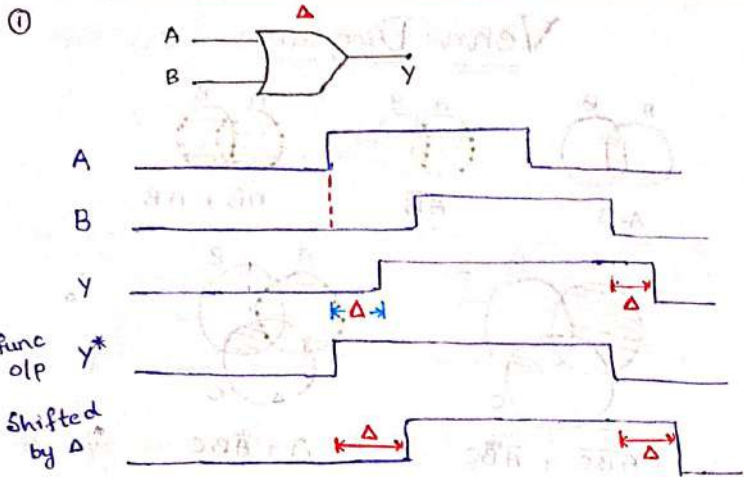
$$= \bar{B}C(\bar{A} + A) + A\bar{B}C$$

$$= \bar{B}C + A\bar{B}C$$

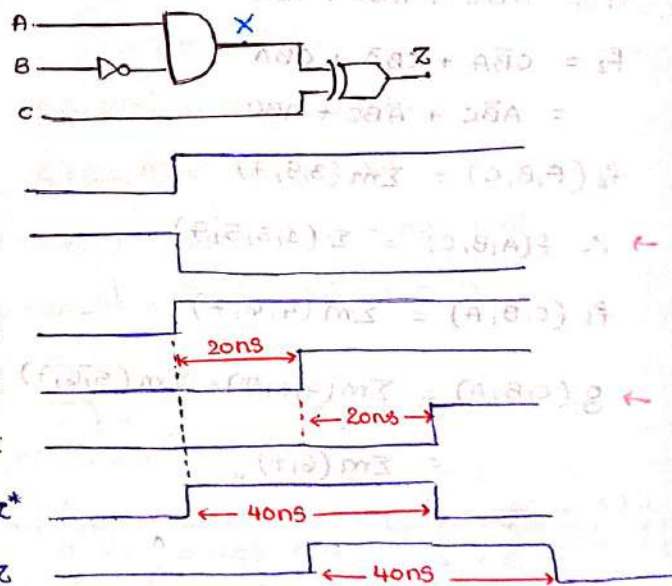
propagation delay :- (Δ or δ)

↳ Delay in propagating changes from input to output

or
If there is a change in o/p due to change in i/p, the change occur



Q. All logic gates shown in the fig have propagation delay of 20ns. let $A=C=0$ & $B=1$ until $t=0$. At $t=0$, all inputs flip ($A=C=1$ & $B=0$) & remain in that state. for $t>0$, o/p $Z=1$, for duration (in ns) of _____.



Q. Two 2's complement numbers having sign bits x and y are added and the sign of bit of result is z. Then the occurrence of overflow is indicated by Boolean function

- (a) xyz (b) $\bar{x}\bar{y}\bar{z}$
(c) $\bar{x}\bar{y}z + xy\bar{z}$ (d) $xy + yz + zx$

$$x \ y \ z \quad f = \bar{x}\bar{y}z + xy\bar{z}$$

0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	0	0	0	0	0
0	1	1	0	0	0	0	0
1	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0
1	1	0	1	0	0	0	0
1	1	1	0	0	0	0	0

eg: overflow

+3	+4	-3	-4
+4	+5	-5	-5
+7	+9	-8	-9
0011	0100	1011	1100
0100	0101	1101	1011
0111	1001	0100	0111
↓	↓	↓	↓
+7	-9	-8	↓

Q. Let $x_1 \oplus x_2 \oplus x_3 \oplus x_4 = 0$ where x_1, x_2, x_3, x_4 are Boolean variables and $\oplus$ is the XOR operator. Which one of the following is true always.

$x_1 \ x_2 \ x_3 \ x_4$

0 0 0 0

1 1 1 1

1 1 0 0

0 0 1 1

1 0 1 0

0 1 0 1

1 0 0 1

0 1 1 0

(a) $x_1 x_2 x_3 x_4 = 0$

(b) $x_1 x_3 + x_2 = 0$

(c) $\bar{x}_1 \oplus \bar{x}_3 = \bar{x}_2 \oplus \bar{x}_4$

(d) $x_1 + x_2 + x_3 + x_4 = 0$

Q. A 3 input majority gate is defined by the logic function $M(a,b,c) = ab + bc + ca$. Which one of the following gates is represented by the function $M(\bar{M}(a,b,c), M(a,b,\bar{c}), c)$?

(a) 3-Input NAND

(c) 3-Input NOR

(b) 3-Input XOR

(d) 3-Input XNOR

a	b	c	$M(a,b,\bar{c})$	$\bar{M}(a,b,c)$	$M(a,b,\bar{c})$	c	f
0	0	0	0	1	0	0	0
0	0	1	0	1	0	1	1
0	1	0	0	1	1	0	1
0	1	1	1	0	0	1	0
1	0	0	0	1	1	0	1
1	0	1	1	0	0	1	0
1	1	0	1	0	1	0	0
1	1	1	1	0	1	1	1

$$f = \sum m(1,2,4,7) = a \oplus b \oplus c$$

K-Map

* for m variable K-Map,

no of variables = m

grouping = 2^n adjacent squares

Eliminating = n variables

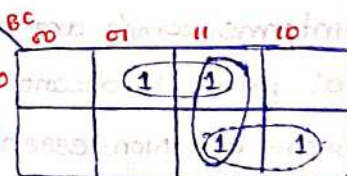
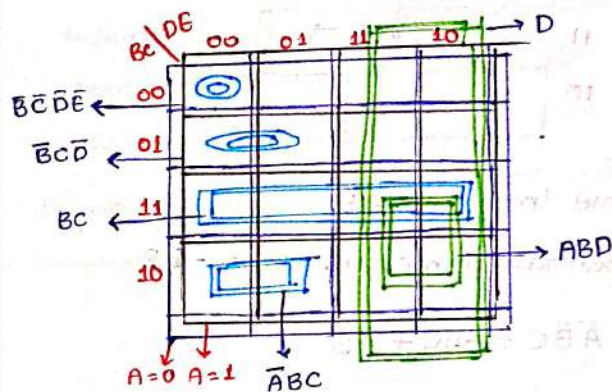
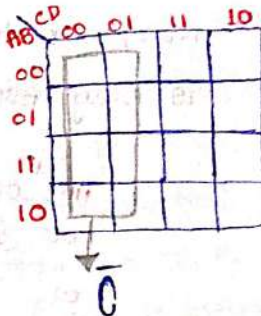
resultant = m-n

eg:- m=4

grouping (8) = 2^3

eliminating = 3

Resultant = $4-3$
= 1



$$\rightarrow \bar{A}C + AB$$

prime Implicant :-

A grouping which can't be expandable is called prime Implicant.

Essential prime implicant :-

If A prime Implicant is covering atleast 1 min/max term which is not covered by any other prime Implicant grouping then it is called as essential prime Implicant.

Redundant:

If a non-essential prime Implicant is covering min terms, which are covered by essential only then it is called redundant.

Note:

If essential prime Implicant covering all min/max terms then minimal expression is only essential prime Implicant.

AB \ CD	00	01	11	10
00	1	1		
01		1	1	
11			1	1
10				

prime Implicant - 5

Essential prime Implicant - 2

$$\rightarrow \bar{A}\bar{B}\bar{C} + ABC + \bar{A}BD$$

Note:

To cover minterms which are not covered by essential prime Implicant we need to consider some of Non-essential prime Implicant in minimal expression with minimum complexity.

Selective prime Implicant:

A prime Implicant is neither essential nor redundant is called selective prime Implicant.

→ Above K-Map,

prime Implicant = 5

Essential prime Implicant = 2

Selective prime Implicant = 3

Redundant = 0

AB \ CD	00	01	10	11
0		1		1
1		1	1	1

→ Ensure whether 00, 01, 11, 10 is in correct order or not

Here order, diff, can't find directly K-Map.

AB \ CD	00	01	11	10
0		1	1	
1		1	1	1

$$C + AB$$

→ prime Implicant = 2

Essential P.I = 2

AB \ CD	00	01	11	10
0		1		1
1		1	1	1

→ P.I = 4

Essential P.I = 2

$$= \bar{B}C + B\bar{C} + AC$$

or

$$\bar{B}C + B\bar{C} + AB$$

These type above expressions are called multiple minimal expressions.

22-07

Find no of P.I, Essential P.I, and minimal expressions.

AB \ CD	00	01	11	10
00	1	1		
01		1	1	
11			1	1
10				1

→ P.I = 6

Essential P.I = 2

There are 3 min terms which are not covered by essential prime implicant. Those can be covered in 3 ways.

$$\bar{A}\bar{B}\bar{C} + AC\bar{D} + \bar{A}\bar{C}D + BCD$$

or

$$\bar{A}BD + ABC$$

or

$$\bar{A}BD + BCD$$

Q. Find Number of P.I, Essential P.I, Miniterms.

AB \ CD	00	01	11	10
00	1		1	1
01	1		1	1
11	1	1	1	1
10	1	1		1

↳ check the possibility of max grouping

8 is octave

↳ (In 8 grouping → inside 4 grouping should not taken)

prime Implicants = 5

Essential prime Implicants = 3

$$\text{Min. Exp} = \bar{D} + A\bar{C} + \bar{A}C + AB$$

or
BC

Q. find no of p.i, essential p.i, min. Exp.

AB \ CD	00	01	11	10
00		1	1	
01	1	1		
11	1	1	1	1
10			1	1

prime Imp = 6

Essential P.I = 2

$$\hookrightarrow AC + B\bar{C} + \bar{A}\bar{B}D$$

redundant = 1 (AB)

Selective p.i = 3

BC	00	01	11	10
0	1	1	1	1
1		1	1	1

$$\bar{A}\bar{B} + AC + B\bar{C}$$

or

$$\bar{A}\bar{C} + \bar{B}C + AB$$

→ prime Implicant = 6
Essential P.I = 0

* No of Inputs of a logic gate - fanning

Q. A function is having "n" no of prime Implicants, (n-1) no of essential prime Implicants then the maximum no of possible min expressions are. (covered by non essential)

Non essentials → (1) → that 1 is redundant whenever 1 non essential is present then it is redundant. It shouldn't consider. The expressions are (1) → that too essential.

* No of outputs of logic gate - fan-out

Minimal expression in POS

BC	00	01	11	10
A	0	0	0	0
1	1	1	1	1

$$(A+B)(\bar{A}+\bar{B})$$

$$\Rightarrow 0 + \bar{A}B + A\bar{B} + 0$$

$$\therefore \text{sop} \cong \text{pos}$$

Q. find pos, sop minimal.

CD	00	01	11	10
AB	00	0	0	0
01	0	0	0	0
11	0	0	0	0
10	0	0	0	0

$$\text{POS} = (\bar{A}+B+C)(\bar{B}+D)(A+D)$$

CD	00	01	11	10
AB	00	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$$\rightarrow \bar{A}BC + BD + \bar{A}D$$

Q.

CD	00	01	11	10
AB	00	0	0	0
01	0	0	0	0
11	0	0	0	0
10	0	0	0	0

$$D(\bar{A}+C)(\bar{B}+\bar{C})$$

CD	00	01	11	10
AB	00	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$$\bar{A}\bar{C}D + \bar{B}CD$$

CD	00	01	11	10
AB	00	0	0	0
01	0	0	0	0
11	0	0	0	0
10	0	0	0	0

3 essential P.2, P.2=5

$$D(\bar{A}+C)(A+\bar{C})(A+\bar{B})$$

$$\text{or } (\bar{B}+C)$$

CD	00	01	11	10
AB	00	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$$\bar{A}\bar{B}\bar{C}D + ACD$$

CD	00	01	11	10
AB	00	0	0	0
01	0	0	0	0
11	0	0	0	0
10	0	0	0	0

$$P.2 = 10$$

$$E.P.2 = 0$$

SOP

$$= \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}CD + \bar{A}B\bar{C}D + \bar{A}B\bar{C}D$$

POS

$$(i) (\bar{B}+\bar{D})(A+C+D)(A+B+\bar{C})(\bar{A}+B+C)(\bar{A}+\bar{C}+D)$$

$$(ii) (B+D)(A+\bar{B}+C)(A+\bar{C}+\bar{D})(\bar{A}+C+\bar{D})(\bar{A}+\bar{B}+\bar{C})$$

Q. The minimal expression of function

$$f(A,B,C,D) = \sum (0,1,3,4,5,7,8,9,11)$$

CD	00	01	11	10
AB	00	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$$\text{POS} = (\bar{A}+\bar{B})(\bar{C}+D)$$

$$\text{SOP} = \bar{A}\bar{C} + \bar{A}D + \bar{B}\bar{C} + \bar{B}D$$

$$(a) \bar{A}\bar{C} + \bar{A}D + \bar{B}\bar{C} + \bar{B}D$$

$$(b) \bar{A}\bar{C} + \bar{A}D + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}D$$

$$(c) \bar{B}\bar{C} + \bar{B}D + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}D$$

$$(d) (\bar{A}+\bar{B})(\bar{C}+D)$$

Don't Care :-

BC	00	01	11	10
A				
0		1	1	
1		x	1	

without D.C $\rightarrow \bar{A}C + BC$

with D.C $\rightarrow C$

BC	00	01	11	10
A				
0	0			0
1	0	x		0

$C(\bar{A} + B) \rightarrow$ With D.C

$C \rightarrow$ without D.C

Q. Find minimal expression for pos and sop.

BCD	00	01	11	10
AB				
00	1	1	x	1
01		x	1	
11			1	
10		x	x	x

$\bar{A}\bar{B} + CD$

CD	00	01	11	10
AB				
00			x	
01	0	x		0
11	0	0		0
10	0	x		x

$(\bar{A} + C)(\bar{B} + D)$

Q. find minimal expression for pos & sop.

CD	00	01	11	10
AB				
00	1			x
01			x	
11	1	1	x	
10	1	1		x

$A\bar{C} + \bar{B}\bar{D}$

CD	00	01	11	10
AB				
00		0	0	x
01	0	0	x	0
11			x	0
10			0	x

$\bar{C}(\bar{A} + \bar{B})(\bar{A} + \bar{D})$

Q. The minimal expression of function $f(A, B, C, D) = \sum m(0, 1, 6, 7, 8, 9) + d(2, 3, 4, 5, 12, 13)$ in pos is _____

(a) $A + C$

(b) $\bar{A} + \bar{C}$

(c) $\bar{A}\bar{C}$

(d) AC

CD	00	01	11	10
AB				
00	0	0	x	x
01	x	x	0	0
11	x	x		
10	0	0		

(Individual sums are multiply)
 $\rightarrow A \cdot C$

1		1	
	1		1

	1		1
1		1	

2^{n-1} Essential P.I $\rightarrow$ max. # of N variables

1		1	
	1		1
1		1	
	1		1

	1		
1		1	
	1		1
1		1	

* If $f(V, W, X, Y, Z) = x + \bar{y}z$

$$2^4 + 2^3 = 24(x)$$

There might be Intersection or common between two. so. find the intersection among min terms.

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

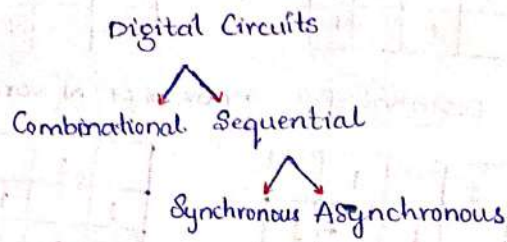
$$= 2^4 + 2^3 - 2^2$$

$$= 16 + 8 - 4$$

$$= 20$$

YZ	00	01	11	10
WX				
00				
01				
11				
10				

Digital Circuits



Combinational :

output depends on present i/p is called combinational circuit.

Sequential :

The output depends on present i/p and past output.

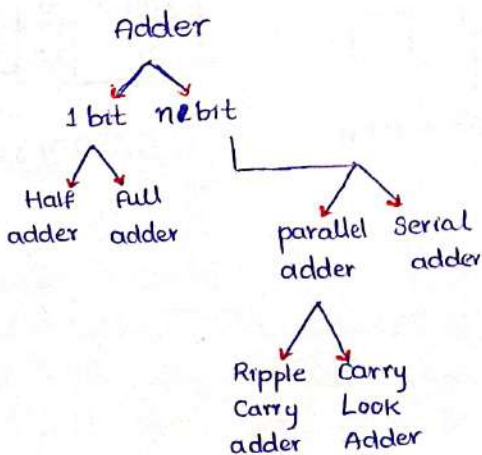
Combinational circuits :-

↳ o/p depend on present i/p.

Procedure to design

1. decide no of i/p's and no of o/p's required to design system.
2. Decide o/p's 0 or 1 or don't care for all possible inputs depending on defined function.
3. find minimal expression of outputs in terms of input.
4. Draw logic diagram.

*



Half Adder :-

A	B	Sum	Carryout
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$$\text{Sum} = \bar{A}B + A\bar{B}$$

$$= A \oplus B$$

$$\text{Carry out} = AB$$



Full adder :-

A	B	C	Sum	Count
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

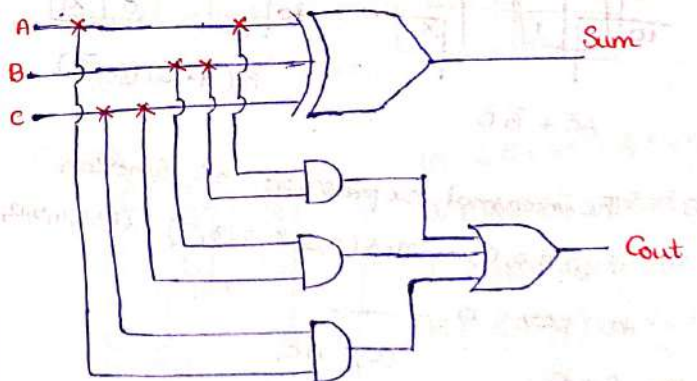
$$\text{Sum} = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

$$= C(\bar{A}\bar{B} + AB) + \bar{A}\bar{C}(B + \bar{B})$$

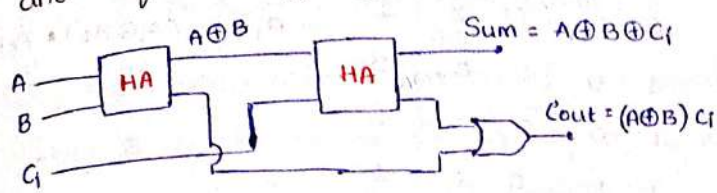
$$= C(A \oplus B) + \bar{A}\bar{C}$$

$$= A \oplus B \oplus \text{Count}$$

$$\text{Count} = AB + BC + CA$$

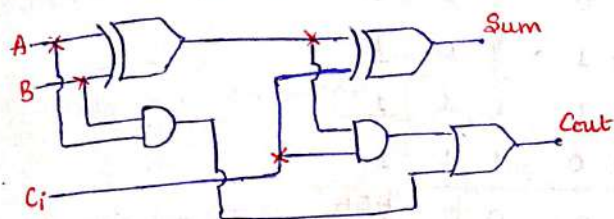


Implement full adder using half adder and required logic gates.



$$C_{out} = \sum m(3, 5, 6, 7)$$

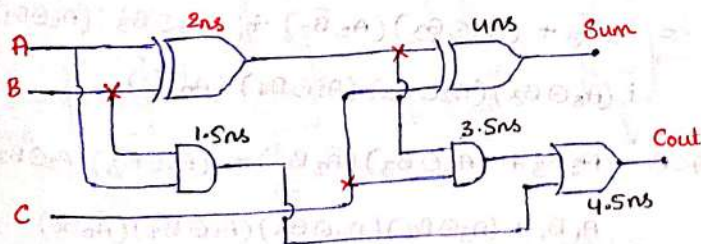
$$\begin{aligned} &= \bar{A}BC_i + A\bar{B}C_i + AB\bar{C}_i \\ &= (\bar{A}B + A\bar{B})C_i + AB(\bar{C}_i + C_i) \\ &= (A \oplus B)C_i + AB \end{aligned}$$



* A full adder is implemented using 2 HA and 2 input OR gate. Half adder is implemented using 1 XOR gate and 2 i/p AND gate.

The propagation delay of XOR, OR, AND gates respectively are 2ns, 1.5ns, 1ns.

The propagation delay of full adder is 4.5 ns.



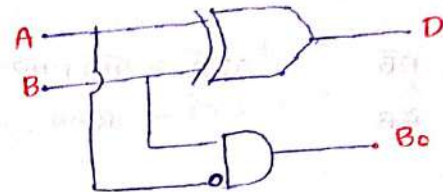
propagation delay = 4.5ns

Half Subtractor :

A	B	D	B ₀
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

$$D = \bar{A}B + A\bar{B} = A \oplus B$$

$$B_0 = \bar{A}B$$



Full Subtractor :

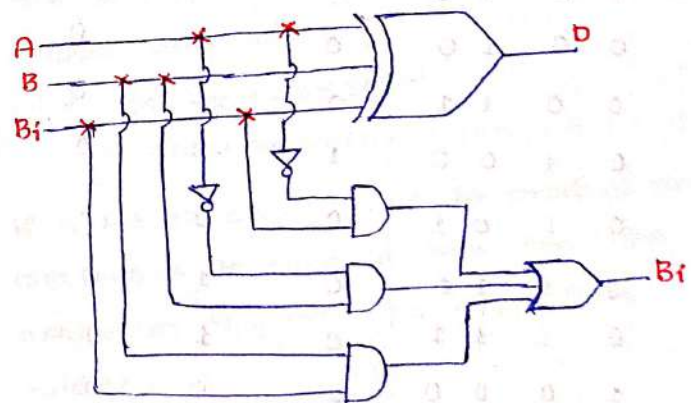
A	B	B _i	D	B ₀
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

A	B	B _i	D	B ₀
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

$$D = A \oplus B \oplus B_i$$

A	B	B _i	D	B ₀
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

$$B_0 = \bar{A}B_i + \bar{A}B + BB_i$$



1 Bit Comparator :-

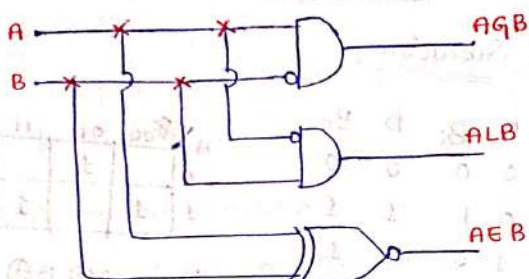
A	B	A > B	A < B	A = B
0	0	0	0	1
0	1	0	1	0
1	0	1	0	0
1	1	0	0	1

$$A < B = A\bar{B}$$

$$A < B = \bar{A}B$$

$$A > B = \bar{A}\bar{B} + AB$$

$$= A \odot B$$



2 Bit Comparator

A ₁	A ₀	B ₁	B ₀	AQB	ALB	AEB
0	0	0	0	0	0	1
0	0	0	1	0	1	0
0	0	1	0	0	1	0
0	0	1	1	0	1	0
0	1	0	0	1	0	0
0	1	0	1	0	0	1
0	1	1	0	0	1	0
0	1	1	1	0	1	0
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	0	0	1
1	0	1	1	0	1	0
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	0	0	1

A₁ A₀ B₁ B₀

A ₁	A ₀	B ₁	B ₀	AQB	ALB	AEB
0	0	0	0	1	1	1
0	0	0	1	1	1	1
0	0	1	0	1	1	1
0	0	1	1	1	1	1
0	1	0	0	1	1	1
0	1	0	1	1	1	1
0	1	1	0	1	1	1
0	1	1	1	1	1	1
1	0	0	0	1	1	1
1	0	0	1	1	1	1
1	0	1	0	1	1	1
1	0	1	1	1	1	1
1	1	0	0	1	1	1
1	1	0	1	1	1	1
1	1	1	0	1	1	1
1	1	1	1	1	1	1

4 Bit Comparator

A₃ A₂ A₁ A₀ B₃ B₂ B₁ B₀

$$A < B = (A_3 > B_3) \vee (A_3 = B_3) \& (A_2 > B_2) \vee$$

$$(A_3 = B_3) \& (A_2 = B_2) \& (A_1 > B_1) \vee$$

$$(A_3 = B_3) \& (A_2 = B_2) \& (A_1 = B_1) \& (A_0 > B_0)$$

$$= A_3 \bar{B}_3 + (A_3 \odot B_3) (A_2 \bar{B}_2) + (A_3 \odot B_3) (A_2 \odot B_2) A_1 \bar{B}_1$$

$$+ (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) (A_0 \bar{B}_0)$$

$$A < B = \bar{A}_3 \bar{B}_3 + (A_3 \odot B_3) (\bar{A}_2 B_2) + (A_3 \odot B_3) (A_2 \odot B_2)$$

$$\bar{A}_1 B_1 + (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) (\bar{A}_0 B_0)$$

$$A > B = (A_3 = B_3) \& (A_2 = B_2) \& (A_1 = B_1) \& (A_0 = B_0)$$

$$= (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) (A_0 \odot B_0)$$

$$2 \times 2 \times 2 \times 2 = 2^4$$

23-07 The total no of combinations for which $A > B$ is true in 5 bit comparator is —.

↳ 5 bit Comparator, the no of combinations be

$A_4 A_3 A_2 A_1 A_0 B_4 B_3 B_2 B_1 B_0$

$$2 \times 2 \times 2 \times 2 \times 2 = 2^5$$

$$G \text{ or } L = \frac{2^{10} - 2^5}{2} = 496$$

$$\text{Equal} = 2^5 = 32$$

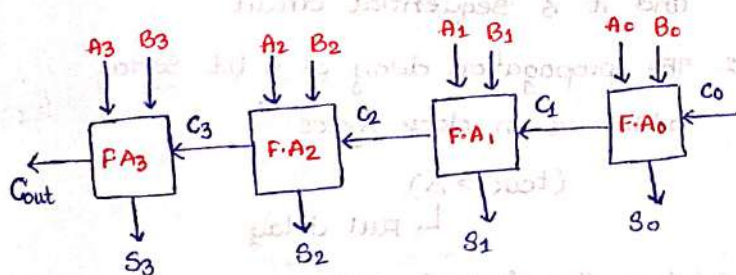
* $A_{n-1} \dots A_0$ $B_{n-1} \dots B_0$
 n n

$$\text{Greater (or Lesser)} = \frac{2^{2n} - 2^n}{2}$$

$$\text{Equal} = 2^n$$

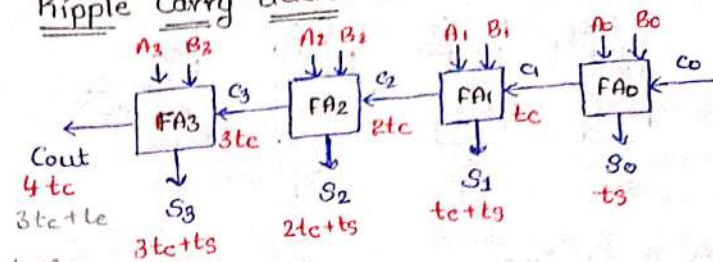
N-Bit Adder

$C_3 \ C_2 \ C_1 \ C_0$ $\downarrow$ parallel adder
 $A_3 \ A_2 \ A_1 \ A_0$
 $B_3 \ B_2 \ B_1 \ B_0$
 Cout $S_3 \ S_2 \ S_1 \ S_0$



↳ It can be also implemented using 1 H.A and 3 F.A. But carry is not taking into account at C_0 . The interlinking of smaller adder with bigger adders is difficult. Hence take all of them as full adders.

Ripple Carry adder:



$t_c = 3n$

$t_s = 4n$

16ns

For 4 bit,

$$t_{4\text{bit}} = 3t_c + \max(t_c, t_s)$$

$$n\text{bit}, t_{n\text{bit}} = (n-1)t_c + \max(t_c, t_s)$$

Note:

If a full adder is taking " t_c " time to produce carry, " t_s " time to produce sum from point to stabilise, the propagation delay of n bit ripple carry adder is $(n-1)t_c + \max(t_s, t_c)$.

Q. If a F.A is taking 12ns to produce carry and 15ns to produce sum then the propagation delay of 16 bit ripple carry adder is —.

$$t_{pd} = (n-1)t_c + \max(t_s, t_c)$$

$$= 15t_c + \max(15, 12) = (15 \times 12) + 15 = 195\text{ns}$$

Q. If a F.A is taking 20ns to produce carry and 40ns to produce sum then the maximum frequency 4 bit ripple carry addition is — MHz.

$$t_{4\text{bit}} = 3t_c + \max(t_c, t_s)$$

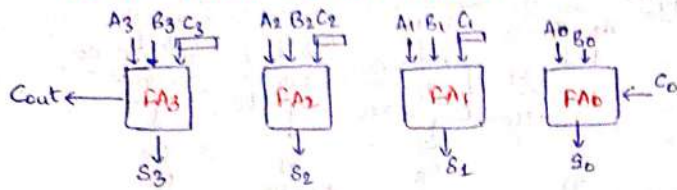
$$= 3(20) + \max(40, 20)$$

$$t_{4\text{bit}} = 60 + 40 = 100\text{ns}$$

$$\text{Maximum freq. } f_{4\text{bit}} = \frac{1}{100\text{ns}} = 10\text{MHz}$$

↳ min slot

Carry Look Ahead Adder :



A	B	C _i	Cout	Sum
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

$$Cout = PC_i + G_i$$

$$C_n = P_{n-1} C_{n-1} + G_{n-1}$$

$$C_1 = P_0 C_0 + G_0$$

$$= (A_0 \oplus B_0) C_0 + A_0 B_0$$

$$C_2 = P_1 C_1 + G_1$$

$$= P_1 (P_0 C_0 + G_0) + G_1$$

$$= P_1 P_0 C_0 + P_1 G_0 + G_1$$

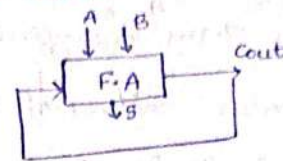
$$C_3 = P_2 C_2 + G_2$$

$$= P_2 (P_1 P_0 C_0 + P_1 G_0 + G_1) + G_2$$

$$= P_2 P_1 P_0 C_0 + P_2 P_1 G_0 + P_2 G_1 + G_2$$

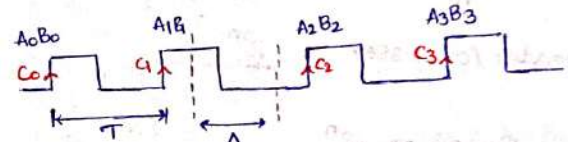
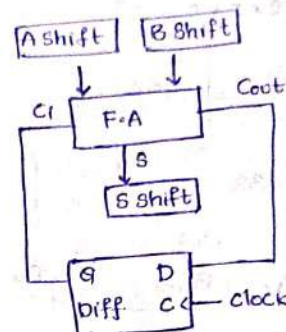
→ Carry look adder may save time for smaller no of bit additions but it may not save time for larger bit no of additions because carry estimation circuit complexity increasing with increasing no of bits.

Serial Adder :



→ Giving FB immediately can cause effect like carry may taken as 1/p.

↳ So, using shift Register



$$t_{nbit} = nT_{clk} \quad (T_{clk} > 0)$$

$$> n\Delta$$

$$t_{clk} = 2n\Delta - t_s - t_{ns}, \quad \Delta = 0$$

$$3 \times 2 \times 4 = 10ns > 10ns$$

Note :

1. It is not a fully combinational circuit and it is sequential circuit.
2. The propagation delay of n bit serial adder is n clock cycles.
($t_{clk} > \Delta$)
↳ Full delay
3. It is the slowest adder.

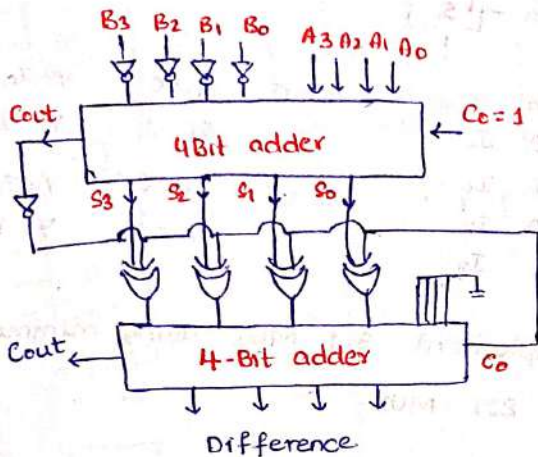
N-Bit Subtractor

1. $A + 2's \text{ of } B = \text{Sum}$

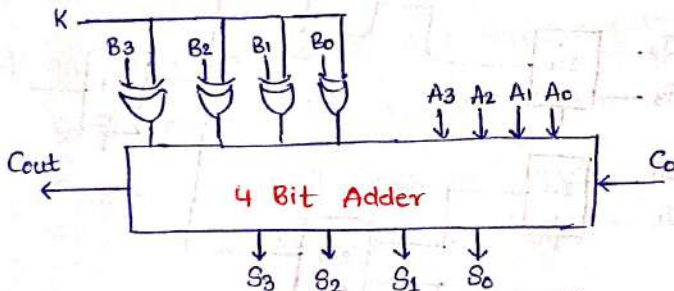
$$A + \bar{B} + 1 = \text{Sum}$$

2. If (Carry = 1)
difference = Sum.

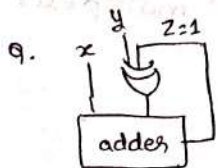
else
difference = 2's of Sum
 $= \overline{\text{Sum}} + 1$



Controlled Adder / Subtraction



K	C ₀	function
0	0	$A+B$
0	1	$A+B+1$
1	0	$A+\bar{B}$
1	1	$A+\bar{B}+1$ ($A-B$)

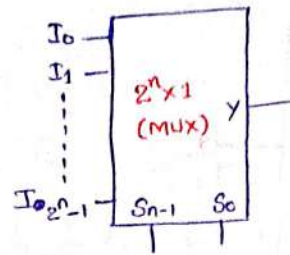


$$\Rightarrow x + \bar{y} + 1$$

$x + 2's \text{ comp of } y$

$x - y$

* Multiplexer * (MUX)

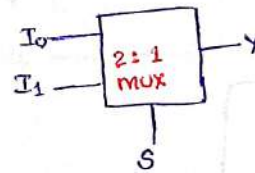


$S_{n-1} - S_0$: Selection lines
(or) control signals.

Inputs : $I_0, I_1, \dots, I_{2^n-1}$
O/p : Y .

↳ Multiplexing data from multiple inputs and giving one output line based on selection line.

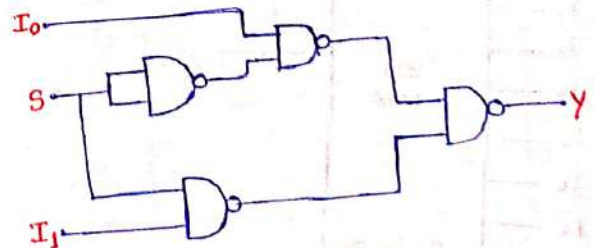
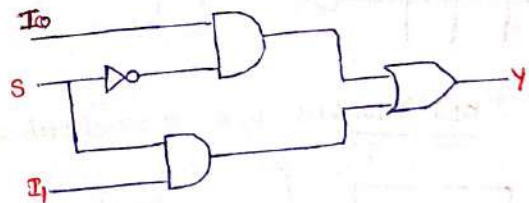
(i) 2:1 MUX



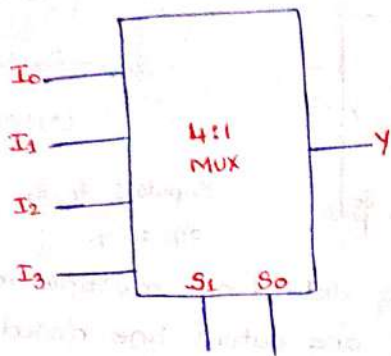
S	I ₀	I ₁	Y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

when
 $S=0 \rightarrow I_0$
 $S=1 \rightarrow I_1$

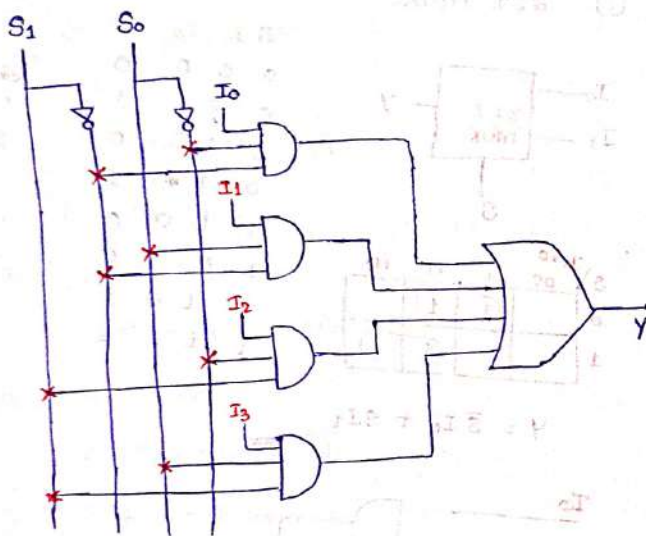
$$Y = \bar{S}I_0 + SI_1$$



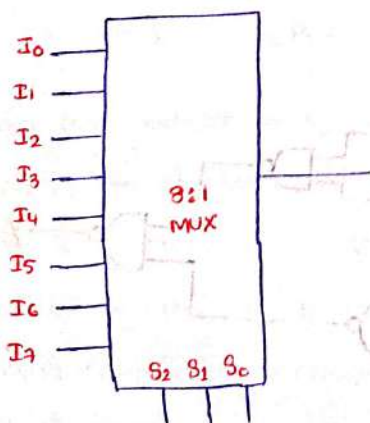
4:1 MUX



$$Y = \bar{S}_1 \bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1 \bar{S}_0 I_2 + S_1 S_0 I_3$$

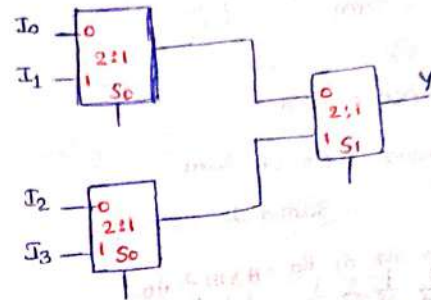


8:1 Mux



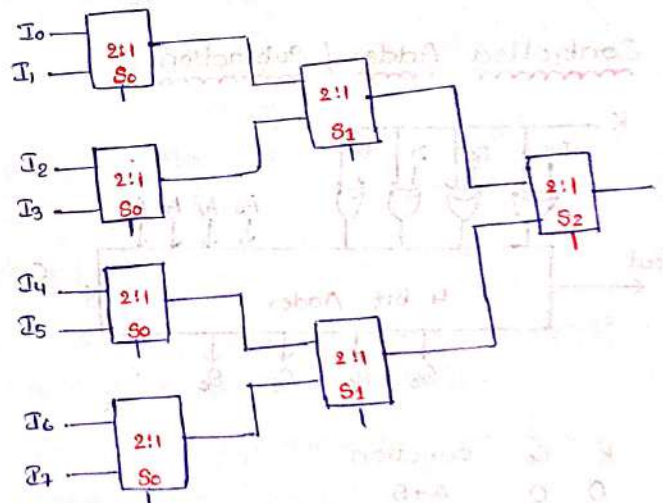
$$Y = \bar{S}_2 \bar{S}_1 \bar{S}_0 I_0 + \bar{S}_2 \bar{S}_1 S_0 I_1 + \bar{S}_2 S_1 \bar{S}_0 I_2 + \bar{S}_2 S_1 S_0 I_3 \\ + S_2 \bar{S}_1 \bar{S}_0 I_4 + S_2 \bar{S}_1 S_0 I_5 + S_2 S_1 \bar{S}_0 I_6 + S_2 S_1 S_0 I_7$$

Q. Implement 4:1 MUX using minimum no of 2:1 MUX.



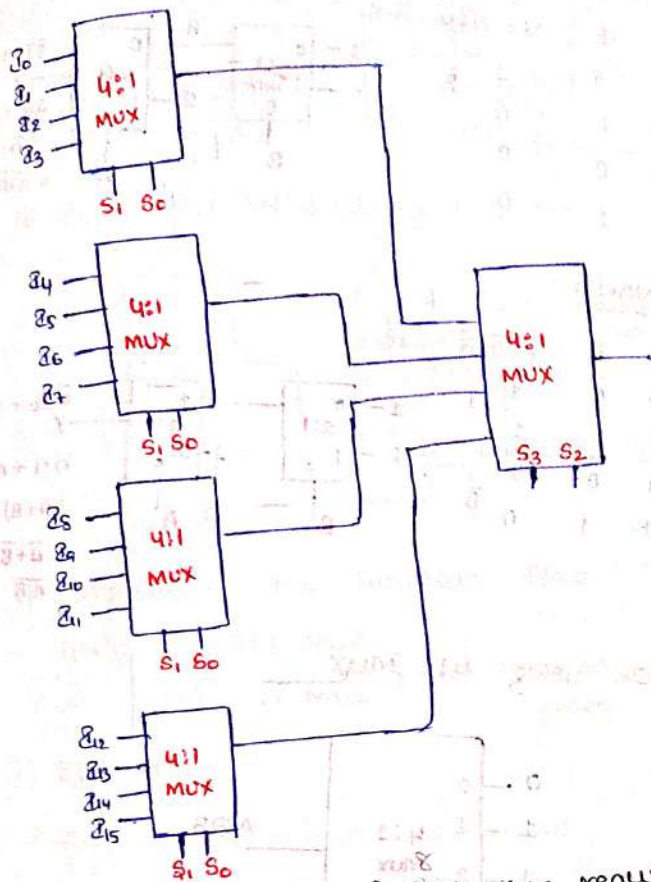
S ₁	S ₀	Y	S ₀ = 0, S ₁ = 0	Y = I ₀
0	0	I ₀	S ₀ = 1, S ₁ = 1	Y = I ₂
0	1	I ₁	S ₀ = 1, S ₁ = 0	Y = I ₁
1	0	I ₂	S ₀ = 1, S ₁ = 1	Y = I ₃
1	1	I ₃		

Q. Implement 8:1 MUX using minimum no of 2:1 MUX



The minimum no of 2:1 multiplexers required to implement 2^n multiplexer is $2^n - 1$

Q. Implement 16:1 Mux using minimum number of 4:1 MUX.



Q. The minimum no of 4:1 Mux required to implement $2^{10}:1$ Mux is _____.

$$2^{10} = 1024 \Rightarrow \frac{1024}{4} = 256 \rightarrow 1^{st} \text{ stage}$$

$$\frac{256}{4} = 64 \rightarrow 2^{nd} \text{ stage}$$

$$\frac{64}{4} = 16 \rightarrow 3^{rd}$$

$$\frac{16}{4} = 4 \rightarrow 4^{th}$$

$$\frac{4}{4} = 1 \rightarrow 5^{th}$$

$$\underline{\underline{341}}$$

Q. The minimum no of 8:1 Mux required to implement $2^{12}:1$ Mux is _____.

$$2^{12} = 4096 \Rightarrow \frac{4096}{8} = 512$$

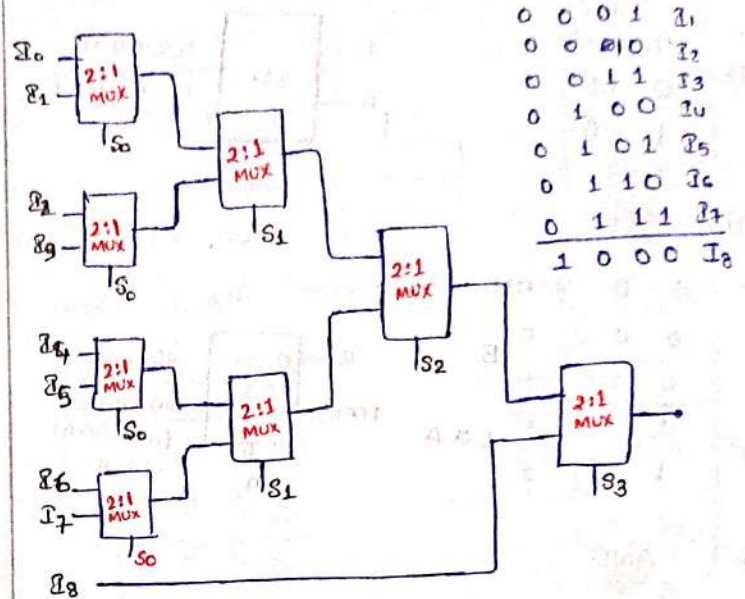
$$\frac{512}{8} = 64$$

$$= 8$$

$$= 1$$

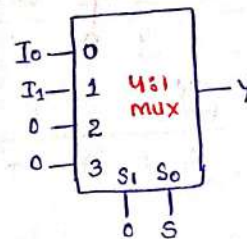
$$\underline{\underline{585}}$$

Q. Implement 9:1 Mux using minimum number of 2:1 MUX.



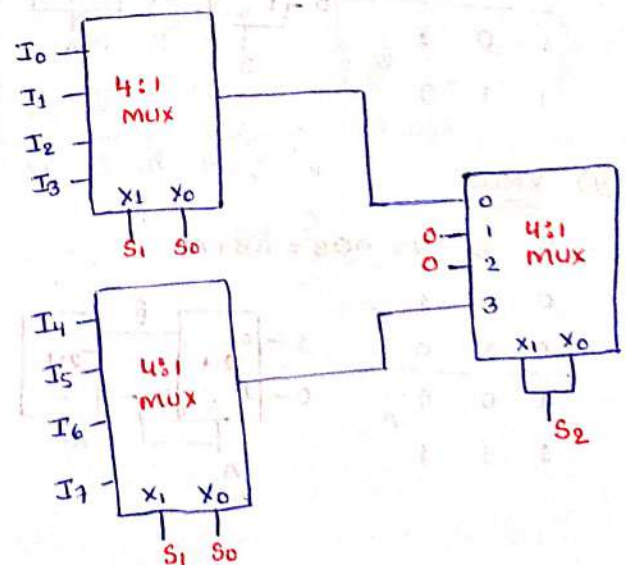
S_3	S_2	S_1	S_0	Y
0	0	0	0	I_0
0	0	0	1	I_1
0	0	1	0	I_2
0	0	1	1	I_3
0	1	0	0	I_4
0	1	0	1	I_5
0	1	1	0	I_6
0	1	1	1	I_7
1	0	0	0	I_8

Q. Implement 2:1 mux using 4:1 MUX



S	Y
0	I_0
1	I_1

Q. Implement 8:1 MUX using 4:1 Mux only.

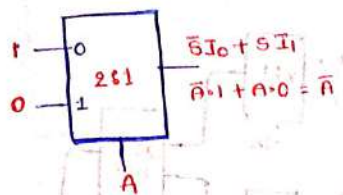


* To prove mux is universal.

(i) NOT

$$A \quad Y = \bar{A}$$

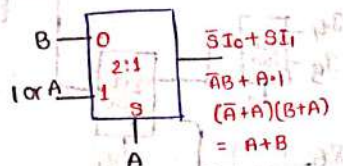
0	1
1	0



(ii) OR

$$A \quad B \quad Y = A + B$$

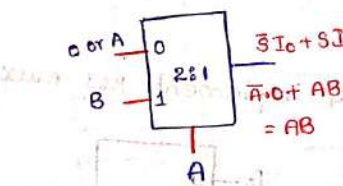
0	0	0
0	1	1
1	0	1
1	1	1



(iii) AND

$$A \quad B \quad Y = A \cdot B$$

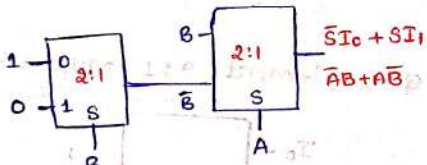
0	0	0
0	1	0
1	0	0
1	1	1



(iv) XOR

$$A \quad B \quad Y = A \oplus B = \bar{A}B + A\bar{B}$$

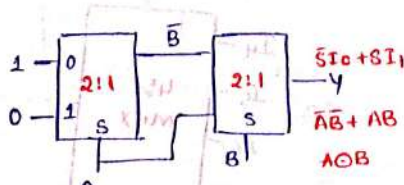
0	0	0
0	1	1
1	0	1
1	1	0



(v) XNOR

$$A \quad B \quad Y = A \odot B = \bar{A}\bar{B} + AB$$

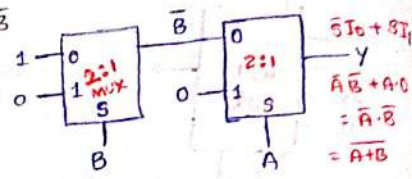
0	0	1
0	1	0
1	0	0
1	1	1



(vi) NOR

$$A \quad B \quad Y = \overline{A+B} = \bar{A} \cdot \bar{B}$$

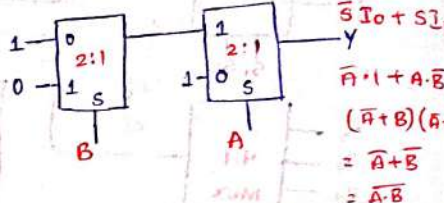
0	0	1
0	1	0
1	0	0
1	1	0



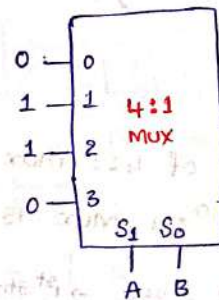
(vi) NAND

$$A \quad B \quad Y = \overline{A \cdot B} = \bar{A} + \bar{B}$$

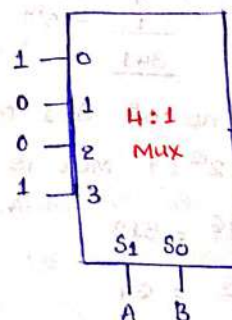
0	0	1
0	1	1
1	0	1
1	1	0



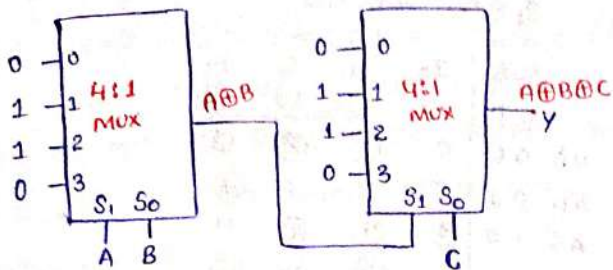
* XOR using 4:1 MUX



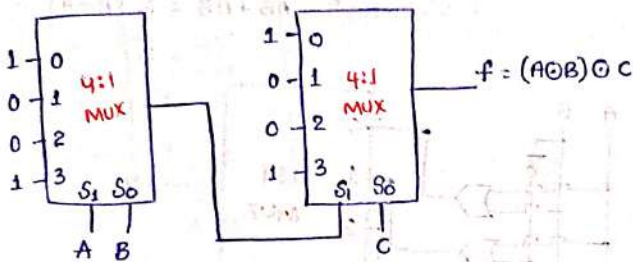
* XNOR using 4:1 MUX



*



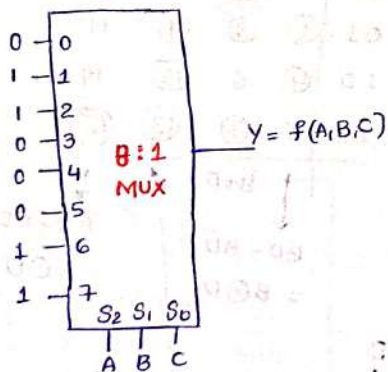
* Three input XNOR using 4:1 MUX



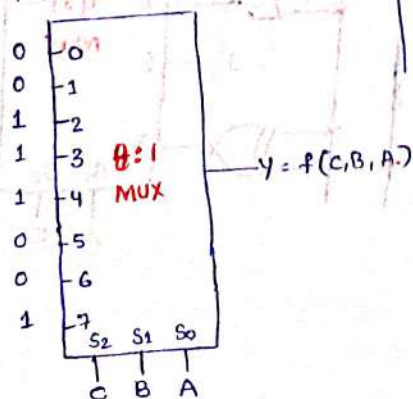
Q. Implement the function $f(A, B, C) = \sum m(1, 2, 6, 7)$ using (i) 8:1 MUX and (ii) 4:1 MUX and required logic gates.

(i) 8:1 MUX

$$f(A, B, C) = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$



$$f^*(C, B, A) = C\bar{B}\bar{A} + \bar{C}B\bar{A} + \bar{C}B\bar{A} + \bar{C}B\bar{A}$$

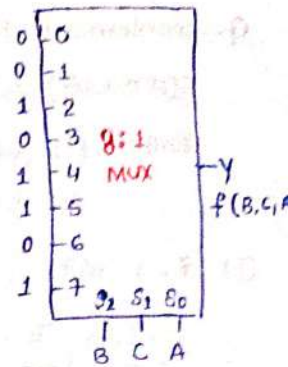


$$f(A, B, C) = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

$$f^*(B, C, A) = \bar{B}\bar{C}\bar{A} + \bar{B}\bar{C}A + B\bar{C}\bar{A} + B\bar{C}A$$

$$= 010 + 0100 + 101 + 111$$

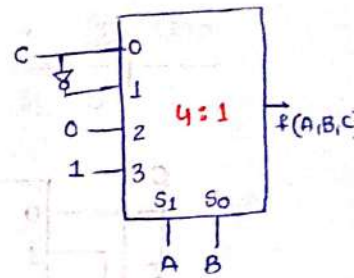
$$= m(2, 4, 5, 7)$$



(ii) 4:1 MUX

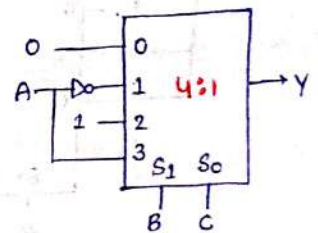
$$S_1 S_0 \rightarrow AB \quad m(1, 2, 6, 7)$$

AB	I ₀	I ₁	I ₂	I ₃
S ₁ S ₀	00	01	10	11
$\bar{C}(0)$	0	②	4	⑥
$\bar{C}(1)$	①	3	5	⑦
	C	$\bar{C}$	0	1



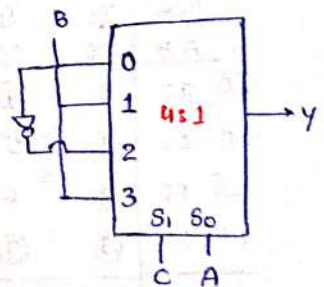
$$b. S_1 S_0 \rightarrow BC \quad m(1, 2, 6, 7)$$

BC	I ₀	I ₁	I ₂	I ₃
S ₁ S ₀	00	01	10	11
$\bar{A}(0)$	0	①	②	3
$\bar{A}(1)$	4	5	⑥	⑦
	0	$\bar{A}$	1	A



$$c. S_1 S_0 \rightarrow CA \quad m(1, 2, 6, 7)$$

CA	I ₀	I ₁	I ₂	I ₃
S ₁ S ₀	00	01	10	11
$\bar{B}(0)$	0	4	①	5
$\bar{B}(1)$	②	⑥	3	⑦
	B	B	$\bar{B}$	B



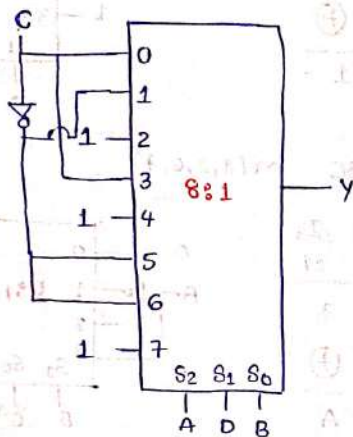
Q. Implement the function

$$f(A, B, C, D) = \sum m(1, 2, 3, 4, 7, 8, 9, 10, 12, 13, 15)$$

using (i) 8:1 MUX } Req. Logic gates.
(ii) 4:1 MUX

(i) 8:1 Mux

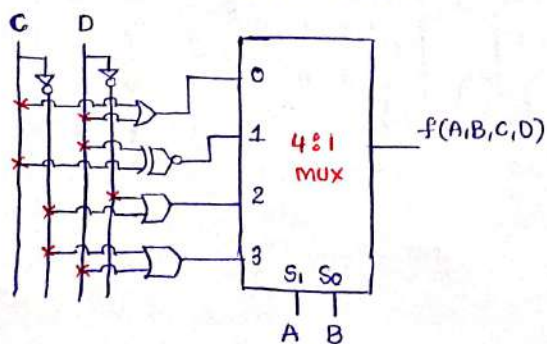
$S_2 S_1 S_0$ A D B	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
$\bar{C}(0)$	0	(4)	(1)	5	(8)	(12)	(9)	(13)
$C(1)$	(2)	6	(3)	(7)	(10)	14	11	(15)
	C	$\bar{C}$	1	C	1	$\bar{C}$	$\bar{C}$	1



(ii) 4:1 MUX

$S_1 S_0$ A B	I_0	I_1	I_2	I_3
$\bar{C}\bar{D}$ 00	0	(4)	(8)	(12)
$\bar{C}D$ 01	(1)	5	(9)	(13)
$C\bar{D}$ 10	(2)	6	(10)	14
CD 11	(3)	(7)	11	(15)
	$C+D$	$\bar{C}+\bar{D}$	$\bar{C}+D$	

$\bar{C}\bar{D} + C\bar{D} = C\bar{D}$

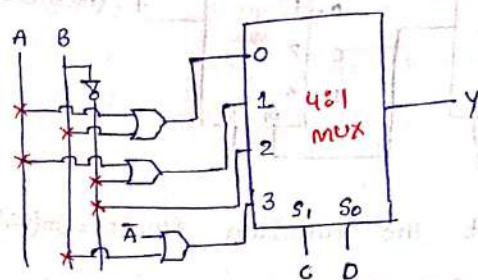


b. $S_1 S_0 \rightarrow CD$

$S_1 S_0$ C D	I_0	I_1	I_2	I_3
$\bar{A}\bar{B}$ 00	0	(1)	(2)	(3)
$\bar{A}B$ 01	(4)	5	6	(7)
$A\bar{B}$ 10	(8)	(9)	(10)	11
AB 11	(12)	(13)	14	(15)

$$A+B \downarrow \quad \bar{A}+B \downarrow$$

$$A\bar{B} + \bar{A}B = \bar{B}(A+\bar{A}) = \bar{B}$$

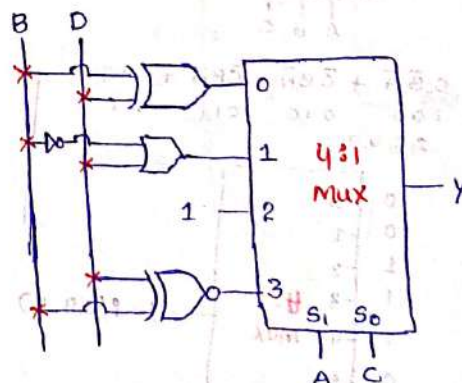


c. $S_1 S_0 \rightarrow AC$

$S_1 S_0$ A C	I_0	I_1	I_2	I_3
$\bar{B}\bar{D}$ 00	0	(2)	(8)	(10)
$\bar{B}D$ 01	(1)	(3)	(9)	11
$B\bar{D}$ 10	(4)	6	(12)	14
BD 11	5	(7)	(13)	(15)

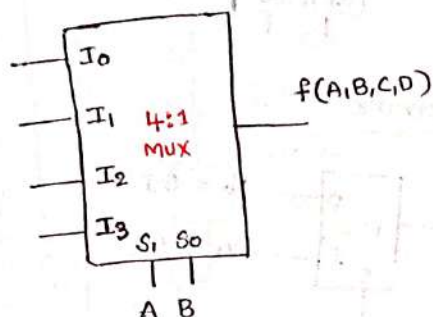
$$\bar{B}+\bar{D} \downarrow \quad \bar{B}+D \downarrow$$

$$\bar{B}\bar{D} + B\bar{D} = \bar{D}$$



29-03

Q. The $f(A, B, C, D) = \sum m(3, 5, 6, 9, 10, 12, 14, 15)$ is implemented using 4:1 Multiplexer shown and 2 input NAND Gates. The min no of two input NAND Gate required is _____.

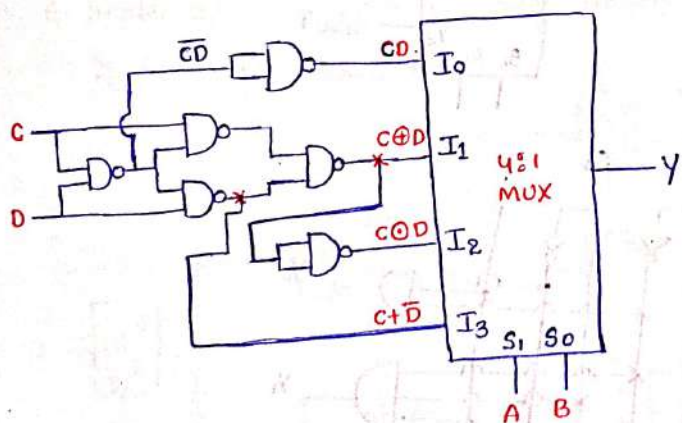


Sol:-

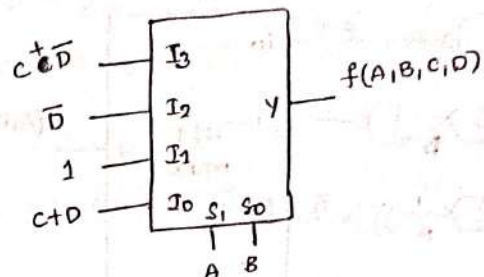
8421

$S_1 S_0$ A B	I_0 00	I_1 01	I_2 10	I_3 11
$\bar{A}\bar{B}\bar{C}\bar{D}$ 00	0	4	8	12
$\bar{A}\bar{B}C\bar{D}$ 01	1	5	9	13
$\bar{A}B\bar{C}\bar{D}$ 10	2	6	10	14
$\bar{A}BC\bar{D}$ 11	3	7	11	15

$\downarrow$ CD $\downarrow$ $C \oplus D$ $\downarrow$ $C \odot D$ $\downarrow$ $C + \bar{D}$



Q. Find minimal expression of function $f(A, B, C, D)$ implemented by following multiplexer circuit in sop and pos.



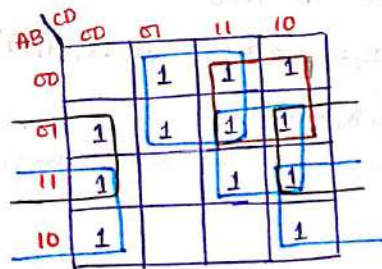
Sol:-

A B $S_1 S_0$	I_0 00	I_1 01	I_2 10	I_3 11
$\bar{C}\bar{D}$ 00	0	4	8	12
$\bar{C}D$ 01	1	5	9	13
$C\bar{D}$ 10	2	6	10	14
CD 11	3	7	11	15

$\downarrow$ $C + D$ $\downarrow$ 1 $\downarrow$ $\bar{D}$ $\downarrow$ $C + \bar{D}$

$$f(A, B, C, D) = \sum m(1, 2, 3, 4, 5, 6, 7, 8, 10, 12, 14, 15)$$

$$\pi M(0, 9, 11, 13)$$



$$= \bar{A}D + BC + A\bar{D} + \bar{A}B + C\bar{D}$$

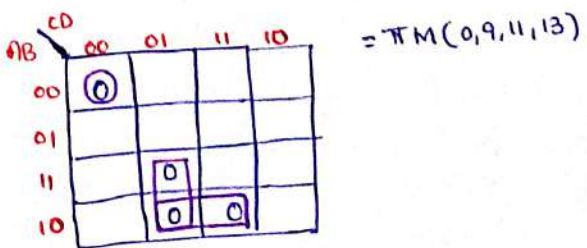
$$(or) \bar{A}B + \bar{A}C$$

$$(or) B\bar{D} + C\bar{D}$$

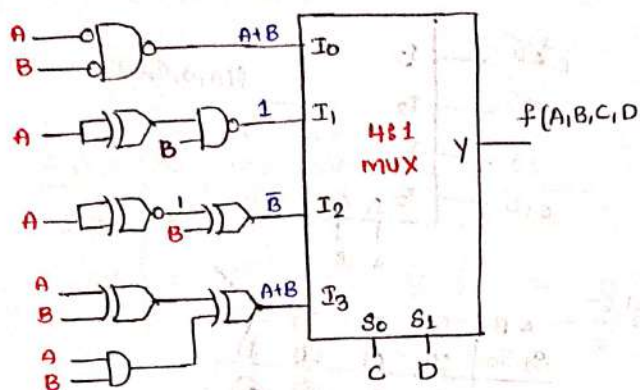
$$(or) B\bar{D} + \bar{A}C$$

pos

$$\hookrightarrow (A+B+C+D)(\bar{A}+C+\bar{D})(\bar{A}+B+\bar{D})$$



Q. find minimal expression of $f(A,B,C,D)$ implemented by following multiplexer circuit in sop and pos.



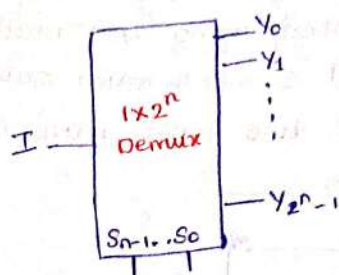
$$I_0 = A+B \quad I_1 = 1 \quad I_2 = \bar{B} \quad I_3 = A+B$$

$\begin{matrix} D & C \\ S_1 & S_0 \end{matrix}$	I_0 00	I_1 01	I_2 10	I_3 11
$\bar{A}\bar{B}$ 00	0	2	1	3
$\bar{A}B$ 01	4	6	5	7
$A\bar{B}$ 10	8	10	9	11
AB 11	12	14	13	15
	$A+B$	1	$\bar{B}$	$A+B$

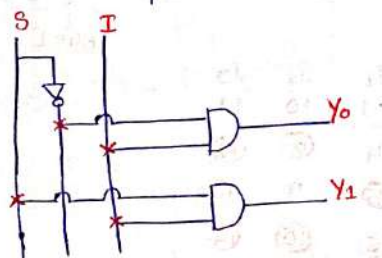
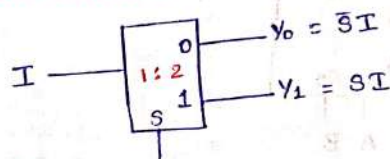
$$f(A,B,C,D) = \sum m(1, 2, 4, 6, 7, 8, 9, 10, 11, 12, 14, 15)$$

$$= \prod M(0, 3, 5, 13)$$

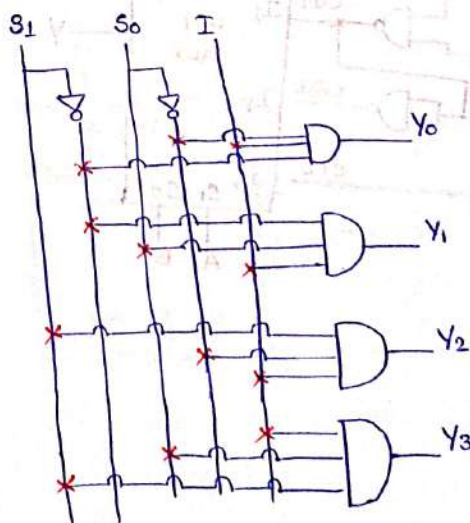
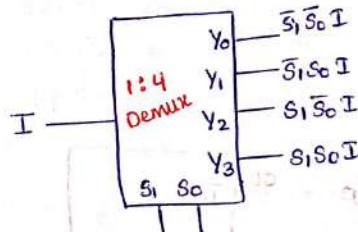
Demultiplexer (Demux) (1×2^n)



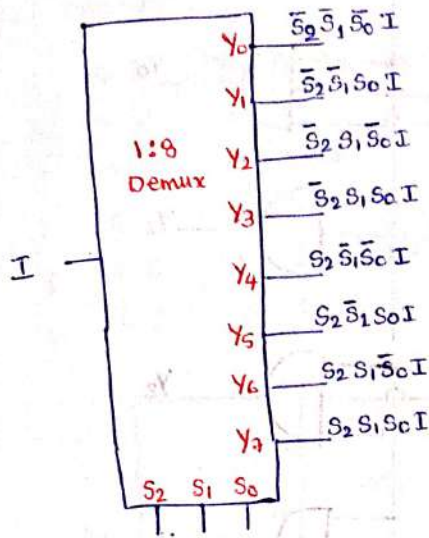
1:2 Demux



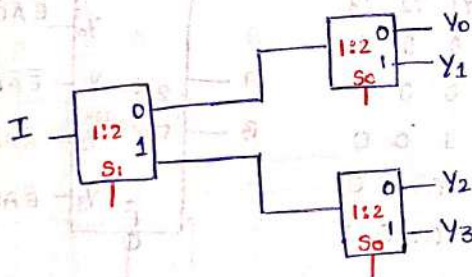
1:4 Demux :



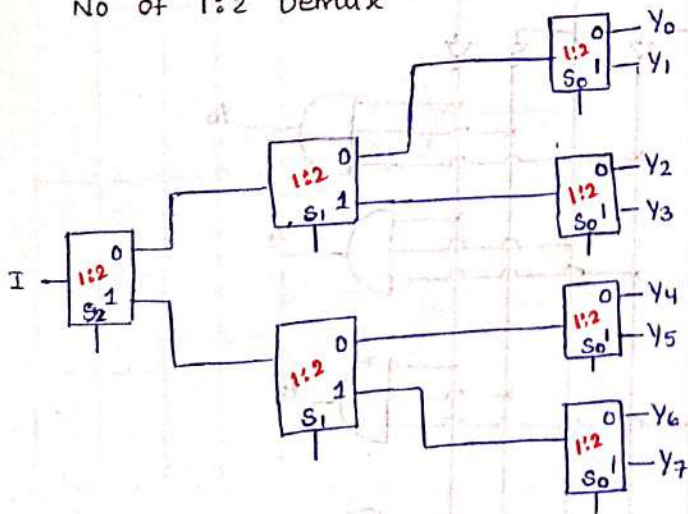
1:8 Demux



Q. Implement 1:8 demux using min no of 1:2 Demux.



Q. Implement 1:8 Demux using Minimum No of 1:2 Demux



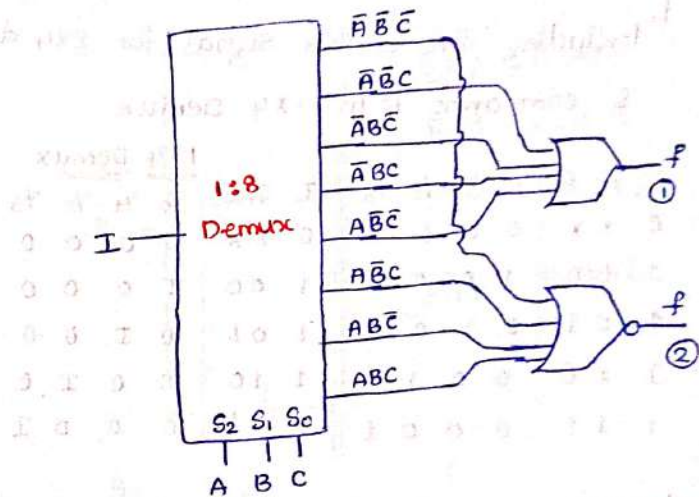
Q. Implement the $f(A,B,C) = \sum m(1,2,3,4)$ using 1:8 demux and required logic gates.

Sol: $f(A,B,C) = \sum m(1,2,3,4)$

$$f = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}C + A\bar{B}\bar{C} \quad \text{--- (1)}$$

$$= \Pi M(0,5,6,7)$$

$$f = \bar{A}\bar{B}\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + ABC \quad \text{--- (2)}$$



Decoder ($n \times 2^n$)

1x2 Decoder

A	Y_0	Y_1
0	1	0
1	0	1

2x4 Decoder

A	B	Y_0	Y_1	Y_2	Y_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

↳ Including the enable signal for 2x4 decoder & comparing with 1x4 Demux.

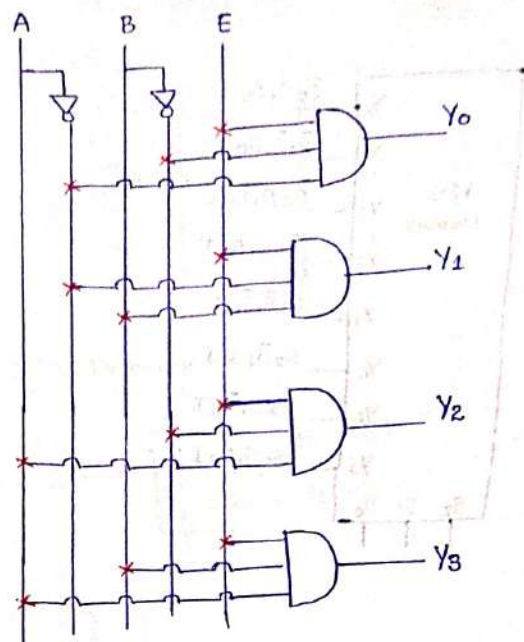
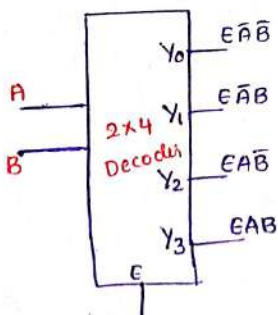
E	A	B	Y_0	Y_1	Y_2	Y_3
0	x	x	0	0	0	0
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1

↳ Decoder + Enable = Demultiplexer
(or)

↳ Demultiplexer can be used as decoder with an Enable input.

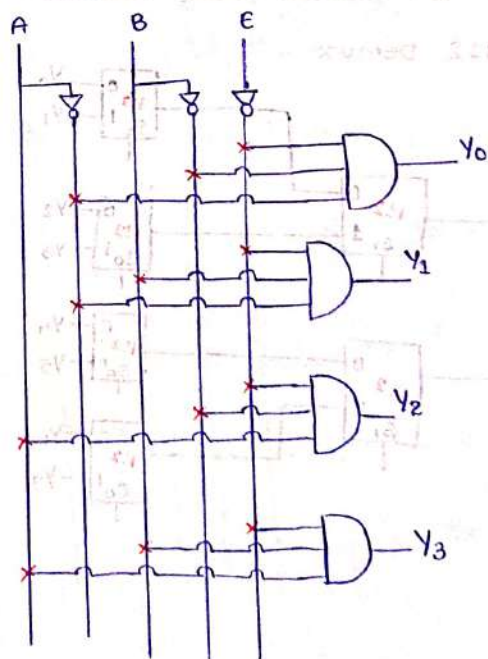
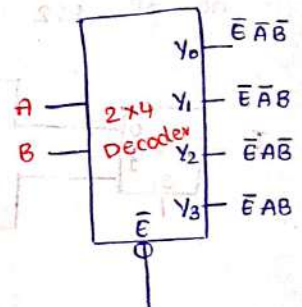
Logic 1 Enable & Logic 1 o/p Decoder

E	A	B	Y_0	Y_1	Y_2	Y_3
0	x	x	0	0	0	0
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1



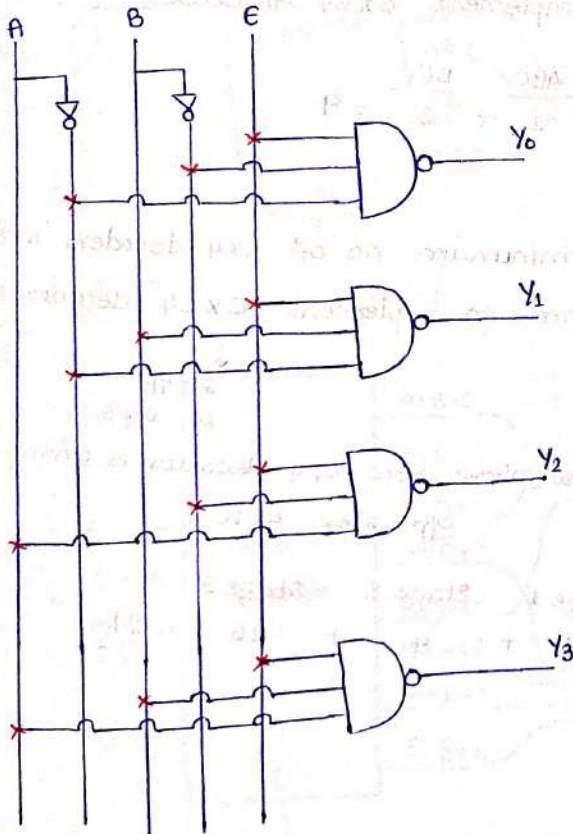
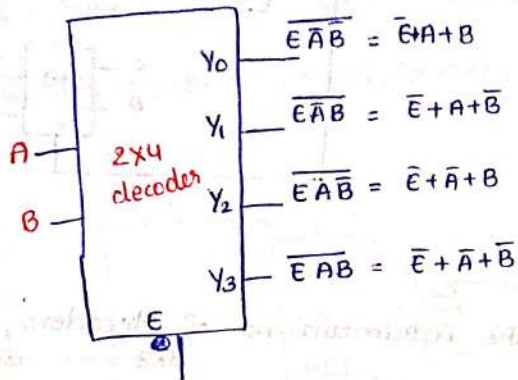
Logic 0 Enable & Logic 1 o/p Decoder

E	A	B	Y_0	Y_1	Y_2	Y_3
1	x	x	0	0	0	0
0	0	0	1	0	0	0
0	0	1	0	1	0	0
0	1	0	0	0	1	0
0	1	1	0	0	0	1



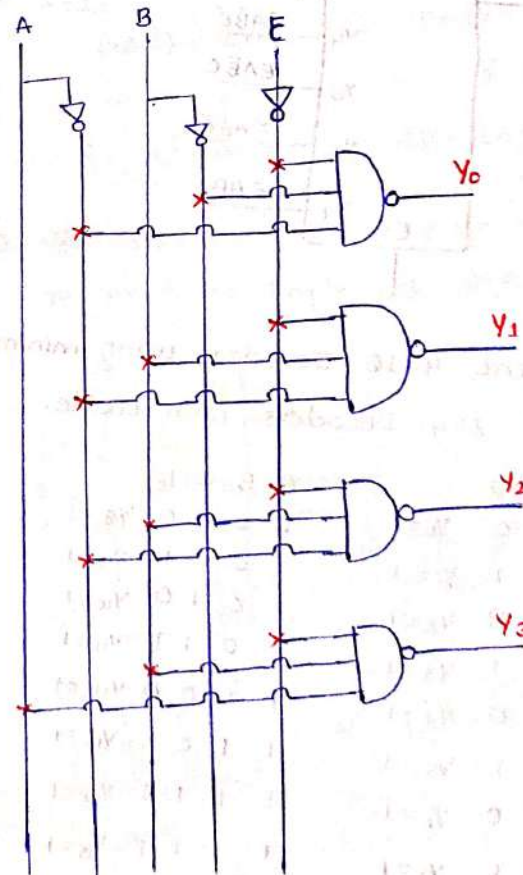
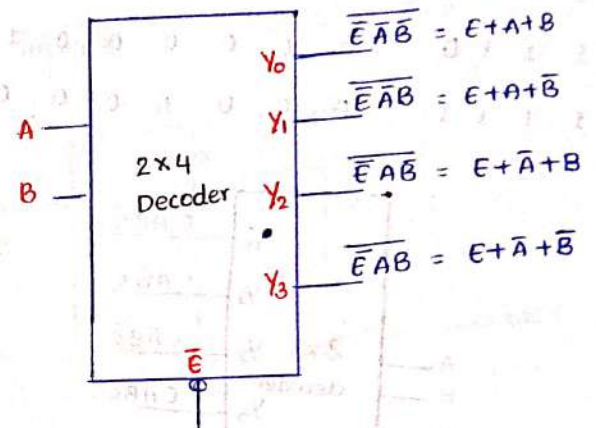
Logic 1 Enable & Logic 0 o/p decoder
2x4 Decoder

E	A	B	Y ₀	Y ₁	Y ₂	Y ₃
0	x	x	1	1	1	1
1	0	0	0	1	1	1
1	0	1	1	0	1	1
1	1	0	1	1	0	1
1	1	1	1	1	1	0



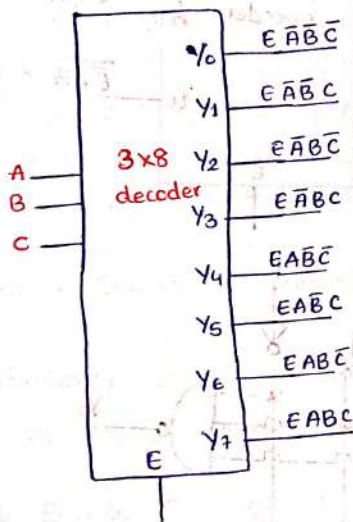
Logic 0 Enable & Logic 0 o/p decoder
2x4 Decoder

E	A	B	Y ₀	Y ₁	Y ₂	Y ₃
1	x	x	1	1	1	1
0	0	0	0	1	1	1
0	0	1	1	0	1	1
0	1	0	1	1	0	1
0	1	1	1	1	1	0



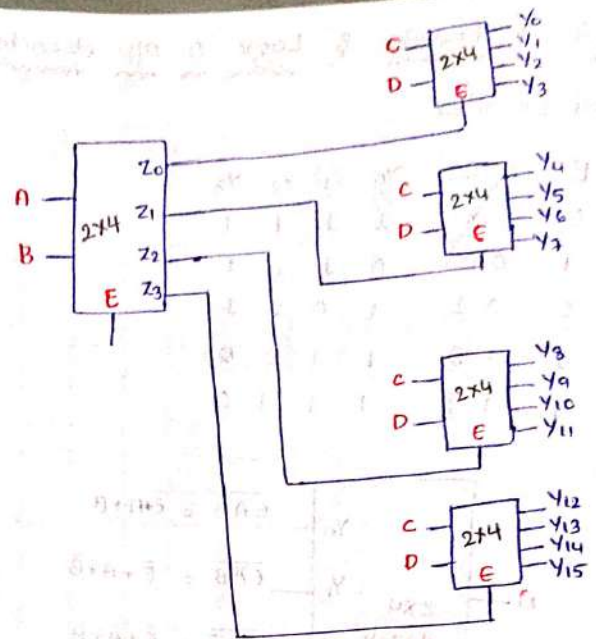
3x8 Decoder

E	A	B	C	Y ₀	Y ₁	Y ₂	Y ₃	Y ₄	Y ₅	Y ₆	Y ₇
0	x	x	x	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	0	0	0	0
1	0	0	1	0	1	0	0	0	0	0	0
1	0	1	0	0	0	1	0	0	0	0	0
1	0	1	1	0	0	0	1	0	0	0	0
1	1	0	0	0	0	0	0	1	0	0	0
1	1	0	1	0	0	0	0	0	1	0	0
1	1	1	0	0	0	0	0	0	0	1	0
1	1	1	1	0	0	0	0	0	0	0	1



Q. Implement 4:16 Decoder using minimum no of 2:4 Decoders with Enable.

A	B	C	D	Y ₀	Y ₁	Y ₂	Y ₃	Y ₄	Y ₅	Y ₆	Y ₇
0	0	0	0	1							
0	0	0	1		1						
0	0	1	0			1					
0	0	1	1				1				
0	1	0	0					1			
0	1	0	1						1		
0	1	1	0							1	
0	1	1	1								1



Q. The minimum no of 2×4 decoders required to implement 6×64 decoder with Enable is 9.

$$\frac{ABC}{1} + \frac{DEF}{8} = 9$$

Q. The minimum no of 2×4 decoders with E required to implement 6×64 decoder is 21.

6 - i/p
64 - o/p's

(See above how 2×4 decoder is giving o/p for 4:16)

$$\text{Stage 1} \quad \text{Stage 2} \quad \text{Stage 3}$$

$$1 + 4 + 16 = 21$$

Q. Implement the function $f(A,B,C) = \sum m(3,5,6,7)$ using 3x8 decoder and required logic gates.

Sol: $f(A,B,C) = \sum m(3,5,6,7)$

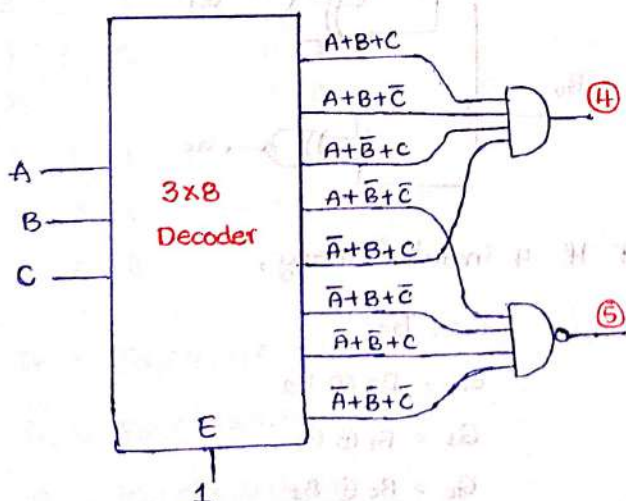
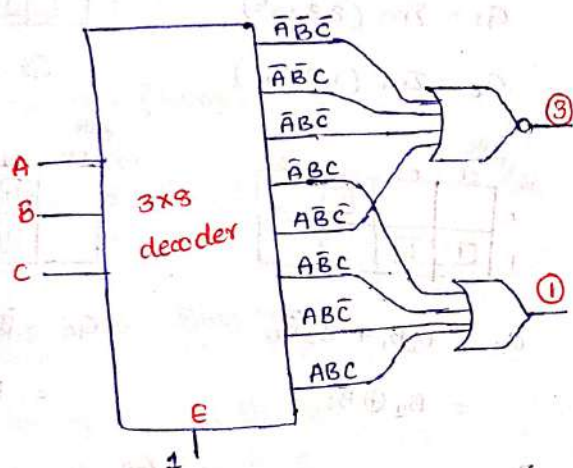
$$\text{sop} = \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC \quad \text{--- (1)}$$

$$\text{sop} = \frac{(A+\bar{B}+\bar{C})(\bar{A}+B+\bar{C})(\bar{A}+\bar{B}+C)(\bar{A}+B+C)}{\text{--- (2)}}$$

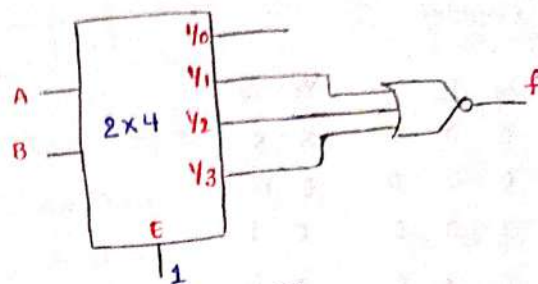
$$= \Pi M(0,1,2,4)$$

$$\text{pos} = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} \quad \text{--- (3)}$$

$$\text{pos} = (A+B+C)(A+B+\bar{C})(A+\bar{B}+C)(\bar{A}+B+C) \quad \text{--- (4)}$$



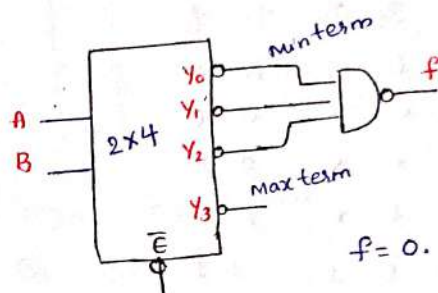
Q. In the following circuit o/p f is ____?



- (a) AB (c) $\bar{A}+\bar{B}$
(b) A+B (d) $\bar{A}\bar{B}$

$$f = \bar{A}B + A\bar{B} + AB = \bar{A}+B = \bar{A} \cdot \bar{B}$$

(ii)



$f = 0$. Disabled

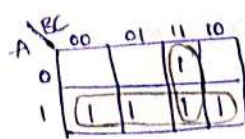
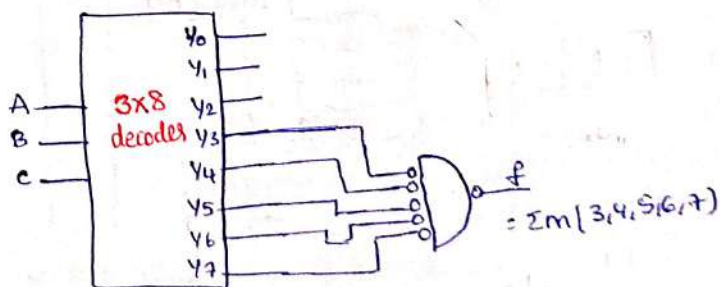
If $E=1$,

$$f(A,B) = \sum m(0,1,2) = \Pi M(3)$$

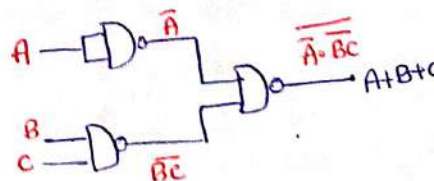
$$= \bar{A}+\bar{B}$$

$$f = \frac{\text{or}}{\bar{A}\bar{B} \cdot \bar{A}\bar{B} + \bar{A}\bar{B}} = \bar{A}\bar{B} + \bar{A}\bar{B} + \bar{A}\bar{B}$$

Q. The minimum no of 2 i/p NAND gates required to implement $f(A,B,C)$



$$f = A+BC$$



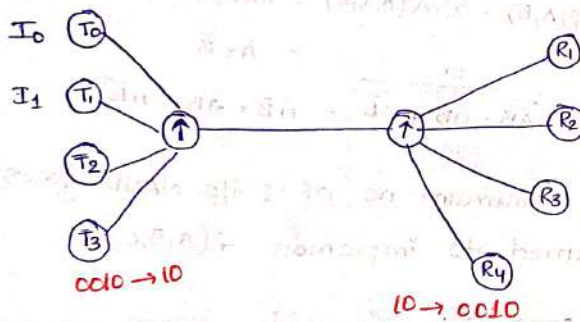
Encoder ($2^n \times n$)

4x2 Encoders

I_0	I_1	I_2	I_3	Y_1	Y_0
0	0	0	0	x	x
1	0	0	0	0	0
0	1	0	0	0	1
0	0	1	0	1	0
0	0	0	1	1	1

Priority Encoder ($I_0 > I_1 > I_2 > I_3$)

I_0	I_1	I_2	I_3	Y_1	Y_0
0	0	0	0	x	x
1	x	x	x	0	0
0	1	x	x	0	1
0	0	1	x	1	0
0	0	0	1	1	1



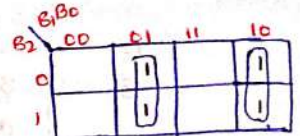
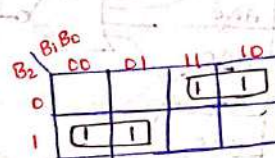
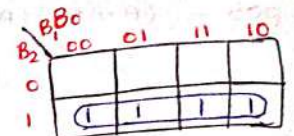
Binary to Gray :

B_2	B_1	B_0	G_2	G_1	G_0
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	1
0	1	1	0	1	0
1	0	0	1	1	0
1	0	1	1	1	1
1	1	0	1	0	1
1	1	1	1	0	0

$$G_2 = \sum m(4, 5, 6, 7)$$

$$G_1 = \sum m(2, 3, 4, 5)$$

$$G_0 = \sum m(1, 2, 5, 6)$$

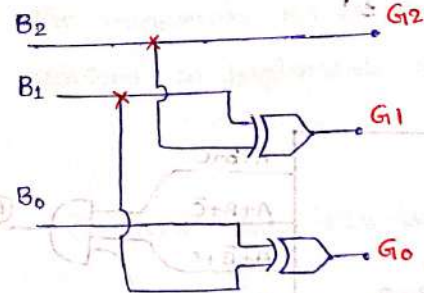


$$G_1 = \bar{B}_2 B_1 + B_2 \bar{B}_1$$

$$= B_2 \oplus B_1$$

$$G_0 = \bar{B}_1 B_0 + B_1 \bar{B}_0$$

$$= B_1 \oplus B_0$$



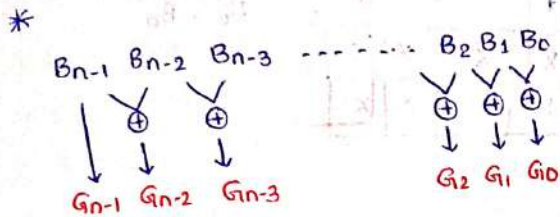
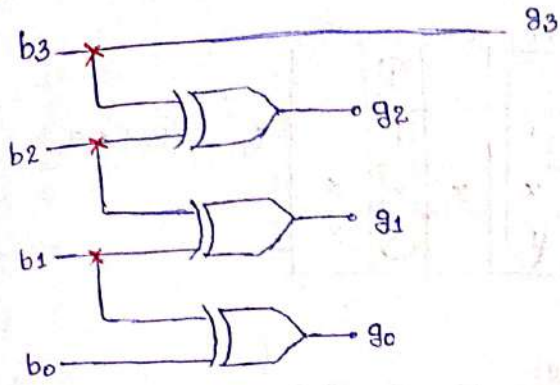
* If 4 input Binary,

$$G_3 = B_3$$

$$G_2 = B_3 \oplus B_2$$

$$G_1 = B_1 \oplus B_2$$

$$G_0 = B_0 \oplus B_1$$



eg: 10110111 (Binary)
 ↓
 11101100 (Gray)

Gray to Binary

	G_2	G_1	G_0	B_2	B_1	B_0
0	0	0	0	0	0	0
1	0	0	1	0	0	1
3	0	1	1	0	1	0
2	0	1	0	0	1	1
6	1	1	0	1	0	0
7	1	1	1	1	0	1
5	1	0	1	1	1	0
4	1	0	0	1	1	1

$$B_2 = \sum m(4, 5, 6, 7)$$

$$B_1 = \sum m(2, 3, 4, 5)$$

$$B_0 = \sum m(1, 2, 4, 7)$$

$G_2/G_1/G_0$

	00	01	11	10
0				
1	1	1	1	1

$$B_2 = G_2$$

$G_2/G_1/G_0$

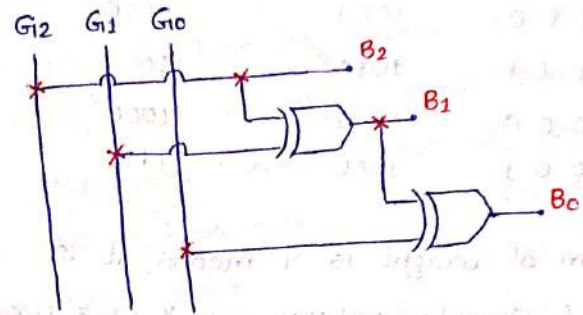
	00	01	11	10
0			1	1
1	1	1		

$$\begin{aligned} B_1 &= G_2 \bar{G}_1 + G_1 \bar{G}_2 \\ &= G_2 \oplus G_1 \\ &= B_2 \oplus G_1 \end{aligned}$$

$G_2/G_1/G_0$

	00	01	11	10
0		1		1
1	1		1	

$$\begin{aligned} B_0 &= G_2 \oplus G_1 \oplus G_0 \\ &= (G_2 \oplus G_1) \oplus G_0 \\ &= B_1 \oplus G_0 \end{aligned}$$



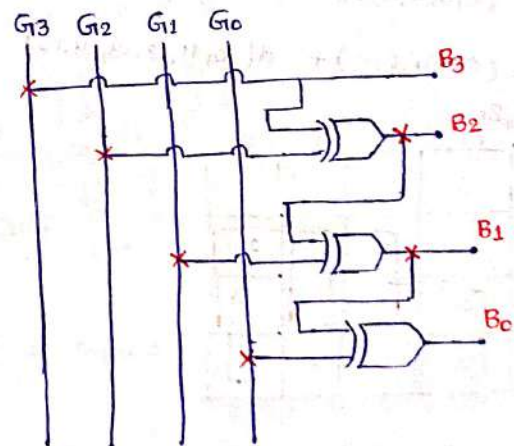
For 4 input Gray code,

$$B_3 = G_3$$

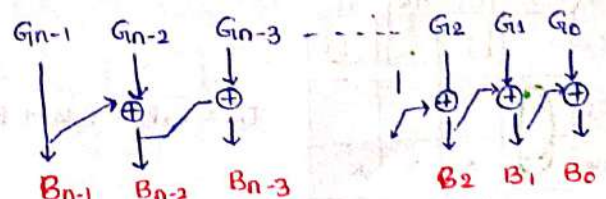
$$B_2 = G_3 \oplus G_2 = B_3 \oplus G_2$$

$$B_1 = G_3 \oplus G_2 \oplus G_1 = B_2 \oplus G_1$$

$$B_0 = G_3 \oplus G_2 \oplus G_1 \oplus G_0 = B_1 \oplus G_0$$



* N-Bit Gray to N-Bit Binary



30-07

8421 to XS-3 (BCD to Excess -3)

$B_3 B_2 B_1 B_0$	$E_3 E_2 E_1 E_0$	84-2-1
0 0 0 0	0011	0000
1 0 0 0	0100	0111
2 0 0 1	0101	0110
3 0 0 1	0110	0101
4 0 1 0 0	0111	0100
5 0 1 0 1	1000	1011
6 0 1 1 0	1001	1010
7 0 1 1 1	1010	1001
8 1 0 0 0	1011	1000
9 1 0 0 1	1100	1111

↳ Sum of weight is 9 means, it is

Self Complementary. $\rightarrow x \rightarrow x_3 x_2 x_1 x_0$

$9's x \rightarrow \bar{x}_3 \bar{x}_2 \bar{x}_1 \bar{x}_0$

↳ 8421 is a natural BCD

$$E_3 = \sum m(5, 6, 7, 8, 9) + d(10, 11, 12, 13, 14, 15)$$

$$E_2 = \sum m(1, 2, 3, 4, 9) + d(10, 11, 12, 13, 14, 15)$$

$$E_1 = \sum m(0, 3, 4, 7, 8) + d(10, 11, 12, 13, 14, 15)$$

$$E_0 = \sum m(0, 2, 4, 6, 8) + d(10, 11, 12, 13, 14, 15)$$

$B_3 B_2$ \ $B_1 B_0$	00	01	11	10
00				
01		1	1	1
10	X	X	X	X
11	1	1	X	X

$$E_3 = B_3 + B_2 B_0 + B_2 B_1$$

$B_3 B_2$ \ $B_1 B_0$	00	01	11	10
00		1	1	1
01	1			
10	X	X	X	X
11		1	X	X

$$E_2 = \bar{B}_2 B_0 + \bar{B}_2 B_1 + B_2 \bar{B}_1 \bar{B}_0$$

$B_3 B_2$ \ $B_1 B_0$	00	01	11	10
00	1		1	
01	1		1	
11	X	X	X	X
10	1		X	X

$$E_1 = \bar{B}_1 \bar{B}_0 + B_1 B_0$$

$B_3 B_2$ \ $B_1 B_0$	00	01	11	10
00	1			1
01	1			1
11	X	X	X	X
10	1		X	X

$$E_0 = \bar{B}_0$$

8 4 -2 -1 BCD to 6 3 +1 +1 BCD

	8	4	-2	-1		6	3	+1	+1
	A	B	C	D		P	Q	R	S
0	0	0	0	0		0	0	0	0
1	0	1	1	1		0	0	1	0
2	0	1	1	0		0	1	0	1
3	0	1	0	1		0	1	0	0
4	0	1	0	0		0	1	1	0
5	1	0	1	1		1	0	0	1
6	1	0	1	0		1	0	1	1
7	1	0	0	1		1	0	1	0
8	1	0	0	0		1	1	0	1
9	1	1	1	1		1	1	1	1

$$P = \sum m(5, 6, 7, 8, 9) + d(10, 11, 12, 13, 14, 15)$$

$$Q = \sum m(2, 3, 4, 8, 9) +$$

$$P = \sum m(8, 9, 10, 11, 15) + d(1, 2, 3, 12, 13, 14)$$

$$Q = \sum m(4, 5, 6, 8, 15) + d(1, 2, 3, 12, 13, 14)$$

$$R = \sum m(4, 7, 9, 10, 15) + d(1, 2, 3, 12, 13, 14)$$

$$S = \sum m(4, 8, 10, 11, 15) + d(1, 2, 3, 12, 13, 14)$$

* The no of unused bits

$$= \text{Total} - \text{used bits}$$

$$\text{For } n \text{ bits} = 2^{4n} - 10^n$$

$$\rightarrow 3 \text{ bits, used} = 0-999$$

$$\text{Unused} = 2^{12} - 1000$$

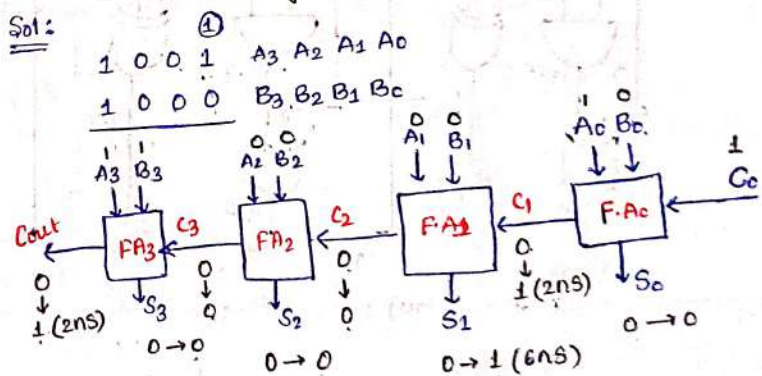
$$2 \text{ bits, Unused} = 2^8 - 100$$

Q. A full adder is taking 2ns to produce carry & 4ns to produce sum from the point input stabilised, Then the propagation delay of 4 bit ripple carry adder with $A_3 A_2 A_1 A_0$ (1001) & $B_3 B_2 B_1 B_0$ (1000) and $C_0 = 1$.

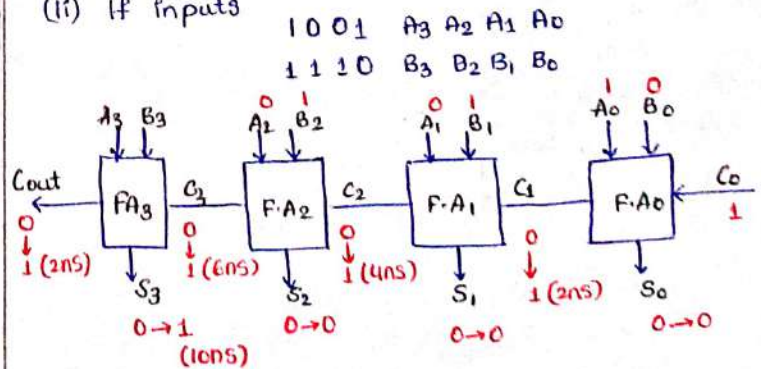
What is the propagation delay?

(Assume all the i/lps and outputs are zero initially):

Sol:

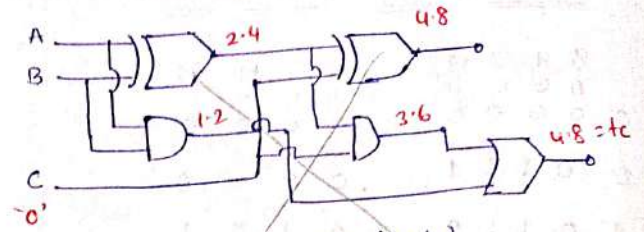


(ii) If inputs



propagation delay = 10ns

Q. A half adder is implemented with XOR and AND gates. A full adder is implemented with two half adders and OR gate (one). The propagation delay of an XOR gate is twice that of an AND/OR gate. The propagation delay of an AND/OR gate is 1.2 microseconds. A 4 bit ripple carry adder binary adder is implemented by using four full adders. The total propagation time of this 4 bit binary adder in microseconds is 12 μ s.

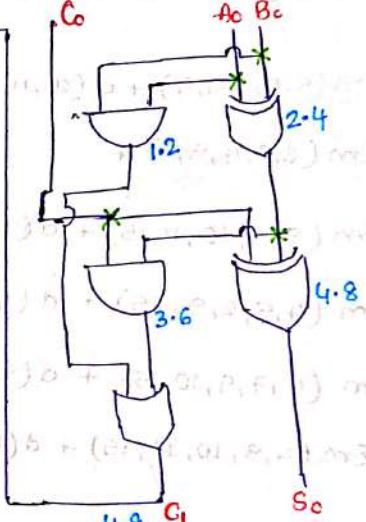
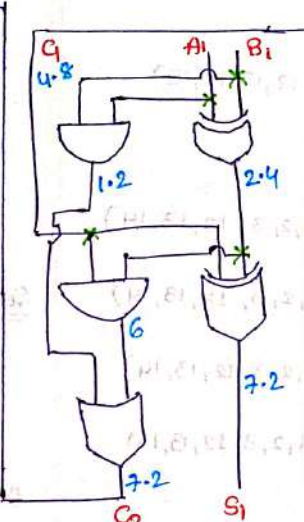
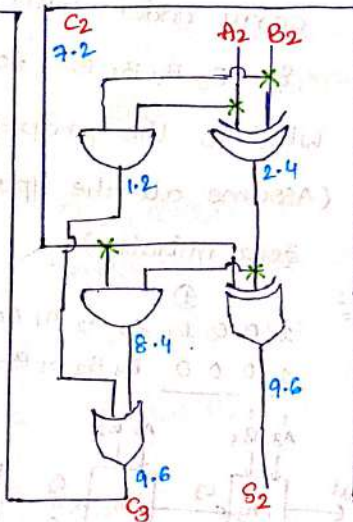
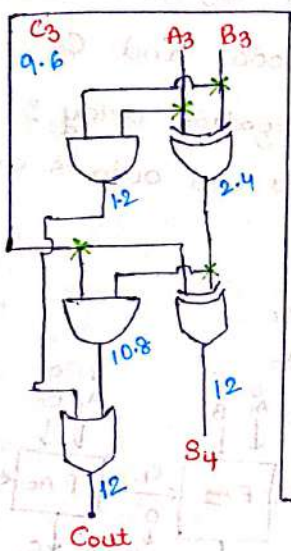


$$t_{4 \text{ bit}} = 3t_c + \max(t_s, t_c)$$

$$= 3(4.8 \mu) + 4.8 \mu$$

$$= 19.2 \mu \text{s}$$

→ Answer can't find so using below



pg 89 Q2: figure 1 shows a 4 bit ripple carry adder realized using full adders and figure 2 shows the ckt of a full adder (FA). The propagation delay of XOR, AND and OR Gates in figures 2 are 20ns, 15ns & 10ns respectively. Assume all inputs to the 4 bit adder are initially reset to 0.

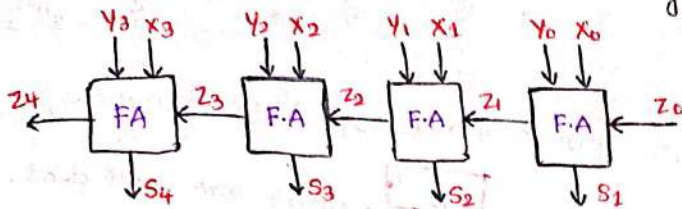


Figure (1)

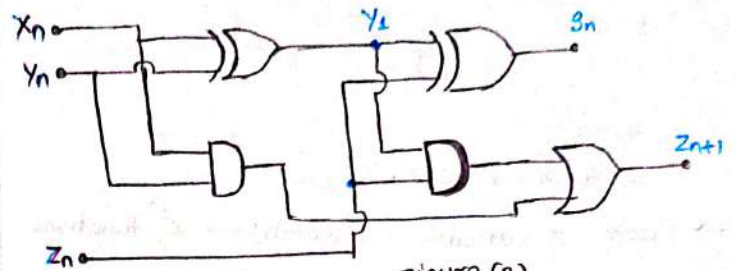
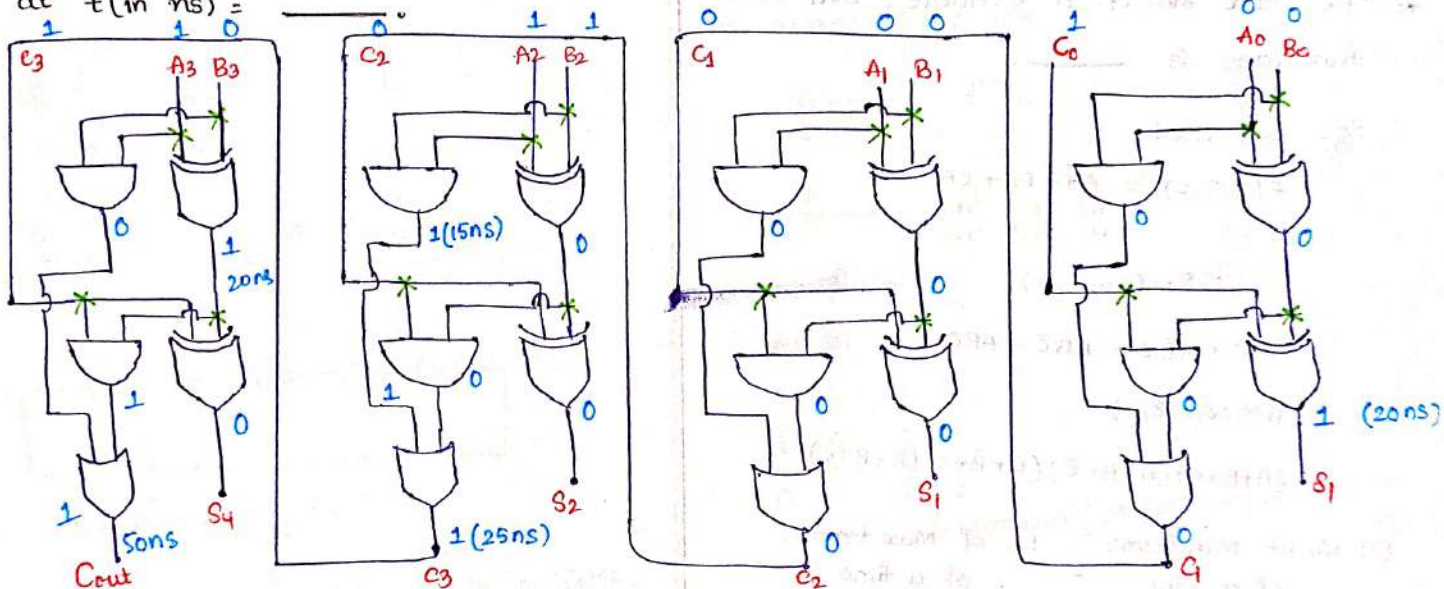


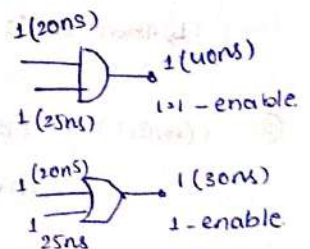
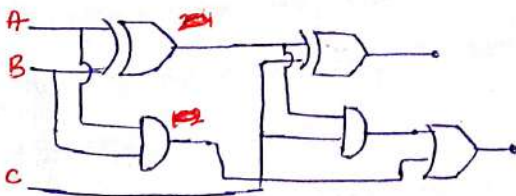
Figure (2)

At $t=0$, the inputs to the 4 bit adder are changed to $x_3x_2x_1x_0 = 1100$, $y_3y_2y_1y_0 = 0100$ and $Z_0 = 1$. The output of the ripple carry adder will be stable

at $t(\text{in ns}) =$



By using formula,



* The maximum no of n variable functions is _____.

A	B	f
0	0	
0	1	
1	0	
1	1	0 1 0 1 ... 1

$$2 - 2^2$$

$$2 \times 2 \times 2 \times 2 = 2^4 = 2^2$$

→ For 3 variable, $2^3(\text{comb}) \rightarrow 2^{2^3}$ functions

↳ For n variable, 2^n combinations
 $2^{2^n} \rightarrow$ Functions.

* The max. No of n variable, self dual functions is _____.

Eg: Self Dual.

$$f(A, B, C) = AB + BC + CA$$

$$\begin{array}{ccc} 110 & 011 & 101 \\ 111 & 111 & 111 \end{array}$$

$$= \Sigma m(3, 5, 6, 7)$$

$$= \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$$

$$= \Pi M(0, 1, 2, 4)$$

$$= (A+B+C)(A+\bar{B}+\bar{C})(A+\bar{B}+C)(\bar{A}+B+C)$$

(i) No of Min terms of a func = No of Max terms of a func.

↳ then it is self-dual. (may / not)

$$\textcircled{2} f(A, B, C) = \bar{A}B + A\bar{B}$$

$$= (A+B)(\bar{A}+\bar{B}) \rightarrow \text{It is not self dual.}$$

(ii) if 000 is in min term, then 111 be in max term.

Similarly,

$$000 \quad 1001 \quad 2010 \quad 3011 \rightarrow \text{Min term}$$

$$7111 \quad 6100 \quad 5101 \quad 4100 \rightarrow \text{Max term}$$

(No priority)

$$\hookrightarrow (0,7) (1,6) (2,5) (3,4)$$

$$2c_1 \quad 2c_1 \quad 2c_1 \quad 2c_1 = 2^4 \text{ (For 3 variable)}$$

For n variables, the combinations be
 $= (0, 2^n-1) (1, 2^n-2) (2, 2^n-3) \dots (2^{n-1}, 2^n)$
 $2c_1 \times 2c_1 \times 2c_1 \dots \times 2c_1$

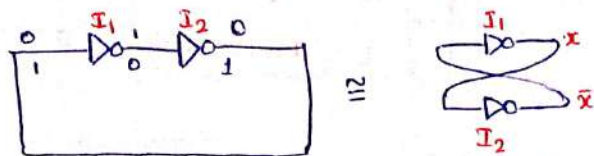
$$= 2^x = 2^{2^{n-1}} \quad \frac{2^n}{2} = 2^{n-1} = x$$

∴ For n variables, 2^n combinations,

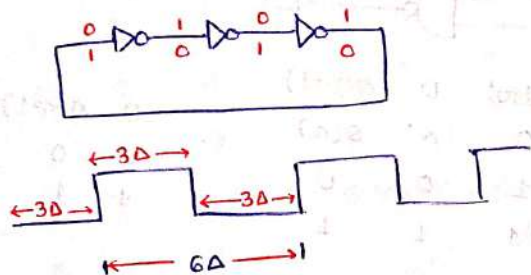
$2^{2^{n-1}}$ pairs are self dual.

Sequential Circuits : (4-6)m

↳ If 2 inverters are connected then it is bistable.

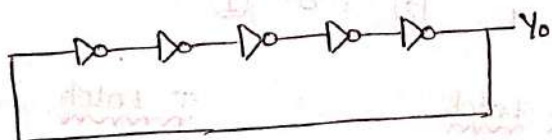


↳ If 3 inverters are connected, it is astable oscillator.



$$f = \frac{1}{6\Delta} = \frac{1}{2 \times 3\Delta} \Rightarrow f = \frac{1}{2 \times n\Delta}$$

Q.



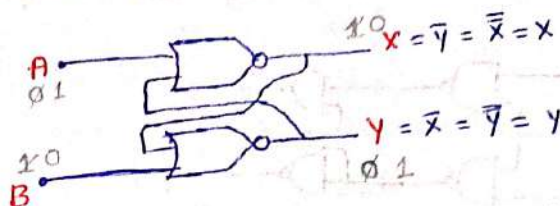
If $\Delta = 10\text{ps}$, $f = ?$

$$f = \frac{1}{2n\Delta} = \frac{1}{2 \times 5 \times 10 \times 10^{-12}} = \frac{1}{10^{-9}} = 1\text{GHz}$$

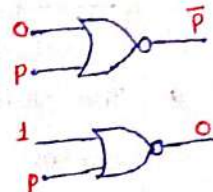
* To hold/store the i/p which i need, there is no feasibility in above ckt. so i/p is added.

↳ i/p conditions for remembering
" " to modify.

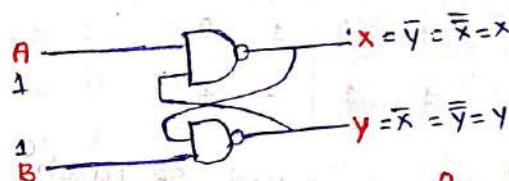
* Let A & B are o/p's



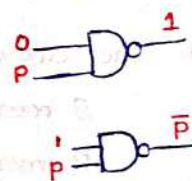
A	B	x	y
0	0	previous	
0	1	1	0
1	0	0	1
1	1	0	0
0	0	Indetermined (01 or 10)	



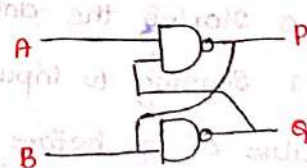
* NAND Based memory element :



A	B	x	y
0	0	previous	
0	1	1	0
1	0	0	1
0	0	1	1
1	1	Indeterminate [01 (or) 10]	



Q. Gate

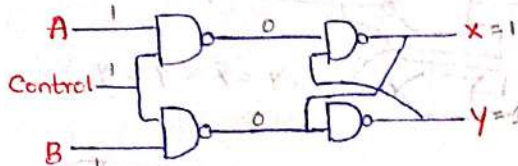


A	B	P	Q
0	1	1	0
1	1	1	0
0	0	1	1
1	1	[01 or 10]	
		Indeterminate	

L

31-07-25

Control signal $\rightarrow$ inputs are applied through control signal.



* The noise is affecting the o/p when control signal not present.

Control	A	B	X	Y
0	X	X	previous	
1	0	0	previous	
1	0	1	0	1
1	1	0	1	0
1	1	1	1	1

* The above is called SR Latch.

S means Set when $S=1, Q=1$

R means Reset when $R=1, Q=0$

S	R
(Set)	(Reset)
$Q=1$	$Q=0$
$S=1$	$R=1$

$\rightarrow$ When Control = 0 storing the data
Control = 1 Sensing to inputs

$Q(n) \rightarrow$ stored value of Q before Sensing to inputs.

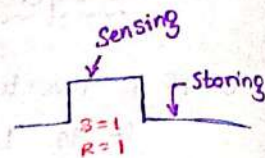
$Q(n+1) \rightarrow$ stored value of Q after sensing to inputs.

$\rightarrow$ At $S=1, R=1, Q(n+1)$ is indeterminate.

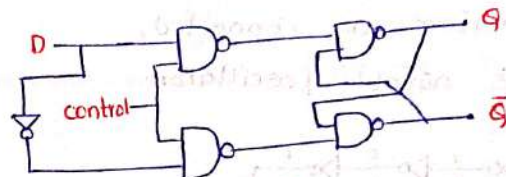
Why?

To know what is stored, we had to make control = 0. And $S=1, R=1$. The output may be 0, 1 or 1, 0. Hence it is indeterminate.

S	R	$Q(n+1)$
0	0	$Q(n)$
0	1	0
1	0	1
1	1	Indeterminate



D-Latch



Control	D	$Q(n+1)$	$Q(n)$
0	X	$Q(n)$	
1	0	0	0
1	1	1	1

Control	D	S	R	$Q(n+1)$
0	X	X	X	$Q(n)$
1	0	0	1	0
1	1	1	0	1

J-K Latch

T-Latch

J	K	$Q(n+1)$	T	$Q(n+1)$
0	0	$Q(n)$	0	$Q(n)$
0	1	0	1	$\overline{Q(n)}$
1	0	1		
1	1	$\overline{Q(n)}$		

Characteristic Tables

(i) SR Latch

S	R	Q(n+1)
0	0	Q(n)
0	1	0
1	0	1
1	1	Indetermined

Truth table

Excitation Table

S	R	Q(n)	Q(n+1)	Q(n)	Q(n+1)	S	R
0	0	0	0	0	0	0	X
0	0	1	1	0	1	1	0
0	1	0	0	1	0	0	1
0	1	1	0	1	1	X	0
1	0	0	1				
1	0	1	1				
1	1	0	X				
1	1	1	X				

$$Q(n+1) = \sum m(1, 4, 5) + d(6, 7)$$

S	R	Q(n)	Q(n+1)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

$$Q(n+1) = S + \bar{R}Q(n)$$

(ii) D-Latch

D	Q(n+1)
0	0
1	1

Truth Table

Excitation table

D	Q(n)	Q(n+1)	Q(n)	Q(n+1)	D
0	0	0	0	0	0
0	1	0	0	1	1
1	0	1	1	0	0
1	1	1	1	1	1

D	Q(n)	Q(n+1)
0	0	0
0	1	0
1	0	1
1	1	1

$$Q(n+1) = D$$

(iii) JK Latch

J	K	Q(n+1)
0	0	Q(n)
0	1	0
1	0	1
1	1	$\bar{Q}(n)$

Truth table

Excitation Table

J	K	Q(n)	Q(n+1)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Q(n)	Q(n+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

J	K	Q(n)	Q(n+1)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

$$Q(n+1) = \bar{K}Q(n) + J\bar{Q}(n)$$

(iv) T Latch

T	Q(n+1)
0	Q(n)
1	$\bar{Q}(n)$

Truth table

Excitation table

T	Q(n)	Q(n+1)
0	0	0
0	1	1
1	0	1
1	1	0

Q(n)	Q(n+1)	T
0	0	0
0	1	1
1	0	1
1	1	0

T	Q(n)	Q(n+1)
0	0	0
0	1	1
1	0	1
1	1	0

$$Q(n+1) = T\bar{Q}(n) + \bar{T}Q(n)$$

$$= T \oplus Q(n)$$

Conversion of Flip Flops / Latches

Procedure :

1. Write truth table for required flip flop and extend with given F.F i/p's.
2. Fill given Flip flop inputs column with the help of excitation table of given flip flop.
3. Find Boolean expression for given Flip flop inputs in terms of required flip flop inputs and state $q(n)$ (Q).
4. Draw logic diagram.

(i) SR to JK

J	K	Q(n)	Q(n+1)	S	R
0	0	0	0	0	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	0	0	1
1	0	0	1	1	0
1	0	1	1	X	0
1	1	0	1	1	0
1	1	1	0	0	1

$$S = \sum m(4, 6) + d(1, 5)$$

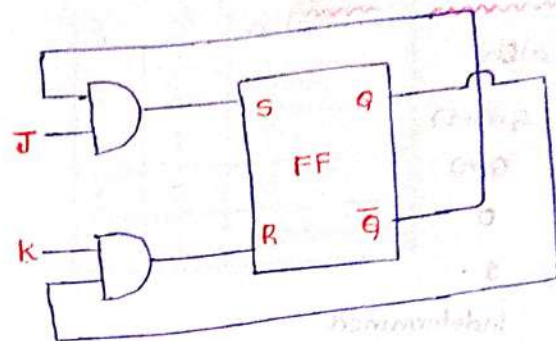
$$R = \sum m(3, 7) + d(0, 2)$$

J \ KQ	00	01	11	10
0		X		
1	1	X		1

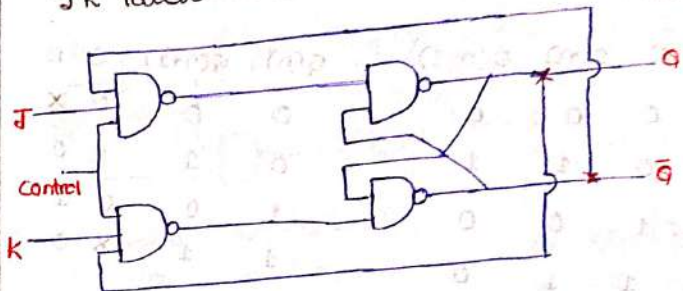
$$S = J\bar{Q}$$

J \ KQ	00	01	11	10
0	X		1	X
1			1	

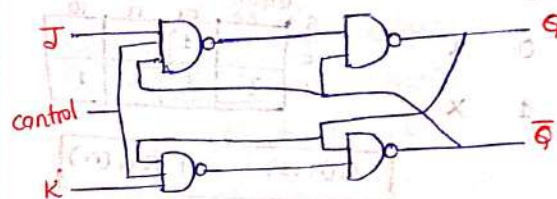
$$R = KQ$$



JK latch



$$\approx$$



(ii) JK to T

T	Q(n)	Q(n+1)	J	K
0	0	0	0	X
0	1	1	X	0
1	0	1	1	X
1	1	0	X	1

$$J = \sum m(\bar{2}) + d(3, \bar{2})$$

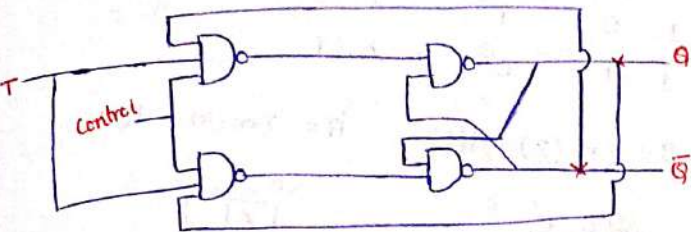
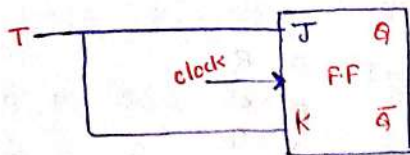
$$K = \sum m(\bar{2}) + d(0, 2)$$

T \ Q	0	1
0		X
1	1	X

$$J = T$$

T \ Q	0	1
0	X	X
1	X	1

$$K = T$$



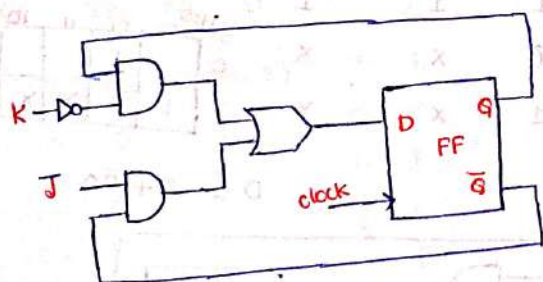
(iii) D to JK

J	K	Q(n)	Q(n+1)	D
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

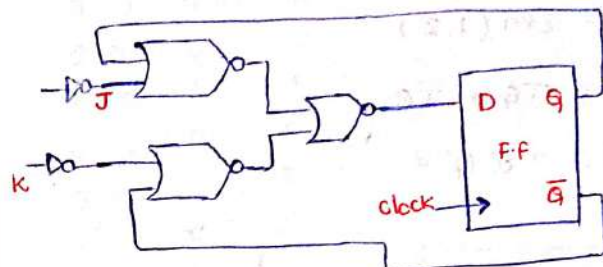
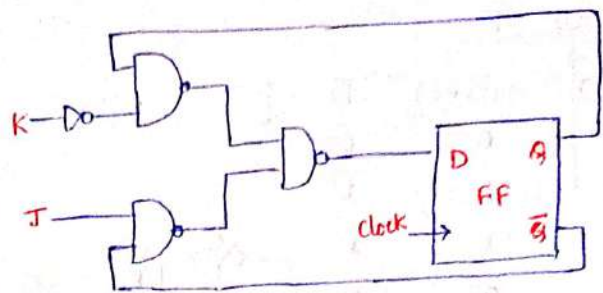
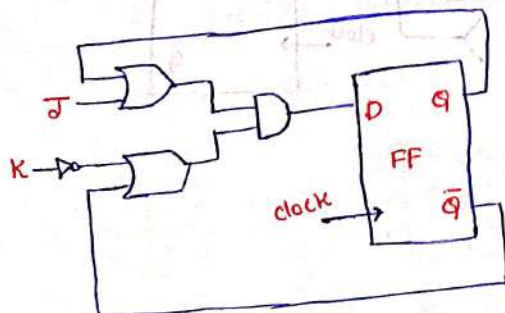
$$D = \sum m(1, 4, 5, 6)$$

$$D = J\bar{Q} + \bar{K}Q$$

J	K	Q	00	01	11	10
0	0	1				
1	1	1	1	1	1	1



$$\text{POS} \Rightarrow D = (J+Q)(\bar{K}+\bar{Q})$$



(iv) T to JK

J	K	Q(n)	Q(n+1)	T
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	1
1	0	1	1	0
1	1	0	1	1
1	1	1	0	1

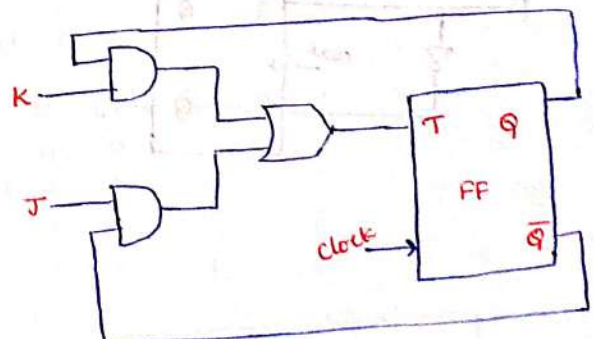
$$T = \sum m(3, 4, 6, 7)$$

$$T = KQ + J\bar{Q}$$

Max

$$T = (J+Q)(K+\bar{Q})$$

J	K	Q	00	01	11	10
0	0	1				
1	1	1	1	1	1	1



(v) D to T

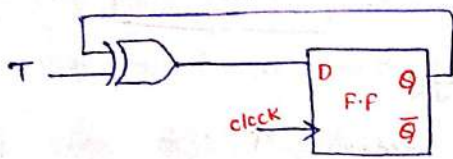
T	Q(n)	Q(n+1)	D
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

T	Q
0	0
0	1
1	0
1	1

$$D = \sum m(1, 2)$$

$$= T\bar{Q} + \bar{T}Q$$

$$D = T \oplus Q$$



(vi) SR to D

D	Q(n)	Q(n+1)	S	R
0	0	0	0	X
0	1	0	0	1
1	0	1	1	0
1	1	1	X	0

$$S = \sum m(2) + d(3)$$

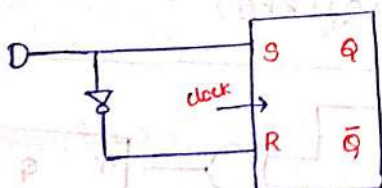
$$R = \sum m(1) + d(0)$$

D	Q
0	0
0	1
1	0
1	1

$$S = D$$

D	Q
0	0
0	1
1	0
1	1

$$R = \bar{D}$$



(vii) SR to T

T	Q(n)	Q(n+1)	S	R
0	0	0	0	X
0	1	1	X	0
1	0	1	1	0
1	1	0	0	1

$$S = \sum m(2) + d(1)$$

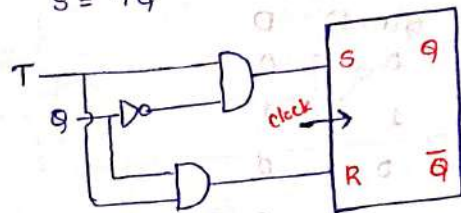
$$R = \sum m(3) + d(0)$$

T	Q
0	0
0	1
1	0
1	1

$$S = T\bar{Q}$$

T	Q
0	0
0	1
1	0
1	1

$$R = TQ$$



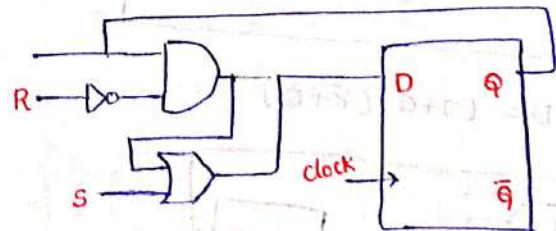
(viii) D to SR

S	R	Q(n)	Q(n+1)	S	R
0	0	0	0	0	0
0	0	1	1	1	0
0	1	0	0	0	1
0	1	1	0	0	1
1	0	0	1	1	0
1	0	1	1	1	0
1	1	0	X	X	X
1	1	1	X	X	X

$$D = \sum m(1, 4, 5) + d(6, 7)$$

S	R
0	0
0	1
1	0
1	1

$$D = S + \bar{R}Q$$



(ix) JK to SR

S	R	Q(n)	Q(n+1)	J	K
0	0	0	0	0	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	0	X	1
1	0	0	1	1	X
1	0	1	1	X	0
1	1	0	X	X	X
1	1	1	X	X	X

$$J = \sum m(4) + d(1,3,5,7,6)$$

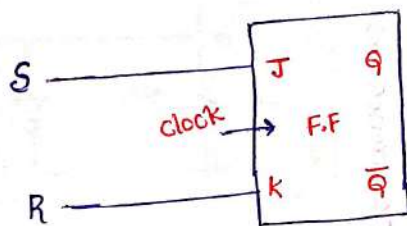
$$K = \sum m(3) + d(0,2,4,6,7)$$

S/R	00	01	11	10
0		X	X	
1	1	X	X	X

$$J = S$$

S/R	00	01	11	10
0	X		1	X
1	X		X	X

$$K = R$$



(x) JK to D

D	Qn	Q(n+1)	J	K
0	0	0	0	X
0	1	0	X	1
1	0	1	1	X
1	1	1	X	0

$$J = \sum m(2) + d(1,3)$$

$$K = \sum m(1) + d(0,2)$$

D	0	1
0		X
1	1	X

$$J = D$$

D	0	1
0	X	1
1		X

$$K = \bar{D}$$

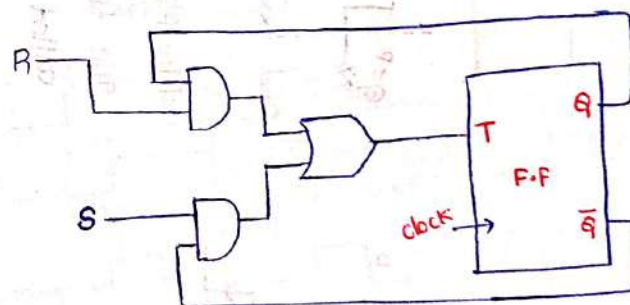
(xi) T to SR

S	R	Q(n)	Q(n+1)	T
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	1
1	0	1	1	0
1	1	0	X	X
1	1	1	X	X

$$T = \sum m(3,4) + d(6,7)$$

S/R	00	01	11	10
0			1	
1	1		X	X

$$T = RQ + S\bar{Q}$$

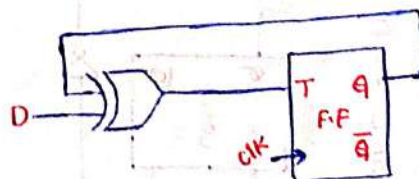


(xii) T to D

D	Q(n)	Q(n+1)	T
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

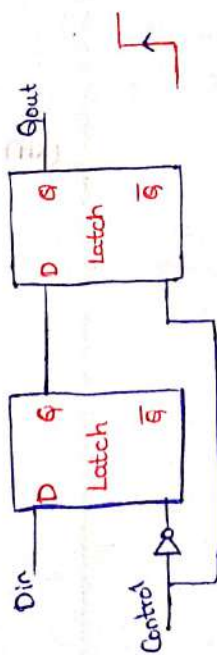
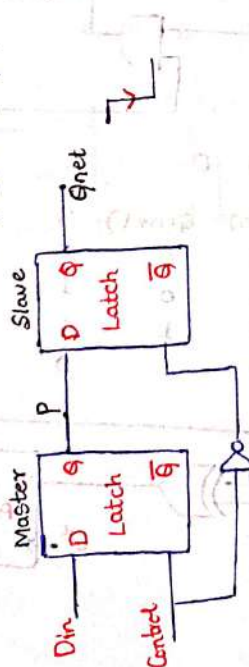
$$T = \sum m(1,2)$$

$$T = \bar{D}Q + D\bar{Q} = D \oplus Q$$

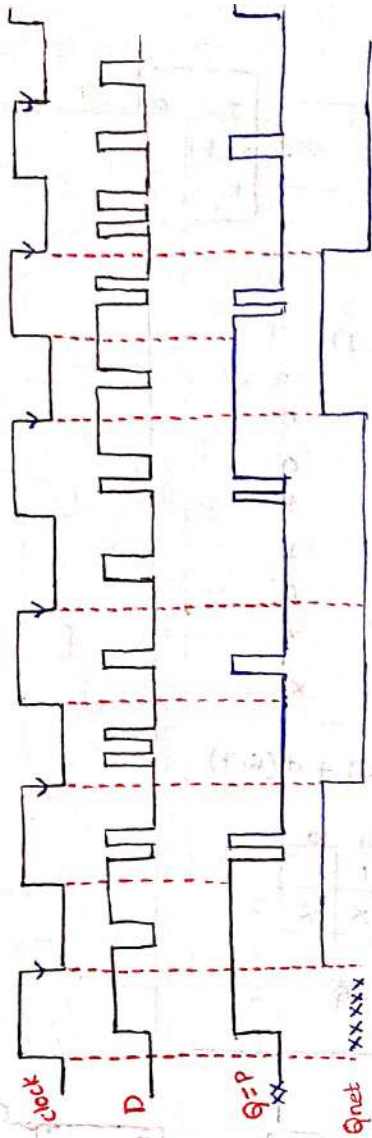


logic 1 enable		logic 0 enable	
Control	D	Q(n+1)	Q(n)
0	x	Q(n)	Q(n)
1	0	0	0
1	1	1	1

* → Disable → Q
 Enable → D
 → Enable - P
 disable Q
 Here Q becomes P.



- * Edge triggered, Edge sensitive - flip flop
- Level triggered, - Latch
- Latch is enable: → o/p is transparent to i/p.
 ↳ D is constant.
- * Master → enable → o/p - P vary with D. -ve edge
- Slave → disabled → no affect with master
 ↳ follow own Q no change in i/p.
- Master → disable → o/p const.



Flip Flop delay / clock to Q delay ($t_{ff} / t_{ck \rightarrow Q}$):

The time required to change flip flop output from clock edge occurred is called flip flop delay or clock to Q delay.

Setup Time (t_s):

The minimum time required for input to be stabilize before occurrence of clock edge is called set up time.

Hold Time (t_{hold}):

The minimum time required input to be stable after clock edge occurrence is called hold time.

Q. If AB Flip flop functionality is defined as shown in table then convert (i) $AB \rightarrow D$

(ii) $AB \rightarrow T$

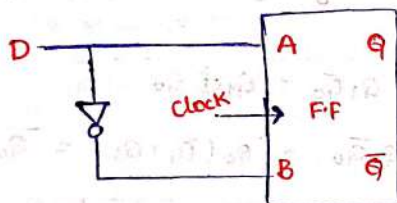
(iii) $D \rightarrow AB$

A	B	$Q(n+1)$
0	0	$\overline{Q(n)}$
0	1	0
1	0	1
1	1	$\overline{Q(n)}$

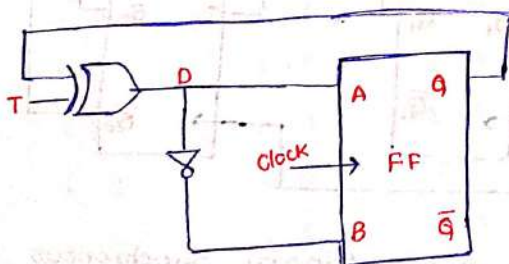
(i) $AB \rightarrow D$

A	B	$Q(n+1)$
0	0	$\overline{Q(n)}$
0	1	0
1	0	1
1	1	$\overline{Q(n)}$

D	$Q(n+1)$
0	0
1	1



(ii) $AB \rightarrow T$



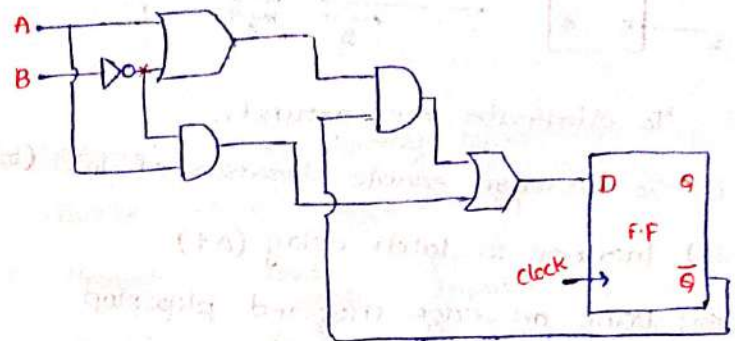
(iii) $D \rightarrow AB$

A	B	$Q(n)$	$Q(n+1)$	D
0	0	0	1	1
0	0	1	0	0
0	1	0	0	0
0	1	1	0	0
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	0	1

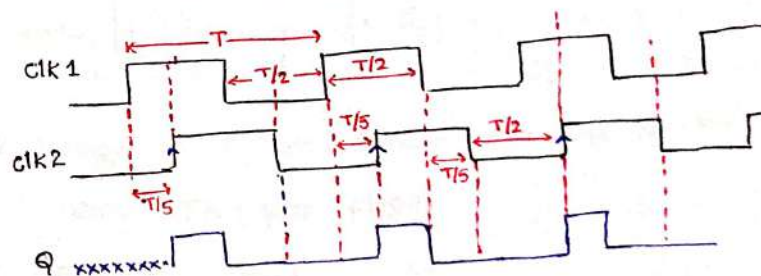
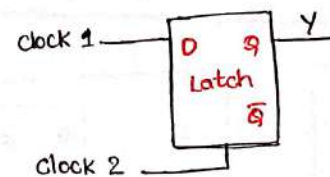
$\overline{A}B$	00	01	11	10
0	1			1
1	1	1		1

$$D = \sum m(0, 4, 5, 6)$$

$$D = \overline{B}\overline{Q} + A\overline{B} + A\overline{Q} \\ = A\overline{B} + \overline{Q}(A + \overline{B})$$



Q. In the following, clock 1 is 50% of duty cycle, clock 2 is delayed version of clock 1 by $t/5$. Assume Latch delay is 0. The % of duty cycle of output wave-form is.



$$T_{on} = \frac{T}{2} - \frac{T}{5} = \frac{3T}{10}$$

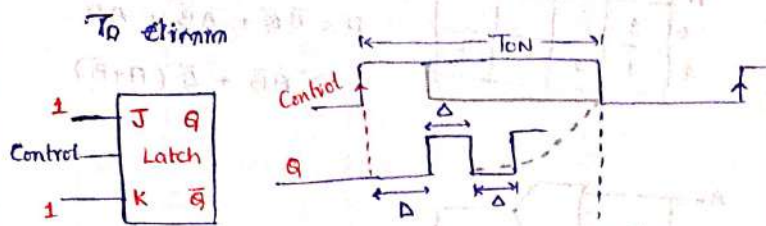
$$T_{off} = \frac{T}{2} + \frac{T}{5} = \frac{7T}{10}$$

$$\% \text{ duty cycle} = \frac{T_{on}}{T_{on} + T_{off}} \times 100$$

$$= \frac{3T/10}{3T/10 + 7T/10} \times 100$$

$$= 30\%$$

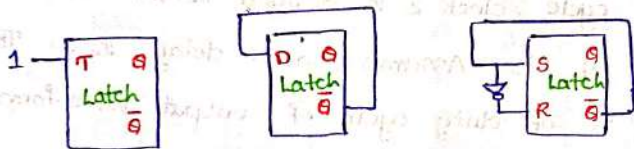
Race Around Condition :



To eliminate race around,

- To decrease enable duration of latch ($T_{ON} \downarrow$)
- Increase in latch delay ($\Delta \uparrow$)
- Using an edge triggered flip flop instead of level triggered latch.

(Latch)
Other Flip Flops also exhibit race around:



* Counters *

↳ Which going through predefined sequence of states is called Counter.

Synchronous
(Ripple counter)

↳ All FF are Synchronous in sensing to i/p.

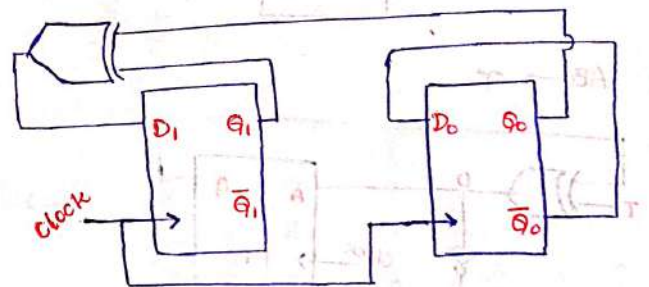
Synchronous Counter

Q. Design a 2 bit binary synchronous up counter using delay Flip flop.

Present state		Next State		FF Input's	
Q_1	Q_0	Q_1	Q_0	D_1	D_0
0	0	0	1	0	1
0	1	1	0	1	0
1	0	1	1	1	1
1	1	0	0	0	0

$$D_1 = \bar{Q}_1 \bar{Q}_0 + Q_1 \bar{Q}_0 = Q_1 \oplus Q_0$$

$$D_0 = \bar{Q}_1 \bar{Q}_0 + Q_1 \bar{Q}_0 = \bar{Q}_0 (\bar{Q}_1 + Q_1) = \bar{Q}_0$$

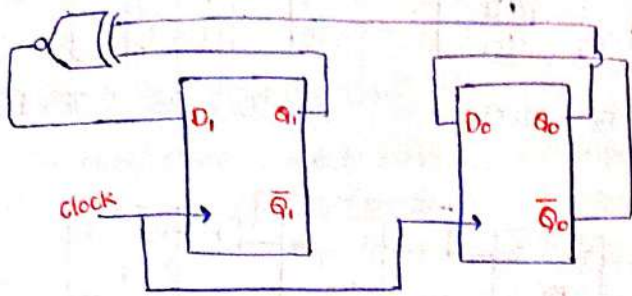


Q. Design 2 bit Binary Synchronous down counter using delay Flip flop.

Present State		Next State		FF i/p's	
Q_1	Q_0	Q_1	Q_0	D_1	D_0
1	1	1	0	1	0
1	0	0	1	0	1
0	1	0	0	0	0
0	0	1	1	1	1

$$D_1 = Q_1 Q_0 + \bar{Q}_1 \bar{Q}_0 = Q_1 \odot Q_0$$

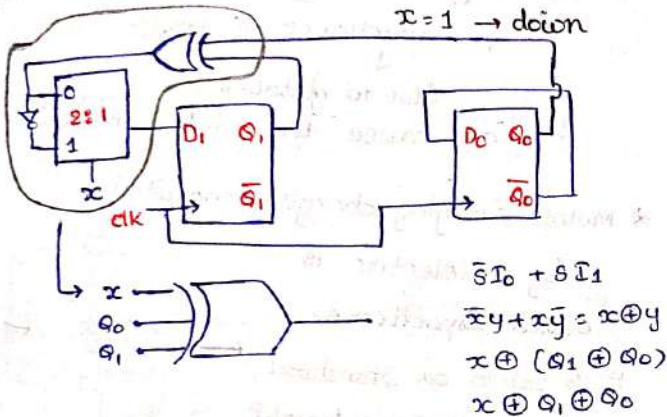
$$D_0 = Q_1 \bar{Q}_0 + \bar{Q}_1 Q_0 = \bar{Q}_0$$



Q. Design a 2 bit binary synchronous up and down counters using delay (D) FF.

With Control Signal $x=0 \rightarrow \text{up}$.

$x=1 \rightarrow \text{down}$



x	Present state		Next state		inputs	
	Q ₁	Q ₀	Q ₁	Q ₀	Q ₁	Q ₀
0	0	0	0	1	0	1
0	0	1	1	0	1	0
0	1	0	1	1	1	1
0	1	1	0	0	0	0
1	0	0	1	1	1	1
1	1	1	1	0	1	0
1	1	0	0	1	0	1
1	0	1	0	0	0	0

$$Q_1 = \sum m(1, 2, 4, 7)$$

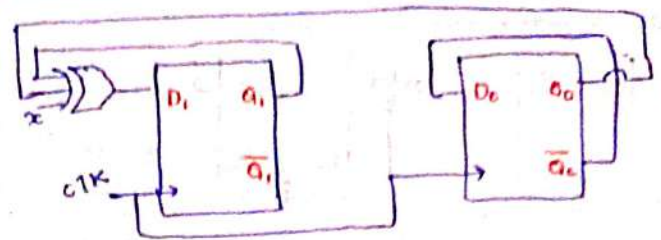
$$Q_0 = \sum m(0, 2, 4, 6)$$

Q ₁ \ Q ₀	0	1
0	1	1
1	1	1

$$D_1 = x_1 \oplus Q_1 \oplus Q_0$$

Q ₁ \ Q ₀	0	1
0	1	1
1	1	1

$$D_0 = \bar{Q}_0$$



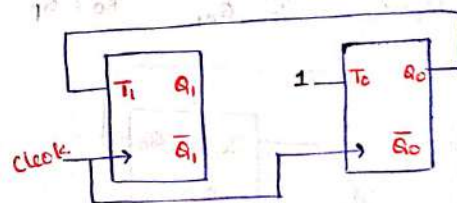
Q. Design a 2 bit binary synchronous up counter using toggle (T) FF.

So!

Present state		Next state		FF Input's	
Q ₁	Q ₀	Q ₁	Q ₀	T ₁	T ₀
0	0	0	1	0	1
0	1	1	0	1	1
1	0	1	1	0	1
1	1	0	0	1	1

$$T_1 = \bar{Q}_1 Q_0 + Q_1 \bar{Q}_0 = Q_0 (\bar{Q}_1 + Q_1) = Q_0$$

$$T_0 = 1$$



Q. Design a 2 bit synchronous up counter using JK (flip flops).

Present State		Next State		F.F Inputs			
Q ₁	Q ₀	Q ₁	Q ₀	J ₁	K ₁	J ₀	K ₀
0	0	0	1	0	X	1	X
0	1	1	0	1	X	X	1
1	0	1	1	X	0	1	X
1	1	0	0	X	1	X	1

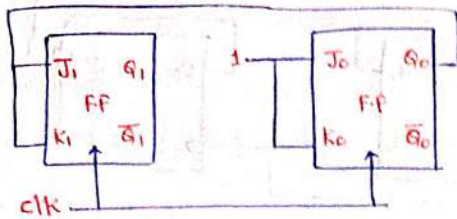
Q ₁ \ Q ₀	0	1
0	1	1
1	1	1

$$J_1 = Q_0$$

$$K_1 = Q_0$$

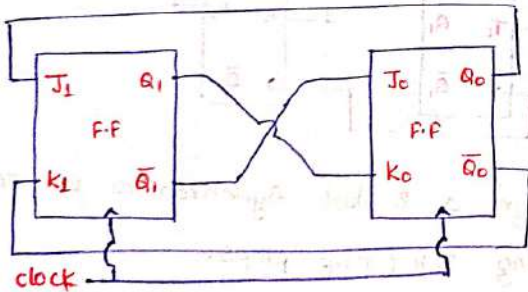
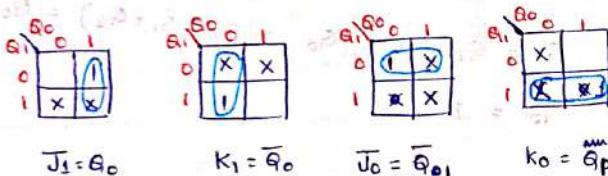
$$J_0 = 1$$

$$K_0 = 1$$



Q. Design 2 bit Synchronous grey counter using JK flip flops.

Present State	Next State	FF Input's
$Q_1 \ Q_0$	$Q_1 \ Q_0$	$J_1 \ K_1 \ J_0 \ K_0$
0 0	0 1	0 x 1 x
0 1	1 0	1 x x 0
1 0	1 1	x 0 x 1
1 1	0 0	x 1 0 x



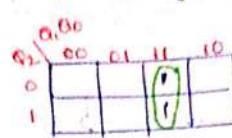
Q. Design 3 bit Binary up counter using Toggle (T) Flip flop.

Present state	Next state	F.F Inputs.
$Q_2 \ Q_1 \ Q_0$	$Q_2 \ Q_1 \ Q_0$	$T_2 \ T_1 \ T_0$
0 0 0	0 0 1	0 0 1
0 0 1	0 1 0	0 1 1
0 1 0	0 1 1	0 0 1
0 1 1	1 0 0	1 1 1
1 0 0	1 0 1	0 0 1
1 0 1	1 1 0	0 1 1
1 1 0	1 1 1	0 0 1
1 1 1	0 0 0	1 1 1

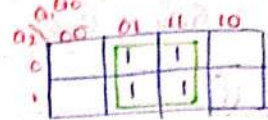
$$T_2 = \sum m(3, 7)$$

$$T_1 = \sum m(1, 3, 5, 7)$$

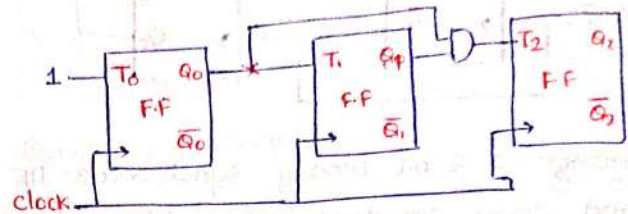
$$T_0 = 1 = \sum m(0, 1, 2, 3, 4, 5, 6, 7)$$



$$T_2 = Q_1 Q_0$$



$$T_1 = \bar{Q}_0 \quad T_0 = 1$$

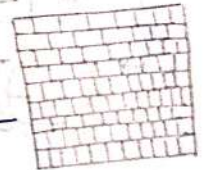


* Net length is more, Net resistance & Capacitance is more.

(due to metals)
↳ Can cause the delay more.

* metals carrying charge separated by dielectric is called Capacitance.

It is called as standard cell. Uniform height



* If 20 nm is present technology, Next technology is

$$A = x \cdot x = x^2$$

$$A' = \frac{x}{\sqrt{2}} \cdot \frac{x}{\sqrt{2}} = \frac{x^2}{2} = \frac{A}{2}$$

$$\rightarrow A' = \frac{20}{\sqrt{2}} = 10\sqrt{2} = 14.14 \rightarrow 14 \text{ nm Next technology}$$

* Comparing among 3-input & 2 input

↳ We can extend upto N bits to get desired.



Q. Design MOD-6 synchronous up counter using toggle flip flop.

MODULUS

No of distinct states counter is repeating is called modulus of the counter.

present state	Next state	F.F Inputs
$Q_2 \ Q_1 \ Q_0$	$Q_2 \ Q_1 \ Q_0$	$T_2 \ T_1 \ T_0$
0 0 0	0 0 1	0 0 1
1 0 0	0 1 0	0 1 1
2 0 1 0	0 1 1	0 0 1
3 0 1 1	1 0 0	1 1 1
4 1 0 0	1 0 1	0 0 1
5 1 0 1	0 0 0	1 0 1

$$T_2 = \sum m(3, 5) + d(6, 7)$$

$$T_1 = \sum m(1, 3, 5) + d(6, 7)$$

$$T_0 = \sum m(0, 1, 2, 3, 4, 5) + d(6, 7)$$

$Q_2 \ Q_1 \ Q_0$	00	01	11	10
0	1	1	1	1
1	1	1	X	X

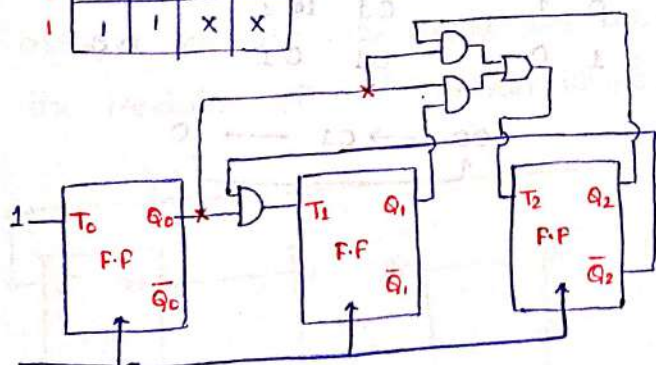
$Q_2 \ Q_1 \ Q_0$	00	01	11	10
0	1	1	1	1
1	1	1	X	X

$$T_2 = Q_1 Q_0 + Q_2 Q_0$$

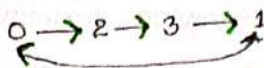
$$T_1 = \bar{Q}_2 Q_0 + Q_1 Q_0$$

$Q_2 \ Q_1 \ Q_0$	00	01	11	10
0	1	1	1	1
1	1	1	X	X

$$T_0 = 1$$



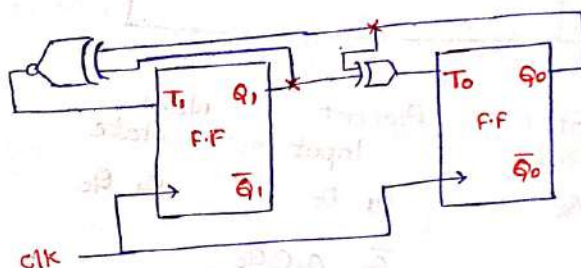
Q. Design synchronous counter using Toggle flip flop which is going through



present state	Next state	F.F 1/p's
$Q_1 \ Q_0$	$Q_1 \ Q_0$	$T_1 \ T_0$
0 0	1 0	1 0
0 1	1 1	0 1
1 1	0 1	1 0
0 1	0 0	0 1

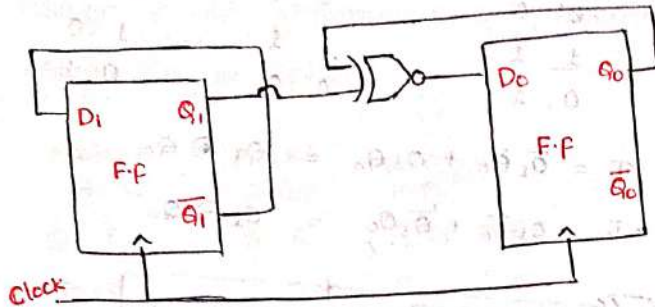
$$T_1 = \bar{Q}_1 \bar{Q}_0 + Q_1 Q_0 = Q_1 \oplus Q_0$$

$$T_0 = Q_1 \bar{Q}_0 + \bar{Q}_1 Q_0 = Q_1 \oplus Q_0$$



Analysing Synchronous

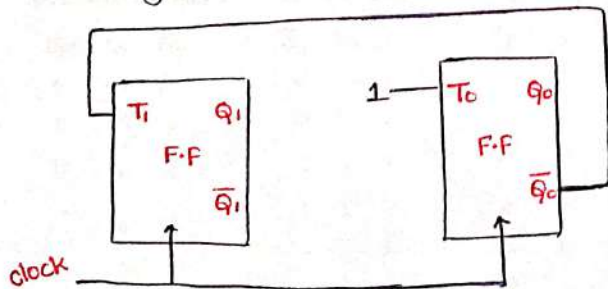
Q. In the following Synchronous counter, assume initially Q_1, Q_0 are 0,0. The Sequence of $Q_1 Q_0$ Counter is giving through _____.



Present State $Q_1 Q_0$	Present Input $D_1 D_0$	Next State $Q_1 Q_0$
	$\bar{Q}_1 \quad Q_1 \oplus Q_0$	
0 0	1 1	1 1
1 1	0 1	0 1
0 1	1 0	1 0
1 0	0 0	0 0

00 → 11 → 01 → 10

Q. In the following synchronous counter assume initially Q_1, Q_0 are 0,0. The Sequence of $Q_1 Q_0$ Counter is giving through _____.

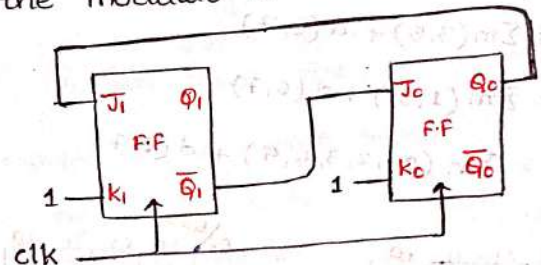


T	$Q(n+1)$
0	$Q(n)$
1	$\bar{Q}(n)$

Present state $Q_1 Q_0$	Present Input $T_1 T_0$	Next state $Q_1 Q_0$
	$\bar{Q}_0 \quad 1$	
0 0	1 1	1 1
1 1	0 1	1 0
1 0	1 1	0 1
0 1	0 1	0 0

00 → 11 → 10 → 01

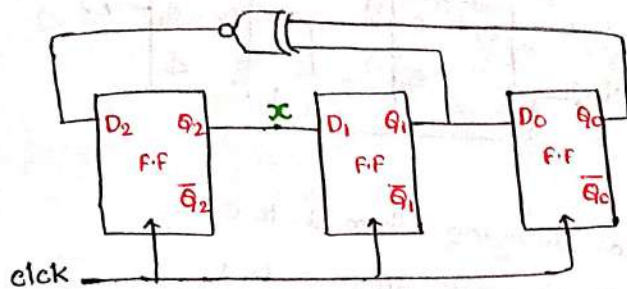
Q. In the following synchronous counter, assume initially Q_1, Q_0 are 0,0. the modulus of Counter is _____.



present State $Q_1 Q_0$	Present Input $J_1 K_1 J_0 K_0$	Next State $Q_1 Q_0$
	$Q_0 \quad 1 \quad \bar{Q}_1 \quad 1$	
0 0	0 1 1 1	0 1
0 1	0 1 1 1	1 0
1 0	0 1 0 1	0 0

00 → 01 → 10

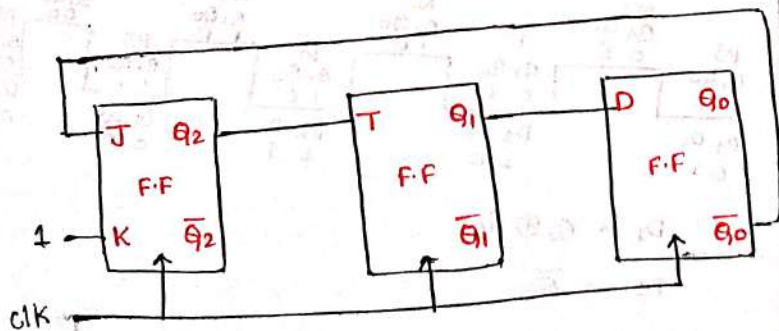
Q. In the following synchronous counter assume initially all the F.F are in reset the sequence observed at x is ____.



Present State	present Input	Next State	($\because$ In DFF $NS, P.I = NS$)
$Q_2 \ Q_1 \ Q_0$	$D_2 \ D_1 \ D_0$	$Q_2 \ Q_1 \ Q_0$	
x	$Q_1 Q_0$	$Q_2 \ Q_1$	
0 0 0	1 0 0	1 0 0	
1 0 0	1 1 0	1 1 0	
1 1 0	0 1 1	0 1 1	
0 1 1	1 0 1	1 0 1	
1 0 1	0 1 0	0 1 0	
0 1 0	0 0 1	0 0 1	
0 0 1	0 0 0	0 0 0	

x is 0 1 1 0 1 0 0 - - -

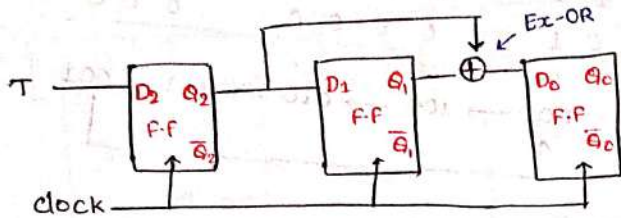
Q. In the following synchronous counter assume initially Q_2, Q_1, Q_0 are 0,0,0 the Modulus of the Counter is 5.



Present State	Present Input	Next State
$Q_2 \ Q_1 \ Q_0$	$T \ K \ T \ D$	$Q_2 \ Q_1 \ Q_0$
	$\bar{Q}_0 \ 1 \ Q_2 \ Q_1$	
0 0 0	1 1 0 0	1 0 0
1 0 0	1 1 1 0	0 1 0
0 1 0	1 1 0 1	1 1 1
1 1 1	0 1 1 1	0 0 1
0 0 1	0 1 0 0	0 0 0
lock out	0 1 1	0 1 1
Restarting	1 0 0	1 0 0
Restarting	1 1 1	1 1 1

Self starting
for every unused, if it is
restart. Means No lock out

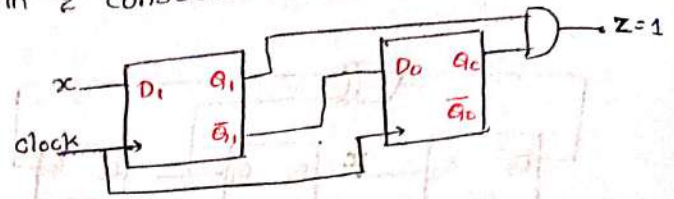
Q. In the following sequential circuit the serial input data $x = 110011011$ is applied in 9 consecutive clock cycle. Assuming initially Q_2, Q_1, Q_0 are 0,0,0 the state Q_2, Q_1, Q_0 after 9th clock cycle is _____.



$x = 110011011$
 $\uparrow$
 start

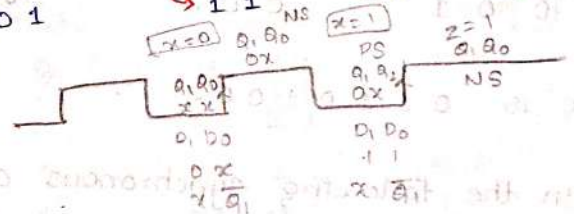
Present State $Q_2 \ Q_1 \ Q_0$	Present Input $D_2 \ D_1 \ D_0$	Next state $Q_2 \ Q_1 \ Q_0$
0 0 0	1 0 0	1 0 0
1 0 0	1 1 1	1 1 1
1 1 1	0 1 0	0 1 0
0 1 0	0 0 1	0 0 1
0 0 1	1 0 0	1 0 0
1 0 0	1 1 1	1 1 1
1 1 1	0 1 0	0 1 0
0 1 0	1 0 1	1 0 1
1 0 1	1 1 1	1 1 1

Q. In the following sequential circuit output $z=1$ when serial data input x in 2 consecutive clock cycles is _____.

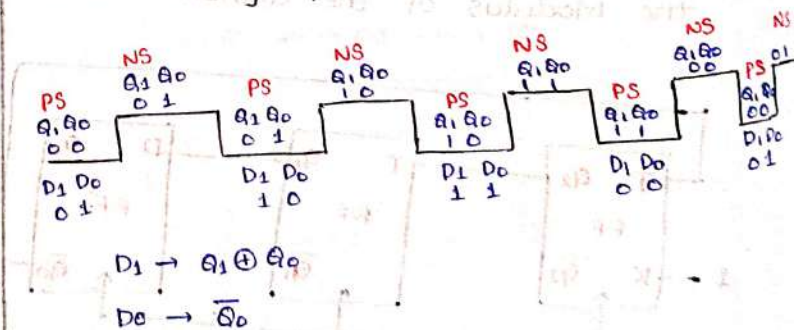


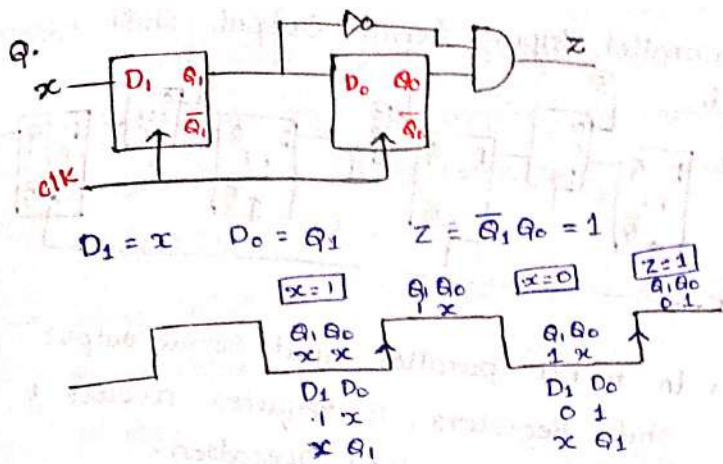
- (a) changing from 1 to 0
- (b) changing from 0 to 1
- (c) constant at 1
- (d) constant at 0

present state $Q_1 \ Q_0$	present Input $D_1 \ D_0$	Next state $Q_1 \ Q_0$	outputs $Q_1 \ Q_0$
0 0	$x \ \bar{Q}_1$	0 1	0
0 1	0 1	1 1	1 ✓
1 1	1 1	1 0	0
1 0	0 0	0 0	0
0 0	0 1	0 1	0
0 1	1 1	1 1	1 ✓



2 Bit Binary up counter w.r.to clock.

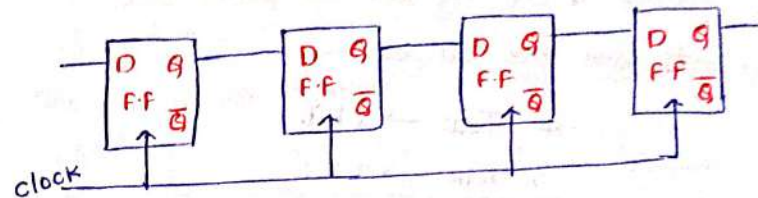




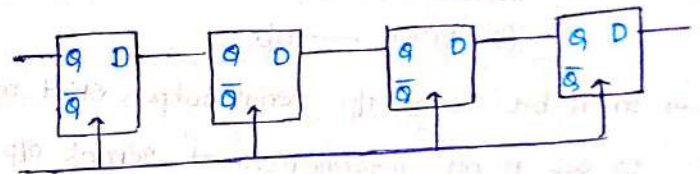
* Register :- *

↳ To store n bit information collectively we use n number of flip flop is called n-bit register.

* Let consider a 4 bit flip flop which connected Synchronously.



1 1 0 0 → 12
 0 1 1 0 → 6 = (12/2)
 0 0 1 1 → 3 = (6/2)
 0 0 0 1 → 1 = (3/2) → Right Shift

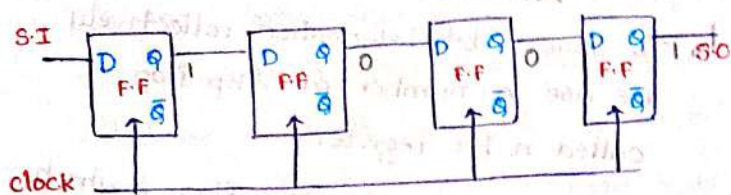


0 0 0 1 → 1
 0 0 1 0 → 2 = (1*2)
 0 1 0 0 → 4 = (2*2)
 1 0 0 0 → 8 = (4*2)

* In shift register, shifting right with 1 bit supplying 0 in MSB is giving division by 2

* In shift register, shifting left in 1 bit supplying 0 in LSB is giving multiply by 2

Serial input Serial output shift register



How many cycles are required for above 4 bit register at serial output or to give the output?

4 Tclk $\rightarrow$ 1 bit

3 Tclk $\rightarrow$ 3 bit

7 Tclk $\rightarrow$ 4 bit

* For n flip flops, are there

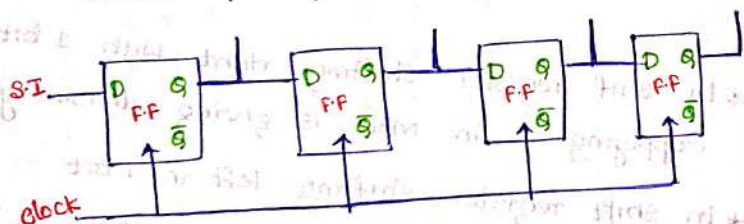
n Tclk $\rightarrow$ 1 bit

(n-1) Tclk $\rightarrow$ (n-1) bit

(2n-1) Tclk $\rightarrow$ n bit

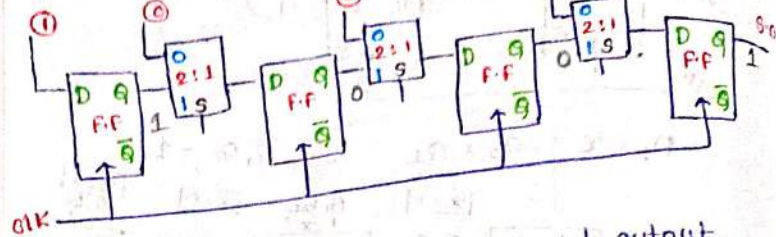
$\rightarrow$ In n bit Serial I/p Serial output shift register to get n-bit information at serial o/p which is supplied serially at serial I/p port requires (2n-1) clock cycles.

Serial Input parallel output shift register



* It requires n Tclk clock cycles for n bit information.

parallel Input Serial Output shift register

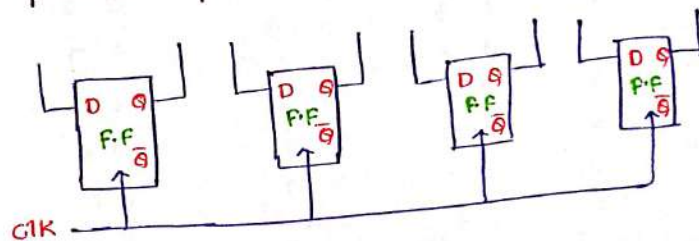


$\rightarrow$ In n bit parallel input Serial output Shift Registers, it requires n clock cycles to complete operation.

* For parallel input data, we make s=0 in each multiplexer.

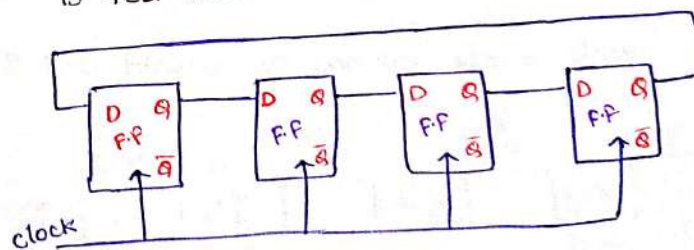
* For Serial shifting, we make s=1 in each multiplexer.

parallel input parallel Output shift register



The No of clock cycles are 1.

* In the 4 bit given shift register, the o/p is fed back to the input. What Happen.



$\Uparrow$
It is called as Ring counter.

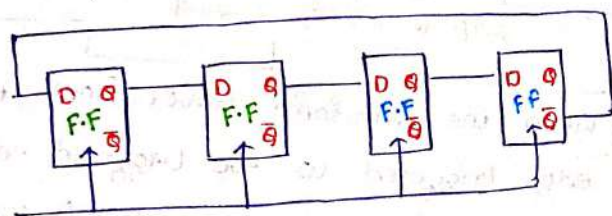
1010	}	1000	} 4
0101		0100	
1010		0010	
	0001		
	1000		

→ In N-bit shift register if output (Q) of last flip flop is applied as input to the first flip-flop resulting n-bit ring counter.

→ Using N-bit ring counter, it is possible to generate maximum n different state.

→ The Maximum modulus of n-bit ring counter is n.

* If the output $\bar{Q}$ is applied (feedback to input, what is called it is.



It is called as twisted ring counter
Johnson counter.
Ring Tail Counter.

0000
1000
1100
1110
1111
0111
0011
0001
0000

8 (2x4)

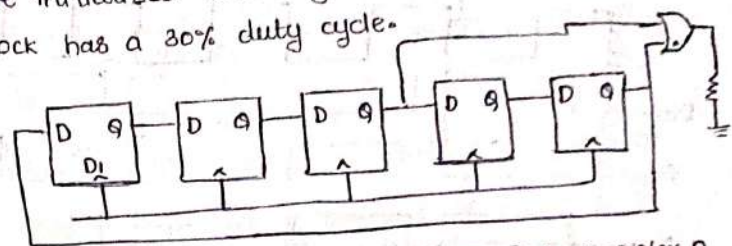
→ In N-bit shift register, if complement of last FF o/p ($\bar{Q}$) is applied as i/p to 1st FF resulting n-bit twisted ring counter.

→ Using n-bit Johnson counter it is possible to generate 2n states.

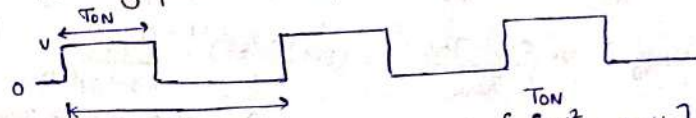
→ So the maximum modulus of n-bit Johnson counter is 2n.

pg 145.

Q8. Assume that all the digital gates in the ckt shown, are ideal, the resistor $R=10K\Omega$. Supply voltage = 5v. The D-Flip Flops D_1, D_2, D_3, D_4, D_5 are initialized with logic values 0, 1, 0, 1 & 0. The clock has a 30% duty cycle.



The avg power dissipated (in mW) in resistor R



$$P_{avg} = \frac{1}{T} \int_0^T f(t) dt$$

$$= \frac{V^2}{R} (0.3)$$

$$= \frac{5^2}{10K} (0.3) = 0.75 \text{ mW} (*)$$

$$P_{avg} = \frac{1}{T} \left[\int_0^{T_{on}} \frac{V^2}{R} + 0 \cdot dt \right]$$

$$= \frac{1}{T} \cdot \frac{V^2}{R} \int_0^{T_{on}} 1 \cdot dt$$

$$= \frac{1}{T} \cdot \frac{V^2}{R} [t]_0^{T_{on}}$$

$$= \frac{V^2}{R} \cdot \frac{T_{on}}{T}$$

Q1	Q2	Q3	Q4	Q5
0	1	0	1	0
0	0	1	0	1
1	0	0	1	0
0	1	0	0	1
1	0	1	0	0
0	1	0	1	0

$$P_{avg} = \frac{V^2}{R} \cdot \frac{T_{on}}{T}$$

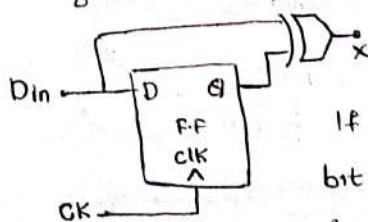
$$= \frac{5^2}{10K} \cdot \frac{3}{5}$$

$$= 1.5 \text{ mW} \times$$

$$= 1.5 \times 0.3 = 0.45 \text{ mW} \times$$

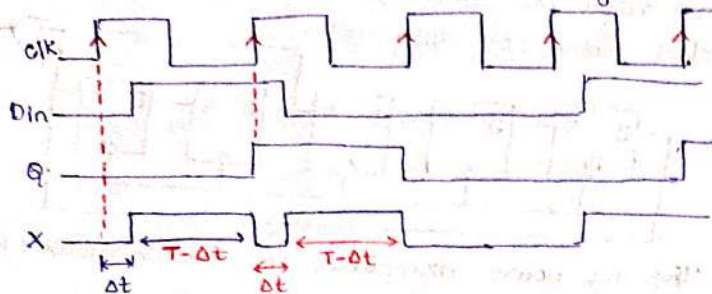
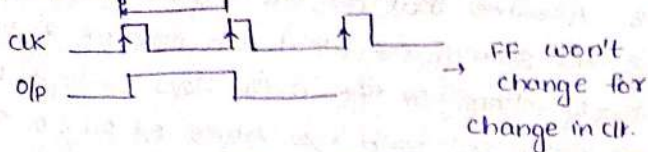
Gate

Q. In the ckt shown below, a +ve edge triggered D F.F used for sampling i/p data D_{in} using clock CK. The XOR gate outputs 3.3 volts for logic high and 0 volts for logic low levels. The data bit & clock periods are equal & value of $\frac{\Delta T}{T_{ck}} = 0.15$, where ΔT & T_{ck} are shown in fig. Assume that the FF & XOR gate are ideal



If probability of i/p data bit (D_{in}) transition in each clock period is 0.3, the avg value of voltage at node x, is _____.

10 clk cycle switching 8 times



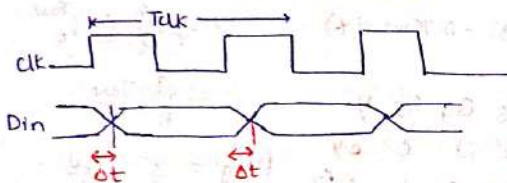
$$V_{avg} = V \cdot \frac{0.85 T_{ck}}{T_{ck}} = (3.3)(0.85) = 2.805$$

$$V_{avg} = (0.3)(2.805) = 0.8415$$

For all Din change to clk signal

$$\frac{\Delta T}{T} = 0.15 \Rightarrow \Delta T = 0.15T$$

$$T - \Delta t = 0.85 T_{ck}$$



$$V_{avg} = \frac{1}{T_{ck}} \int_0^{T_{ck}} V dt = \frac{1}{T_{ck}} \left[\int_0^{0.15} 0 \cdot dt + \int_{0.15}^{1-T} V \cdot dt \right]$$

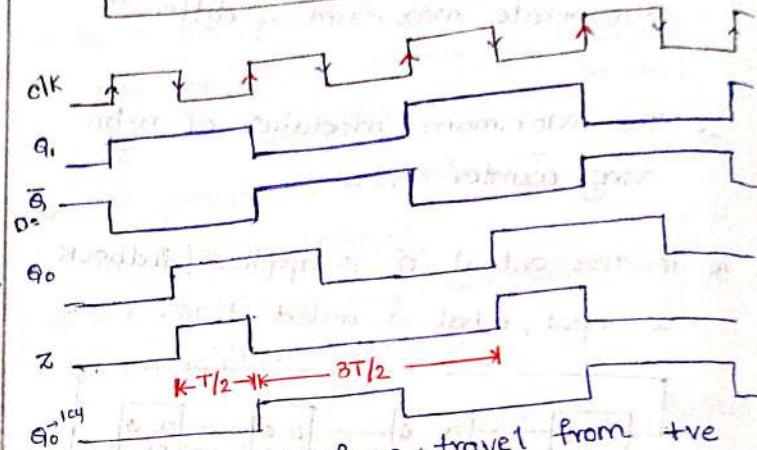
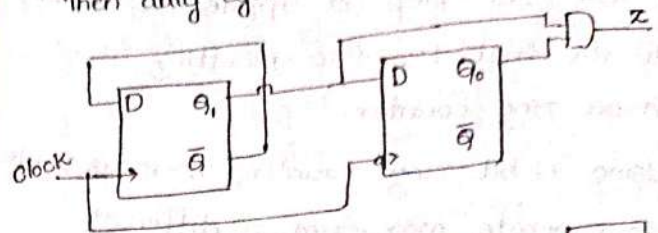
$$= \frac{1}{T_{ck}} \cdot V \int_{0.15}^{T_{ck}} 1 \cdot dt$$

$$= \frac{1}{T_{ck}} \cdot V [t]_{0.15}^{T_{ck}}$$

$$= \frac{V(T_{ck} - 0.15)}{T_{ck}} =$$

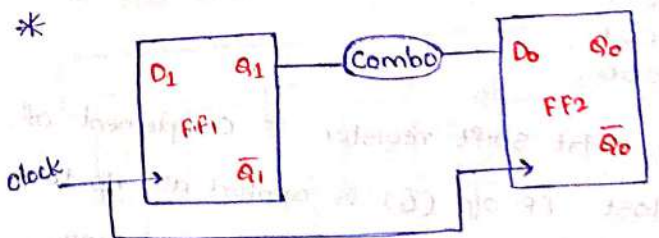
Q2-Q3

Assume the duty cycle of clock is 50%.
Then duty cycle of Z is 50%.



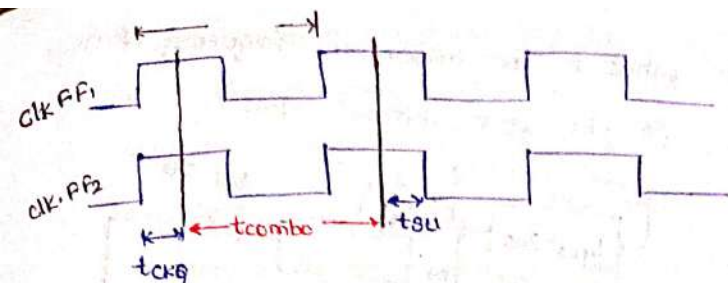
* When the waveform travel from +ve edge triggered to -ve triggered edge there will be a delay of $\frac{1}{2}$ cycle.

* When the waveform travel from +ve edge triggered to another +ve edge triggered, the delay would be of an 1 cycle.



FF1 → Launching data

FF2 → Capturing the data.



- The data must reach on or before to the t_{su} , then there is no problem,
- If the data is reaching late means, it will be due to providing delay time late. So minimum delay has to give such that it will reach on correct.

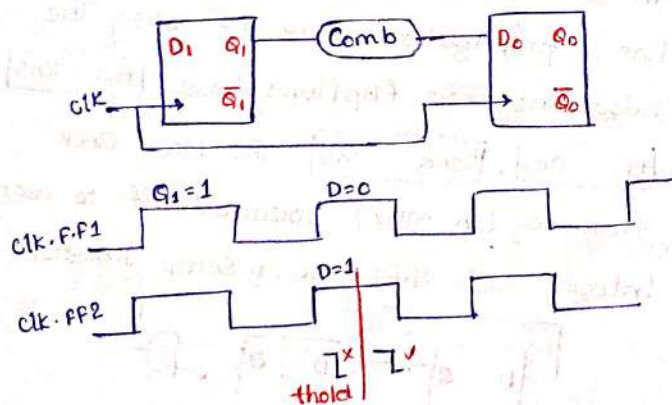
$$T_{min} = t_{ckq} + t_{comb} + t_{su}$$

$F.F1$ $F.F2$

- The maximum frequency will be,

$$f_{max} = \frac{1}{T_{min}} \rightarrow \text{Max. clock frequency.}$$

- * Suppose sequentially data is flown through Flip flops.

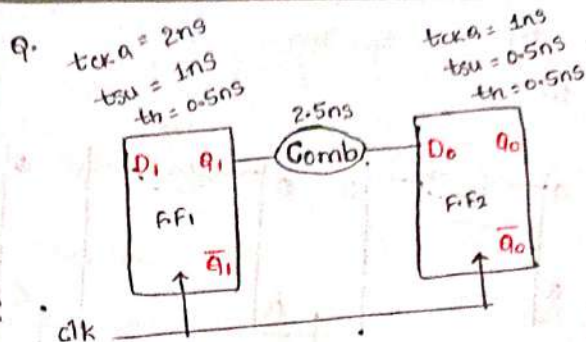


- D_0 is capturing simultaneously D_1 is launching. Time delay can occur if D_0 didn't receive immediately & D_1 launch faster.

Then,

$$t_{ckq} + t_{comb} \geq t_{hold}$$

- * Hold is not affecting the performance.



- What is the maximum clock frequency so that circuit work without fail?

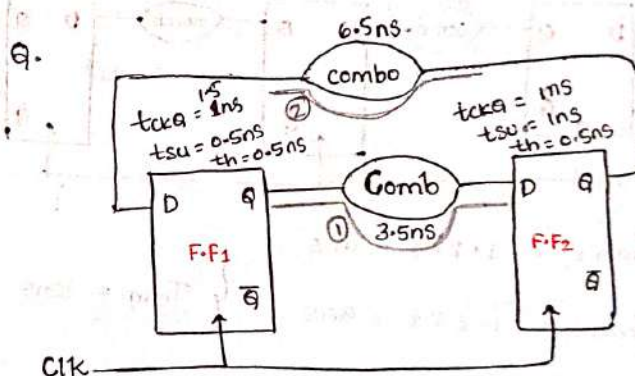
Launching time FF1 Capturing time FF2

$$T_{min} = t_{ckq} + t_{comb} + t_{su}$$

$$= 2ns + 2.5ns + 0.5ns$$

$$= 5ns$$

$$f_{max} = \frac{1}{5ns} = 0.2 \text{ GHz} = 200 \text{ MHz.}$$



- The Max. clock frequency at which the ckt work without fail _____.

- * In synchronous, we will see from 1 FF to another FF & we won't see another 1/p FF at same time.

- Hence, we have to consider 2 paths here and analyse. In a single clock cycle it won't possible.

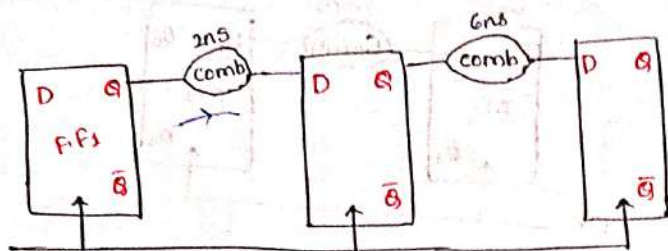
$$T_{min1} = 1.5ns + 3.5ns + 1ns = 6ns \quad \} T_{min} = 8ns$$

$$T_{min2} = 1ns + 6.5ns + 0.5ns = 8ns$$

$$f_{max} = \frac{1}{8ns} = 125 \text{ MHz}$$



Q. $t_{ckq} = 1\text{ns}$ $t_h = 1\text{ns}$
 $t_{su} = 1\text{ns}$

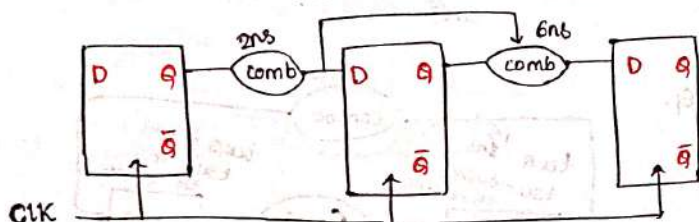


$$T_{min1} = 1 + 2 + 1 = 4\text{ns} \quad \left. \vphantom{T_{min1}} \right\} T_{min} = 8\text{ns}$$

$$T_{min2} = 1 + 6 + 1 = 8\text{ns}$$

$$f_{max} = 125\text{MHz}$$

Q. $t_{ckq} = 1\text{ns}$ $t_{su} = 1\text{ns}$ $t_h = 1\text{ns}$



$$T_{min1} = 1 + 2 + 1 = 4\text{ns}$$

$$T_{min2} = 1 + 6 + 1 = 8\text{ns} \quad \left. \vphantom{T_{min2}} \right\} T_{min} = 10\text{ns}$$

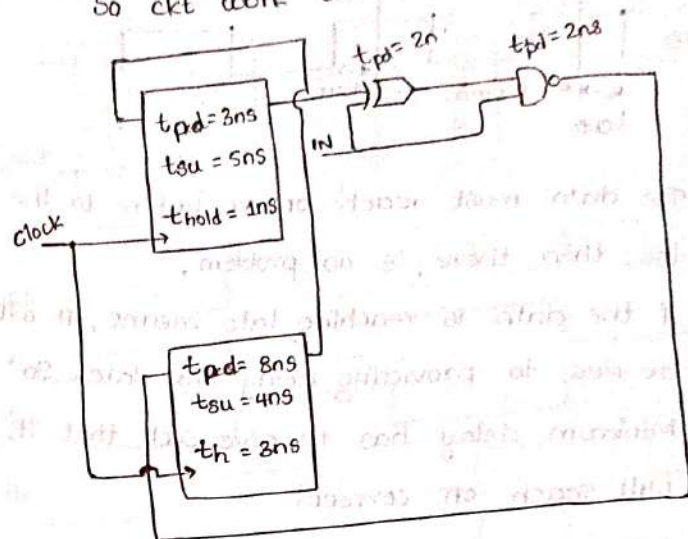
$$T_{min3} = 1 + 2 + 6 + 1 = 10\text{ns}$$

$$f_{max} = \frac{1}{10\text{n}}$$

$$= 0.1\text{GHz}$$

$$= 100\text{MHz}$$

Q. What is the maximum frequency of clk
 So ckt work without fail?

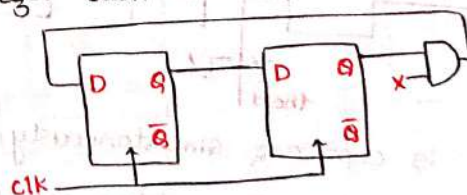


$$T_{min1} = 3 + 4 + 4 = 11\text{ns}$$

$$T_{min2} = 8 + 5 = 13\text{ns}$$

$$f_{max} = \frac{1}{13\text{n}} = 76.9\text{MHz} = 77\text{MHz}$$

Q. In the circuit shown below, the AND gate has a propagation delay of 1ns. The edge triggered flipflops have $t_{su} = 2\text{ns}$, $t_h = 0\text{ns}$, $t_{ckq} = 2\text{ns}$. The max. clock frequency (in MHz) rounded off to nearest Integer such that No v setup violation.



$$T_{min1} = 2 + 2 = 4\text{ns}$$

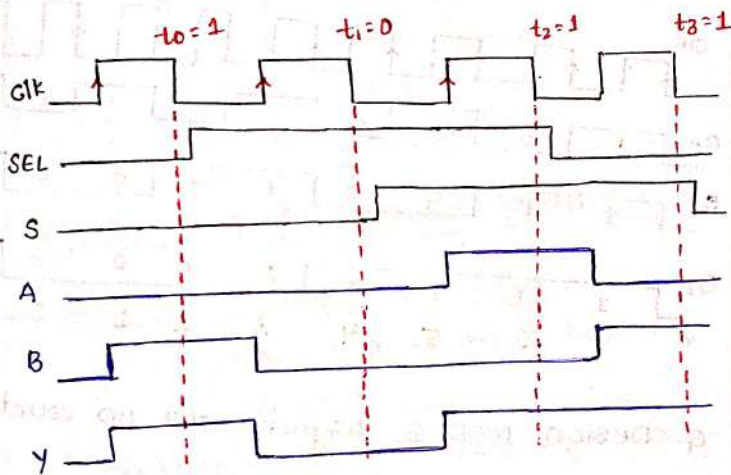
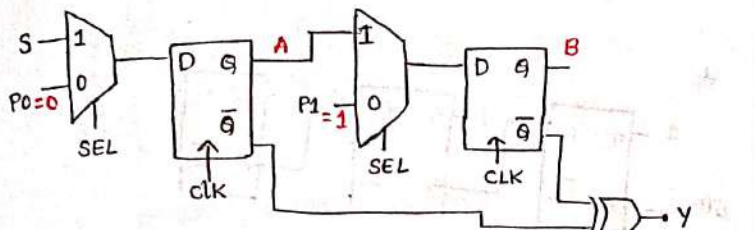
$$T_{min2} = 2 + 2 + 1 = 5\text{ns} \quad \left. \vphantom{T_{min2}} \right\} T_{min} = 5\text{ns}$$

$$f_{max} = \frac{1}{5\text{n}} = 200\text{MHz}$$

Q. A positive edge triggered flip flop Sequential ckt is shown below. There are no timing violations in ckt. Input P0 is set to logic 0 and P1 is set to logic 1 at all times.

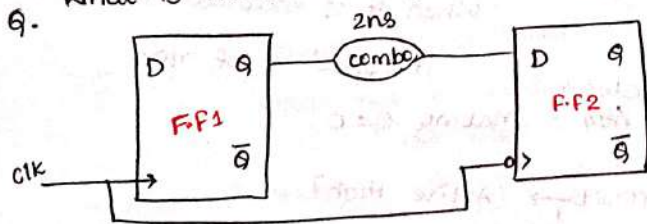
The timing diagram of inputs SEL and S are shown below.

The sequence of output Y from time T₀ to T₃ is _____.



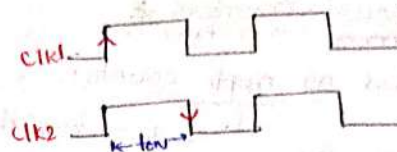
→ 1011

Q. What is max. clk freq, which ckt works.



$t_{clk} = 1ns$ $t_{su} = 1ns$ $t_h = 1ns$

* Launching at one FF and passing through combinational & receiving at -ve edge triggered is possible when it reach early than settling time (su).



$$(T_{on})_{min} = t_{clk} + t_{comb} + t_{su}$$

(i) 40% Duty cycle

$$(0.4) T_{min} = 1 + 2 + 1$$

$$T_{min} = \frac{4}{0.4}$$

$$T_{min} = 10ns$$

$$f_{max} = \frac{1}{10n} = 100MHz$$

(ii) 60% Duty cycle

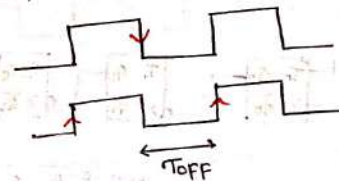
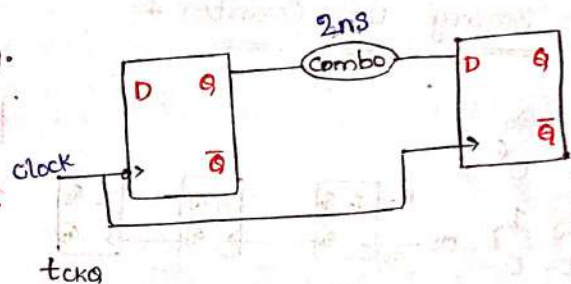
$$(0.6) T_{min} = 4ns$$

$$T_{min} = \frac{4n}{0.6}$$

$$= 6.66ns$$

$$f_{max} = 150MHz$$

Q.



$$(T_{off})_{min} = t_{clk} + t_{comb} + t_{su}$$

(i) 60% D.C

$$(0.6) T_{min} = 1 + 2 + 1$$

$$= 4ns$$

$$T_{min} = \frac{4}{0.6} = 6.66ns$$

$$f_{max} = \frac{1}{6.66 \times 10^{-9}}$$

$$f_{max} = 100MHz$$

(ii) 40% DC

$$(0.4) T_{min} = 4ns$$

$$T_{min} = \frac{4n}{0.4}$$

$$= 6.66ns$$

$$f_{max} = \frac{1}{6.66 \times 10^{-9}}$$

$$= 150MHz$$

* Asynchronous Counters *

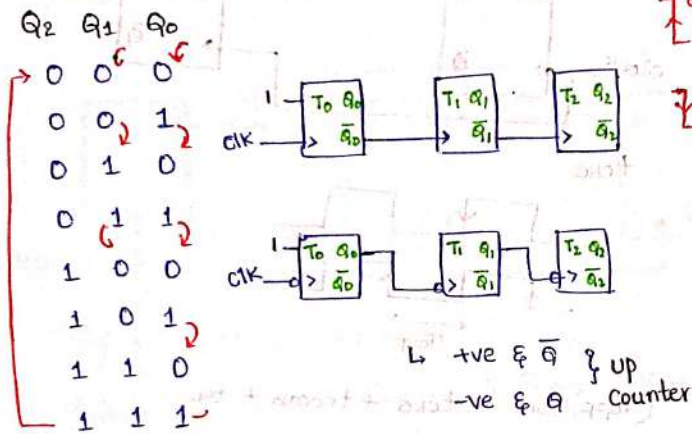
↳ It is also called as ripple counter.

→ clock is not same for every F.F.
 ↳ clock is rippled from 1 stage to another

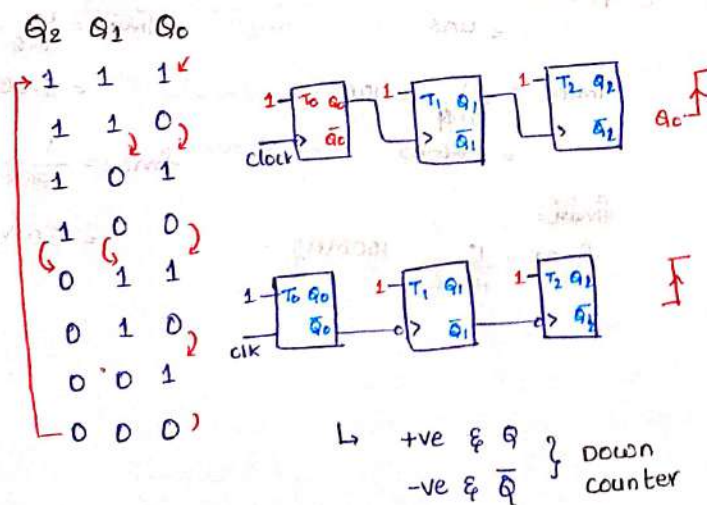
→ If clk is same for every F.F then it is Synchronous.

→ The clock at 1st flip flop is called as reference or primary clock. In remain it is considered as input.

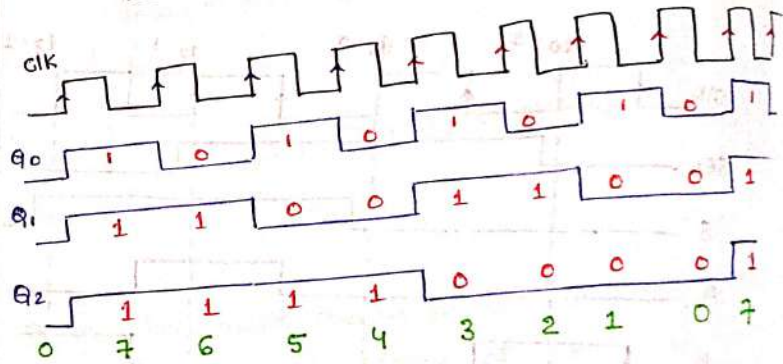
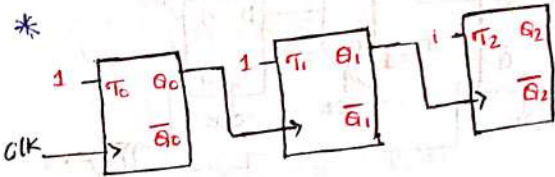
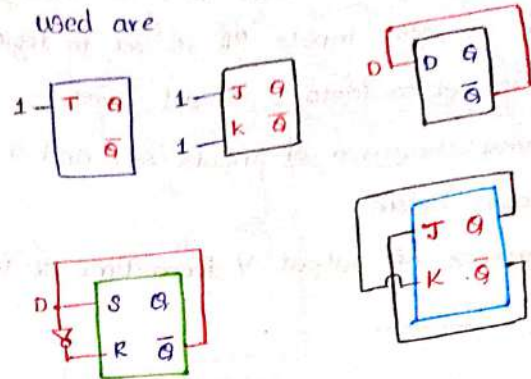
* 3 Bit Binary Up Counter *



* 3 Bit Binary Down Counter *



The possible flip flops or which can be used are



Q. Design MOD-6 Asynchronous up counter using JK flip flops.

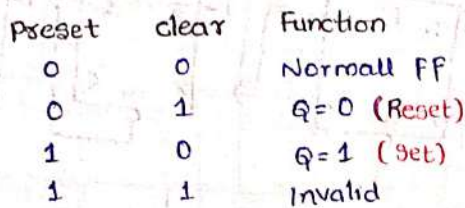
preset - make Q=1

(Set) When it is enabled, it makes Q=1 irrespective of Jp.

clear - making Q=0.

preset (Active High)
Active Low.

Active High (or) logic 1 enable preset & clear:-



Active Low (or) logic 0 enable preset & clear :-



preset	clear	function.
1	1	Normal FF
0	1	Q=1 (preset)
1	0	Q=0 (Reset)
0	0	Invalid

Synchronous & Asynchronous preset & clear:

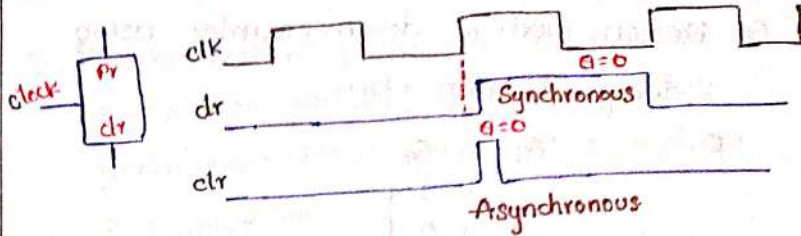
- * Synchronous preset and Clear :-
 - When it is enable & occurring clk edge.

↳ operation is synchronized with clock edge.

* Asynchronous preset and clear :-

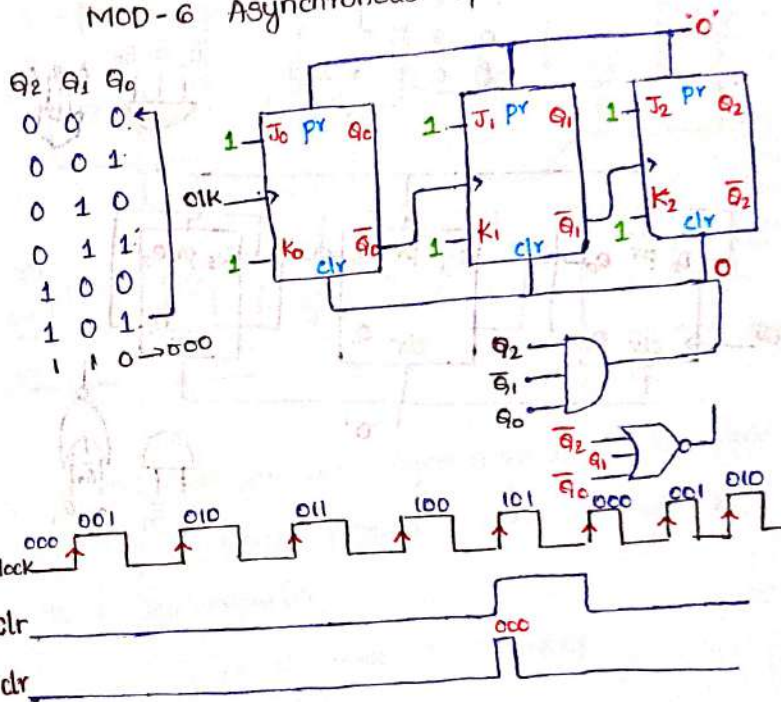
- It should be enabled and need not to wait for clk edge.

- And irrespective of clock edge occurring, it performs the operation.



வினா :-

MOD-6 Asynchronous up counter :-

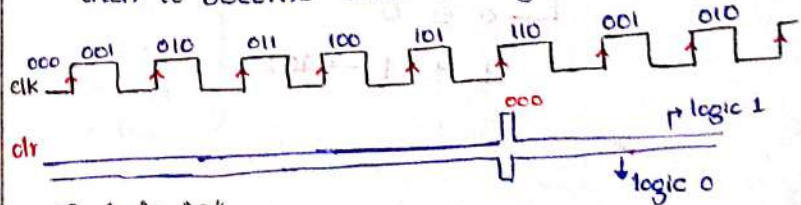


↓
Since it is asynchronous

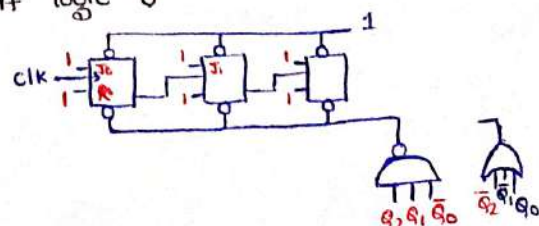
→ To clear all Flip flop at a common point we have to go with FF with Asynchronous preset & clear.

→ Here we can see above, the FF start & end at 1001, then last state is 100, it will become the mod-5 Asynchronous.

→ So use the next state 110 as temporary.
then it become mod-6 Asynchronous.



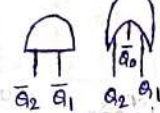
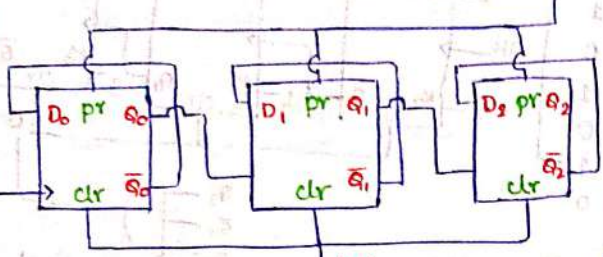
If logic "0"



Q. Design mod-6 down counter using delay-D flip flop.

Q_2	Q_1	Q_0
1	1	1
1	1	0
1	0	1
1	0	0
0	1	1
0	1	0
0	0	1

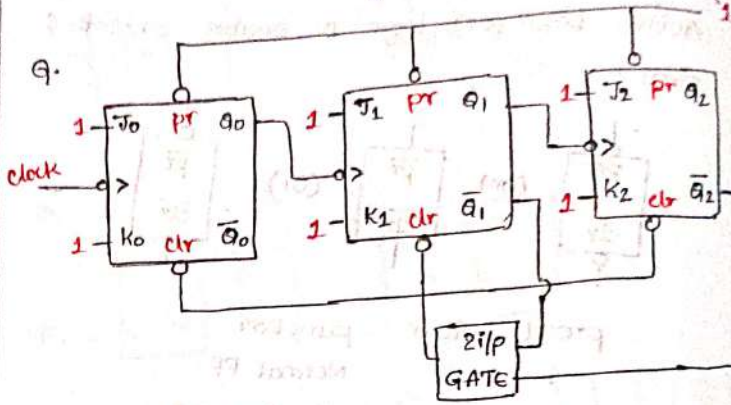
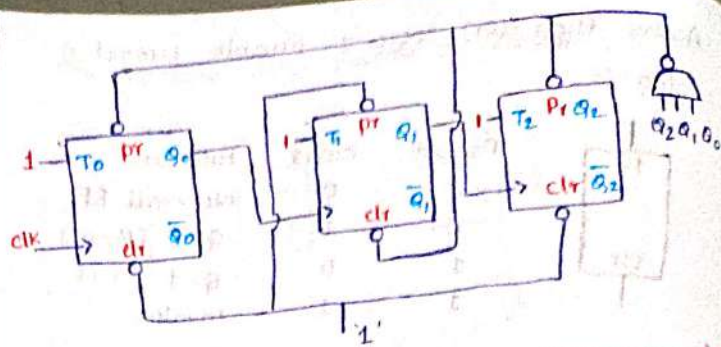
0 0 1 → 111



Q. Design mod-6 Down counter using Toggle (T) flip flop.

Q_2	Q_1	Q_0
1	0	1
1	0	0
0	1	1
0	1	0
0	0	1
0	0	0
1	1	1

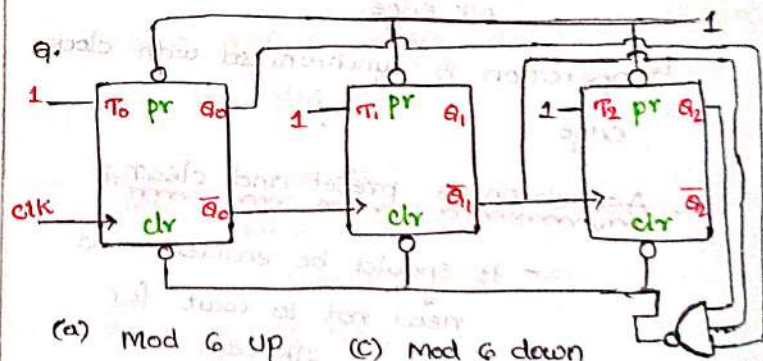
1 1 1 → 101



(a) AND (b) NAND (c) OR (d) NOR

Q_2	Q_1	Q_0
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

1 1 0 → 000

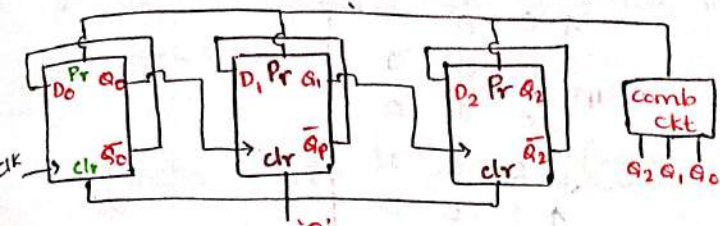


(a) Mod 6 up (c) Mod 6 down
(b) Mod 5 up (d) Mod 5 down

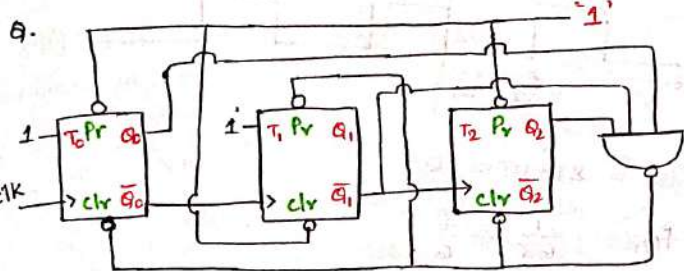
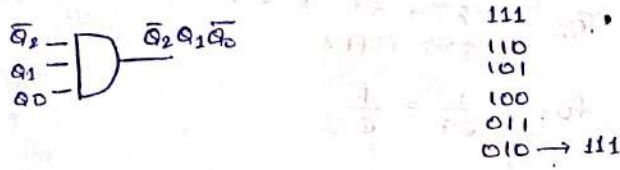
- * ① check whether up counter / down counter
- ② check where to start / stop.
↳ o/p of logic gate.
- ③ Next check Increment / Decrement
↳ Counter sequence.
- ④ Stop when if cond. are satisfied.

03-08

If the following counter is MOD-5 counter then Boolean Expression of y is — .



- (a) $Q_2 \bar{Q}_1 Q_0$ (c) $\bar{Q}_2 + Q_1 + \bar{Q}_0$
 (b) $Q_2 + \bar{Q}_1 + Q_0$ (d) $\bar{Q}_2 Q_1 \bar{Q}_0$

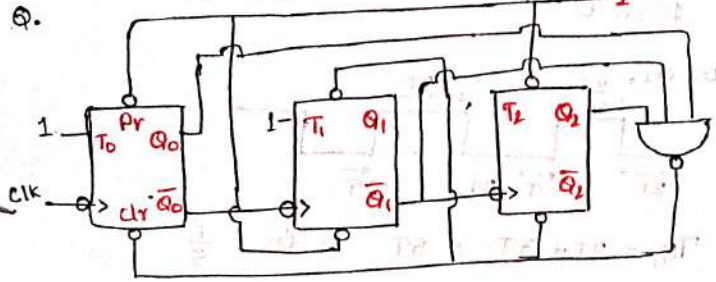


The Modulus of the counter is — .

- ↳ upcounter
- ↳ Increment
- ↳ Stop where getting '111'

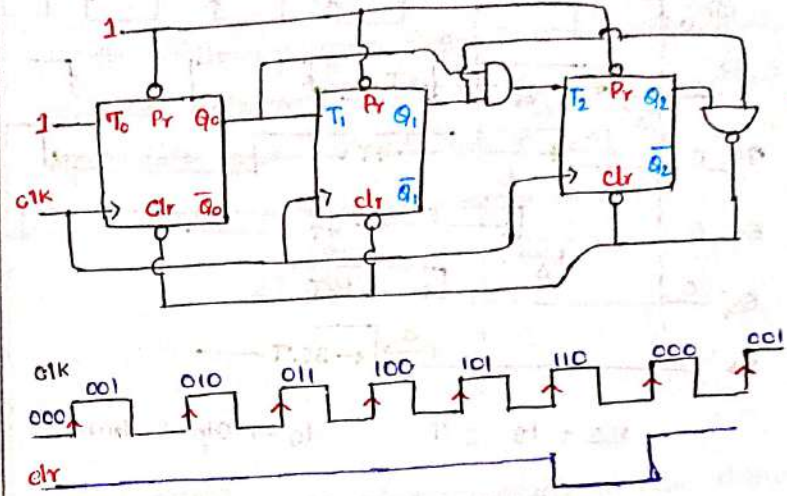
$Q_2 Q_1 Q_0$
010
011
100
101
110
111

$Q_2 \bar{Q}_1 Q_0 \Rightarrow 101$
 $\Rightarrow \text{MOD-3 counter.}$



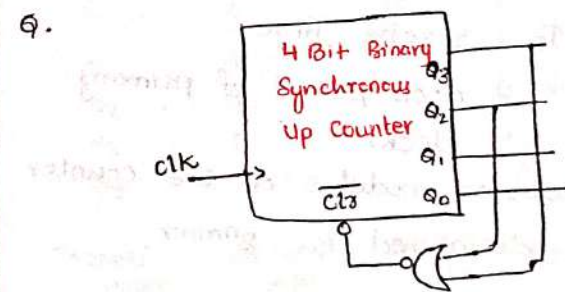
↳ Down counter
 MOD-5

Q. In the following synchronous counter, assume all the flip flops are having Synchronous clear counter. The modulus of Counter is — .



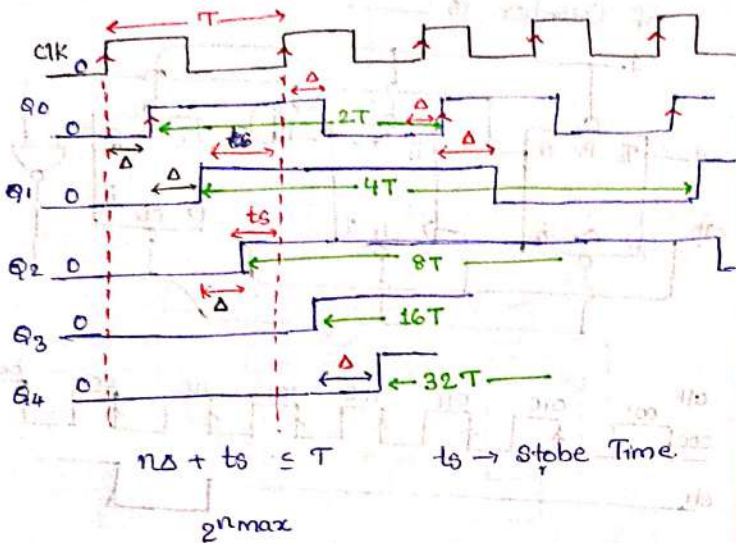
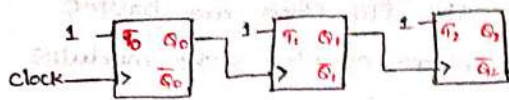
In this, we can use synchronous preset clear or asynchronous present & clear. Since it is a Synchronous.

∴ It is mod-7 counter.



Logic 0 disable.

$Q_3 Q_2 Q_1 Q_0$
0000
0001
0010
0011
0100



→ The max. No of bit ripple counter can be designed for a given clock max n value which can satisfy the equation $n\Delta + t_s \leq T$

Δ = each flip flop delay.

T_s = Strobe Time

T = clock period of primary clock.

→ The maximum modulus of the counter can be designed is $2^{n_{\max}}$.

→ Frequency (f) = $\frac{1}{T}$

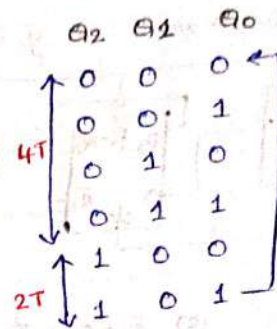
$$T_{Q_0} = 2T \Rightarrow f_{Q_0} = \frac{1}{2T} = \frac{f}{2}$$

$$T_{Q_1} = 4T \Rightarrow f_{Q_1} = \frac{1}{4T} = \frac{f}{4}$$

$$T_{Q_2} = 8T \Rightarrow f_{Q_2} = \frac{1}{8T} = \frac{f}{8}$$

$$T_{Q_{n-1}} = 2^n T \Rightarrow f_{Q_{n-1}} = \frac{f}{2^n}$$

Q. Let take mod-6 counter.



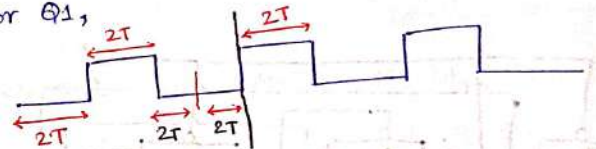
The frequencies of MSB to LSB are,

$$T_{Q_2} = 2T + 4T = 6T$$

(ON) (OFF)

$$f_{Q_2} = \frac{1}{6T} = \frac{f}{6}$$

For Q_1 ,

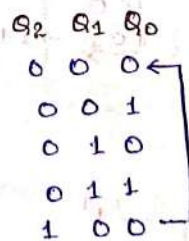


$$T_{Q_1} = 2T + 4T = 6T$$

$$f_{Q_1} = \frac{1}{6T} = \frac{f}{6}$$

$$T_{Q_0} = T + T = 2T \Rightarrow f_{Q_0} = \frac{1}{2T} = \frac{f}{2}$$

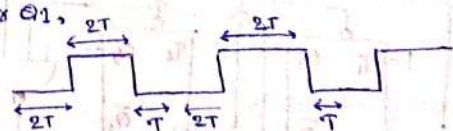
Q. For mod-5 counter find frequencies of Q_2 Q_1 Q_0 .



$$T_{Q_2} = 4T + T = 5T$$

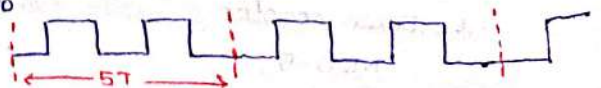
$$f_{Q_2} = \frac{1}{5T} = \frac{f}{5}$$

For Q_1 ,



$$T_{Q_1} = 2T + 3T = 5T \Rightarrow f_{Q_1} = \frac{f}{5}$$

For Q_0

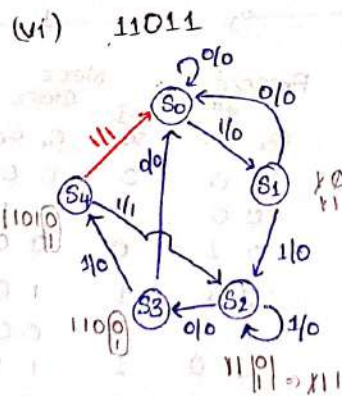
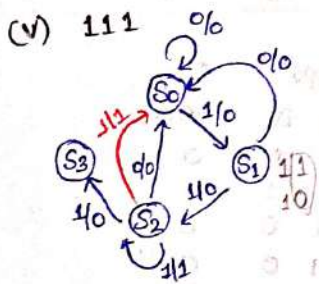
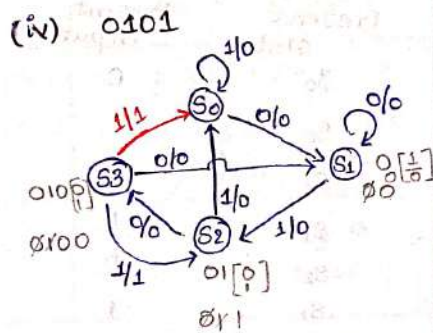
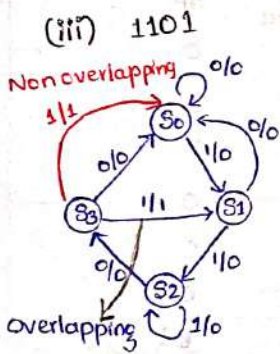
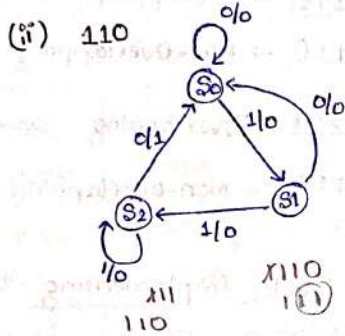
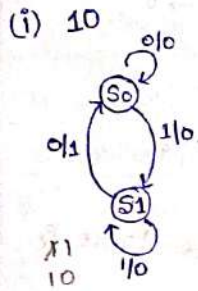


$$T_{Q_0} = 5T \Rightarrow f_{Q_0} = \frac{f}{5}$$

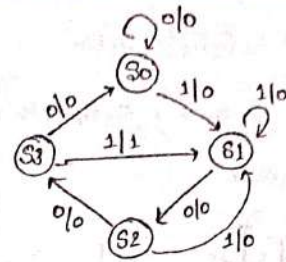
* Counters are frequency dividing circuits.

* These frequencies are applicable for Synchronous counter also.

Finite State Machines (FSM) → 2M



Q.



(a) 11011011

(b) 01100110

(c) 10010010

(d) 10000001

If the following state diagram is representing a Sequence detection which of the following Serial input data give output result $z=1$ for twice

1001 → overlapping twice in
10010010

Q. For 1101 state diagram given beside, draw

State table.

Present State	Present Next State	Next Input	output
S0	0	S0	0
S0	1	S1	0
S1	0	S0	0
S1	1	S2	0
S2	0	S3	0
S2	1	S2	0
S3	0	S0	0
S3	1	S1	1

Present State	Present Input	Next State	1/p's	o/p
Q1 Q0	x	Q1 Q0	Q1 Q0	z
0 0	0	0 0	0 0	0
0 0	1	0 1	0 1	0
0 1	0	0 0	0 0	0
0 1	1	1 0	1 0	0
1 0	0	1 1	1 1	0
1 0	1	1 0	1 0	0
1 1	0	0 0	0 0	0
1 1	1	0 1	0 1	1

Q1 Q0

0 0 → S0

0 1 → S1

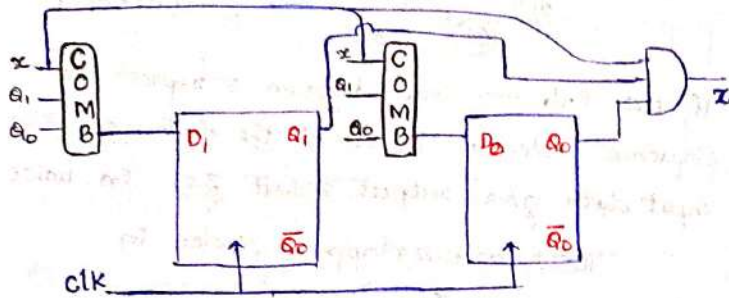
1 0 → S2

1 1 → S3

$$D_1 = \sum m(3,4,5) = \bar{a}_1 \bar{a}_0 x + a_1 \bar{a}_0$$

$$D_0 = \sum m(1,4,7) = \bar{a}_1 \bar{a}_0 x + a_1 \bar{a}_0 \bar{x} + a_1 a_0 x$$

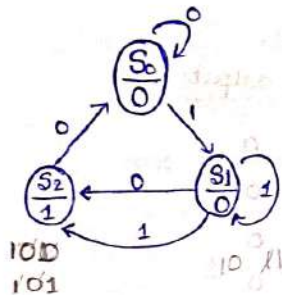
$$Z = \sum m(7) = a_1 a_0 x$$



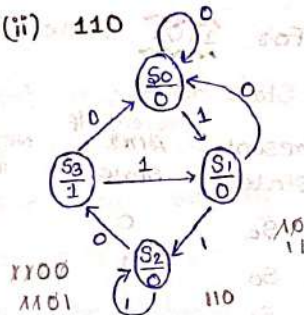
↳ Above is called Mealy FSM.

* Moore FSM *

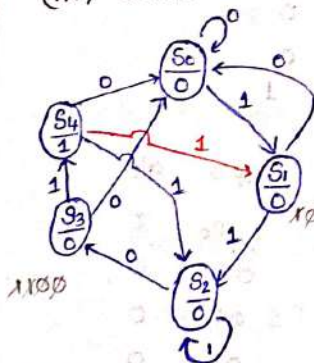
(i) 10



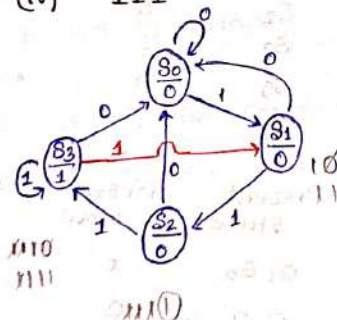
(ii) 110



(iii) 1101

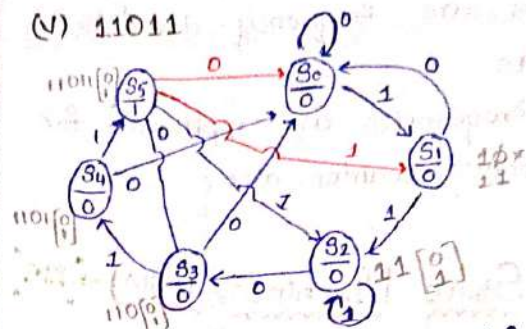


(iv) 111



↳ $S_4 \rightarrow S_2$
non-overlapping
{ discarding be
only until 1101
Not discard }

(v) 11011



110110 → overlapping $S_5 - S_3 \rightarrow 0$

110110 → Non-Overlapping $S_5 - S_0 \rightarrow 0$

110111 → overlapping $S_5 - S_1$

110111 → Non-overlapping $S_5 - S_0$

Q. For 111, implementing the state table &

ckt.

Present State	Present Input	Next State	o/p
S_0	0	S_0	0
S_0	1	S_1	0
S_1	0	S_0	0
S_1	1	S_2	0
S_2	0	S_0	0
S_2	1	S_3	0
S_3	0	S_0	1
S_3	1	S_3	1

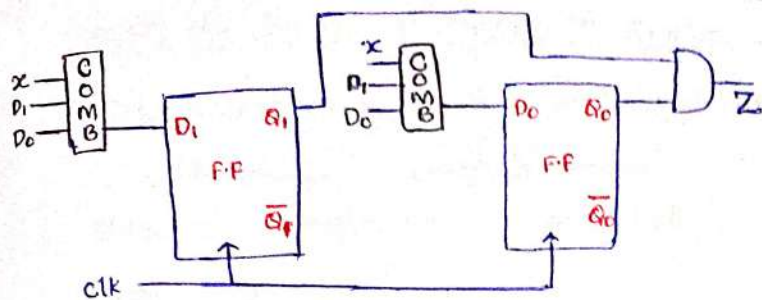
Present State	Next State	i/p	o/p
$Q_1 Q_0$	$Q_1 Q_0$	$D_1 D_0$	$x Z$
0 0	0 0	0 0	0
0 0	0 1	0 1	0
0 1	0 0	0 0	0
0 1	1 0	1 0	0
1 0	0 0	0 0	0
1 0	1 1	1 1	0
1 1	0 0	0 0	1
1 1	1 1	1 1	1

$Q_1 Q_0$
0 0 → S_0
0 1 → S_1
1 0 → S_2
1 1 → S_3

$$D_1 = \sum m(3,5,7) = (a_1 + a_0)x$$

$$D_0 = \sum m(1,5,7) = (a_1 + \bar{a}_0)x$$

$$Z = \sum m(6,7) = a_1 a_0$$



Mealy	Moore
1. output depends on state as well as input	1. o/p depend only on state
2. It requires less no of states	2. It require more no of states
3. It require less no of flip flops.	3. May require more no of flip flop.
4. i/p variations can affect the o/p immediately.	4. i/p can affect the o/p after next clk edge occurring.
5. Noise Immunity is less in Mealy.	5. Noise Immunity is more in Moore
6. Stability is less.	6. Stability is more.

Present State	Next State, Z	
	x=0	x=1
A	E, 0	G, 0
B	F, 0	C, 1
C	E, 0	A, 0
D	F, 0	A, 1
E	A, 0	D, 0
F	D, 0	E, 1

Rewriting the above table into NS & o/p Z at $x=0, x=1$.

Present State	Next State $x=0, x=1$	output $x=0, x=1$
A	E C	0 0
B	F C	0 1
C	E A	0 0
D	F A	0 1
E	A D	0 0
F	D E	0 1

The outputs of A and C are {0,0}, which both are same, so it can be eliminated either A or C. Replacing A with C or C with A in Next state at $x=1$, should be done.

Similarly, the output of B & D are same. Repeat the same as done with A & C. The new state table will be.

Present State	Next state $x=0, x=1$	output $x=0, x=1$
A	E A	0 0 $A \leftrightarrow C$
B	F A	0 1 $B \leftrightarrow D$
E	A B	0 0
F	B E	0 1

① output {0,0} same. Replacing C with A.

② Removing one of B & D $\rightarrow$ properties same.

$\therefore$ There are 2 redundant states

No of states after minimisation are 4.

Q. It is required to design FSM to control traffic light at which Green is on for 70s, yellow for 5s, Red for 75s. The clock pulse edge is 5s. Then How many flip flops required to design FSM.

Light	Duration(s)	Clock cycle (5s)	pulse Needed
Green	70	5	$\frac{70}{5} = 14$
Yellow	5	5	$5/5 = 1$
Red	75	5	$75/5 = 15$

Total no of states = $14 + 1 + 15$

$N = 30$ clock cycles, states.

No of Flip flop required,

$$N \leq 2^n$$

$$\Rightarrow 30 \leq 2^n$$

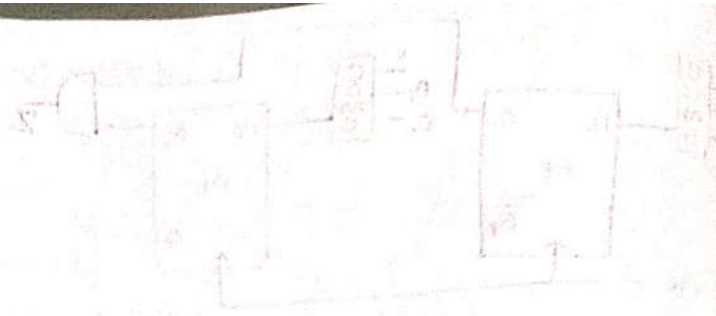
$$n=4 \Rightarrow 30 \leq 2^4 \Rightarrow 30 \leq 16$$

$$\boxed{n=5} \Rightarrow 30 \leq 2^5 \Rightarrow \boxed{30 \leq 32}$$

$\therefore$ 5 flip flops are needed.

It counts from 0 to 29. Hence it will be

MOD-30 counter.



Analog to Digital & Digital to Analog

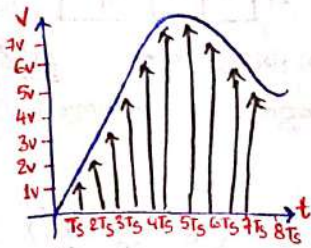
Analog:- Continuous in time and amplitude

↳ Continuous in amplitude means

there is no restriction to amplitude.

Sampled Signal:

discrete in time, continuous in amplitude.



→ Analog signal is sampled & its sampled version of analog is discrete in it.

2. Quantization :

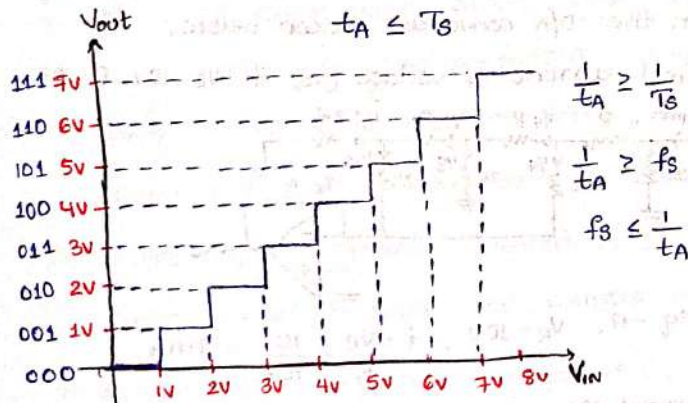
It is process of converting continuous range of amplitude level into finite set of discrete levels.

$$\Delta = \frac{V_{\max} - V_{\min}}{2^n}$$

let $V_{\min} = 0$

$V_{\max} = 8V$

$$\Delta = \frac{8-0}{2^3} = 1V$$



$$f_{\max} = \frac{1}{t_A}$$

→ Each Quantization is coded with Binary vector called encoding process.

→ The time required to convert Analog to discrete then it is called Conversion time.

Digital to Analog

$$110 \rightarrow 6V = 1V(2^2 \cdot 1 + 2^1 \cdot 1 + 2^0 \cdot 0) = 1(4+2) = 6V$$

$$V_0 = K (\text{Decimal equivalent of Digital})$$

↓ step voltage

$$000 \rightarrow 0V$$

$$001 \rightarrow 1V \rightarrow \text{Resolution (LSB voltage)} \rightarrow \text{Smallest possible change}$$

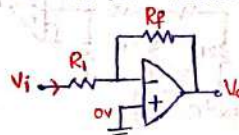
$$100 \rightarrow 4V \rightarrow \text{Mid voltage (MSB only)}$$

$$111 \rightarrow 7V \rightarrow \text{Full Scale voltage}$$

$$\text{Resolution} = \frac{V_{FS}}{2^n - 1}$$

1. Weighted Resistor type D/A

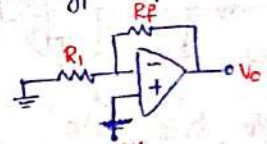
2. R-2R Resistor Network type D/A



$$\frac{V_i - 0}{R_1} = \frac{0 - V_o}{R_f}$$

$$\frac{V_i}{R_1} = -\frac{V_o}{R_f}$$

$$V_o = -\frac{R_f}{R_1} \cdot V_i$$



$$\frac{V_i}{R_1} = \frac{V_o - V_i}{R_f}$$

$$\frac{R_f}{R_1} \cdot V_i = V_o - V_i$$

$$V_o = \left[1 + \frac{R_f}{R_1} \right] V_i$$

open loop op-Amp :

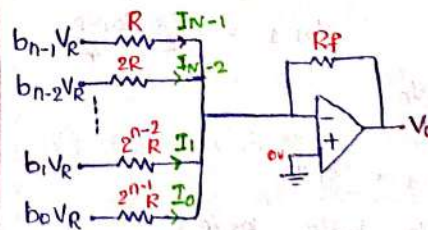
$$V_2 \rightarrow V_1 \rightarrow V_o = A(V_2 - V_1)$$

If $V_2 > V_1 \rightarrow V_o = +V_{sat} (1)$

$V_2 < V_1 \rightarrow V_o = -V_{sat} (0)$

$V_2 = V_1 \rightarrow V_o = 0V (\text{Ideally})$

1. Weighted Resistor DAC



$$I_{n-1} + I_{n-2} + \dots + I_1 + I_0 = I$$

$$\frac{b_{n-1}VR}{R} + \frac{b_{n-2}VR}{2R} + \dots + \frac{b_1VR}{2^{n-2}R} + \frac{b_0VR}{2^{n-1}R} = \frac{0 - V_o}{R_f}$$

$$\frac{VR}{2^{n-1}R} [b_{n-1}2^{n-1} + b_{n-2}2^{n-2} + \dots + b_12^1 + b_02^0] = -\frac{V_o}{R_f}$$

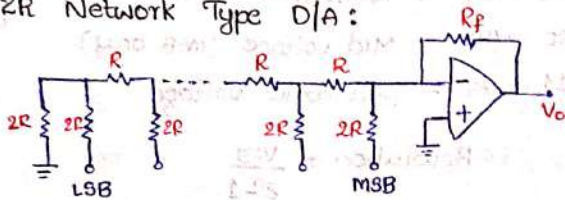
$$V_o = -\frac{VR}{2^{n-1}R} \times \frac{R_f}{R} [b_{n-1}2^{n-1} + b_{n-2}2^{n-2} + \dots + b_12^1 + b_02^0]$$

$$V_o = K (\text{Decimal equivalent of digital})$$

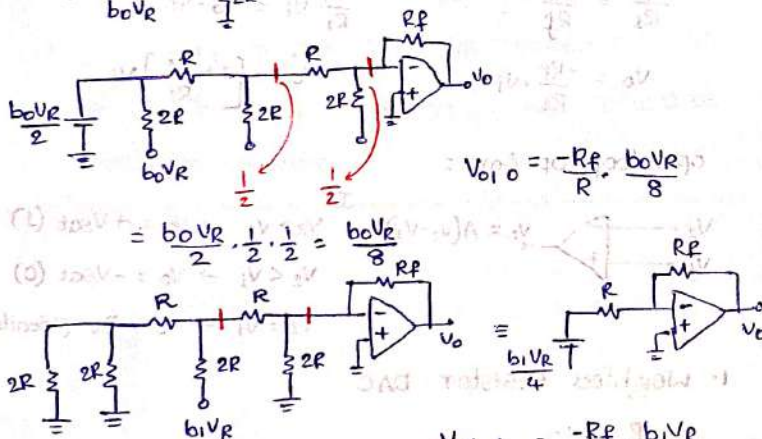
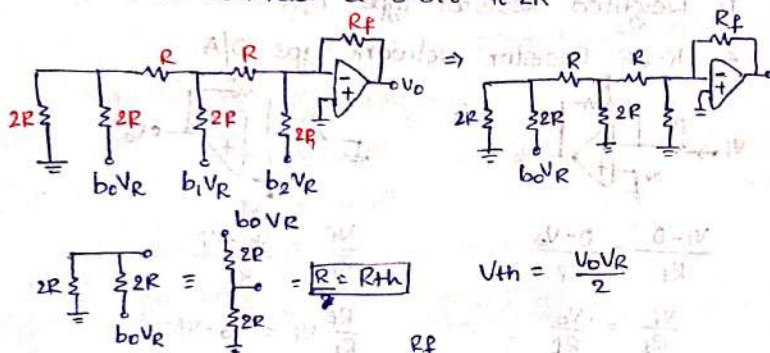
1. For weighted Resistor type D/A.

- It requires N different resistor for N -bit conversion.
- It requires very large resistance ($2^{n-1}R$) when n is large.
- The linearity is missing in o/p voltage.

2. R-2R Network Type D/A:



Let us consider a 3-bit R-2R



$$V_{o10} = -\frac{R_f}{R} \cdot \frac{b_0 V_R}{8}$$

$$V_{o12} = -\frac{R_f}{R} \cdot \frac{b_2 V_R}{2}$$

$$\therefore V_o = V_{o10} + V_{o11} + V_{o12}$$

$$= -\frac{R_f}{R} \left(\frac{b_0 V_R}{8} + \frac{b_1 V_R}{4} + \frac{b_2 V_R}{2} \right)$$

$$= -\frac{V_R}{8} \cdot \frac{R_f}{R} (b_2 \cdot 4 + b_1 \cdot 2 + b_0)$$

$$= K [\text{decimal equivalent of digital}]$$

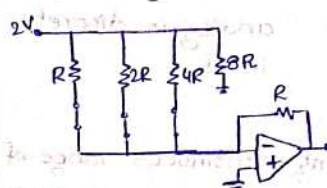
$$\text{Resolution} = \frac{V_R}{2^n}$$

1. R-2R Network type is using 2 different resistances (R & $2R$).

2. Linearity is improved.

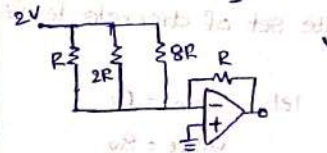
3. It is using approximately $2 \times n$ no of resistors for n -bit conversion.

Q. A 4 bit weighted resistor DAC with inputs is designed. The switches are closed when inputs are logic bit '1'. otherwise '0'. If Input $b_3 b_2 b_1 b_0$ changes from 1110 to 1101, the magnitude of change in o/p voltage is:



$$V_o = -R \left(\frac{2V}{R} + \frac{2V}{2R} + \frac{2V}{4R} \right)$$

$$= \frac{8+4+2}{4} = \frac{14}{4} = -3.5V$$



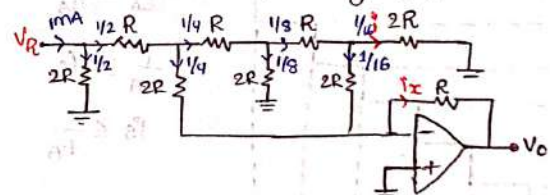
$$V_o = -R \left(\frac{2V}{R} + \frac{2V}{2R} + \frac{2V}{8R} \right)$$

$$= \frac{16+8+2}{8} = \frac{26}{8} = -3.25V$$

$$\text{change in o/p voltage} = |-3.5 + 3.25| = 0.25V = 250mV$$

Q. In the D/A converter shown below,

find current i , voltage (V_o) if $V_R = 10V$ $R = 10k$



$$R_{eq} = R, V_R = 10V, i = \frac{V_R}{R} = \frac{10}{10k} = 1mA$$

$$\text{Current } (i) = \frac{1}{16} mA = 0.0625 mA = 62.5 \mu A$$

$$V_o = -i_x R,$$

$$\text{But } i_x = \frac{1}{4} + \frac{1}{16} = \frac{4+1}{16} = \frac{5}{16} = \frac{5}{16} mA$$

$$\text{Voltage } (V_o) = -\left(\frac{5}{16} \times 10^3\right)(10 \times 10^3)$$

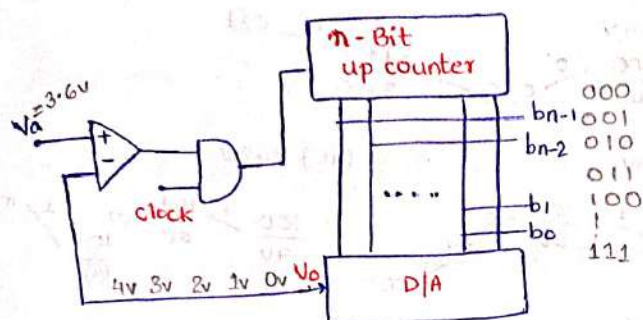
$$= -(0.3125)(10)$$

$$V_o = -3.125V$$

Analog to Digital Converter Method

1. counter type A/D
2. Flash type A/D
3. Successive Approximation A/D
4. Dual slope type A/D

1. Counter Type A/D :



In Counter type A/D,

1. It is used n-bit binary up counter in which count sequence is increased by 1 at every clock pulse.
2. At every clock pulse (cycle), count sequence is converted to analog output voltage V_o using digital to Analog converter & compared with analog sample voltage (V_a).
3. If ($V_a > V_o$) Comparator output is 1, AND Gate output is clock and counter increments.
4. If ($V_a < V_o$) Comparator output is 0, AND Gate output is 0 and counter stops incrementing as it is not getting clock cycles.
4. The Count Sequence on counter when it stop incrementing is considered as a approximate digital o/p for analog sample voltage.

5. The Conversion time is depending on analog sample voltage V_a & maximum Conversion time is 2^{n-1} clock cycles for n bit conversion.

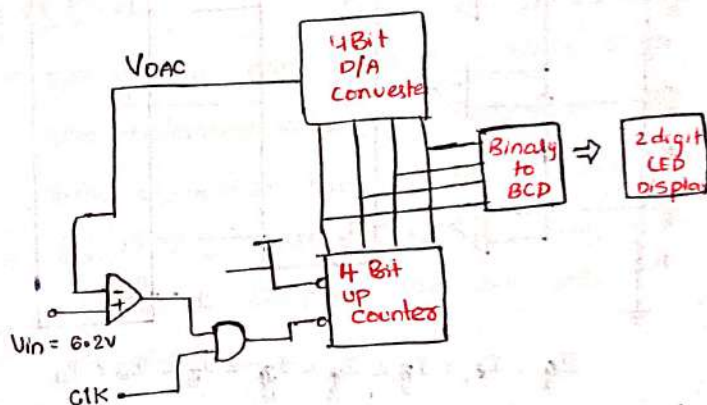
pg 172.

Q. In the following ckt, the comparator o/p logic is 1 if $V_{in} > V_{DAC}$ & 0 otherwise.

D/A conversion as per relation $V_{DAC} = \sum_{n=0}^3 2^{n-1} b_n V$

where b_3 (MSB), b_2 , b_1 , b_0 (LSB) are counter outputs.

The Counter starts from clear state.



(i) The stable reading of the LED display is

$$V_o = \sum_{n=0}^3 2^{n-1} b_n \text{ volts} = 2^2 b_3 + b_2 2^1 + b_1 2^0 + b_0 2^{-1}$$

To get step size, take only LSB = 1

$$\text{Step Size} = 0 + 0 + 0 + 1 \cdot 2^{-1} = \frac{1}{2} = 0.5 \text{V}$$

$$\begin{matrix} 1 & 2 & & 12 & 13 \\ 0001 & 0010 & & & \\ 0.5, & 1, & 1.5, & 2, & \dots, & 6, 6.5 \end{matrix}$$

$$1101 \text{ in BCD} = 0001 \ 0011 = \underline{13} \ 1101$$

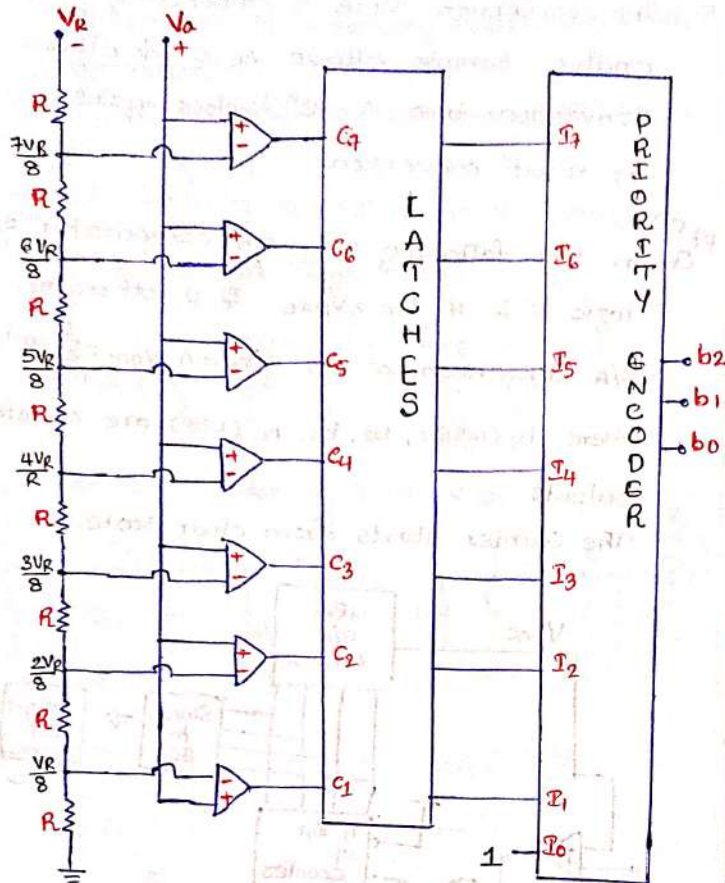
(ii) The magnitude of the error between V_{DAC} and V_{in} at steady state in volts is

$$= \text{Final stopped value} - V_{in} \text{ error}$$

$$= 6.5 - 6.2$$

$$= 0.3$$

2. Flash Type A/D :- (Parallel Comparator)



$$I_0 < I_1 < I_2 < I_3 < I_4 < I_5 < I_6 < I_7$$

$$V_a < \frac{V_R}{8} \quad \begin{matrix} C_1 & C_2 & C_3 & C_4 & C_5 & C_6 & C_7 & b_2 & b_1 & b_0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix}$$

$$\frac{V_R}{8} < V_a < \frac{2V_R}{8} \quad \begin{matrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{matrix}$$

$$\frac{2V_R}{8} < V_a < \frac{3V_R}{8} \quad \begin{matrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 10 \end{matrix}$$

$$\frac{3V_R}{8} < V_a < \frac{4V_R}{8} \quad \begin{matrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 11 \end{matrix}$$

$$\frac{4V_R}{8} < V_a < \frac{5V_R}{8} \quad \begin{matrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 100 \end{matrix}$$

$$\frac{5V_R}{8} < V_a < \frac{6V_R}{8} \quad \begin{matrix} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 101 \end{matrix}$$

$$\frac{6V_R}{8} < V_a < \frac{7V_R}{8} \quad \begin{matrix} 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 110 \end{matrix}$$

$$V_a > \frac{7V_R}{8} \quad \begin{matrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 111 \end{matrix}$$

1. It is fastest analog to digital converter.
2. Its conversion time is ns [flash time]
3. The complexity is very high as it is using $2^n - 1$ op-Amp comparators for n bit conversion.

3. Successive Approximation Register (SAR):

(i) 3.4V

$$\frac{100}{4V} \xrightarrow{0} \frac{010}{2V} \xrightarrow{1} \frac{011}{3V} \xrightarrow{1} 011$$

(ii) 4.3V

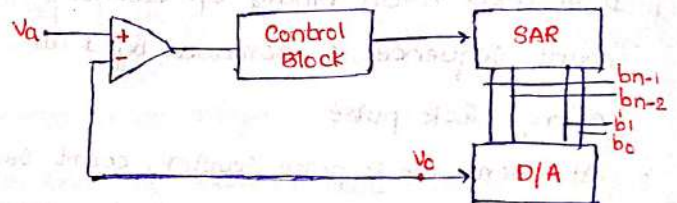
$$\frac{100}{4V} \xrightarrow{1} \frac{100}{6V} \xrightarrow{0} 101 \xrightarrow{0} 100$$

(iii) 5.2V

$$\frac{100}{4V} \xrightarrow{1} \frac{110}{6V} \xrightarrow{0} 101 \xrightarrow{1} 101$$

(iv) 6.7V

$$\frac{100}{4V} \xrightarrow{1} \frac{110}{6V} \xrightarrow{1} 111 \xrightarrow{0} 110 \xrightarrow{0} 110$$



Q. In a 5 bit approximate SAR, ADC is used digital to analog converter, logic 1 voltage reference of 8v. If Analog Sample voltage $V_a = 3.65V$, the SAR contents in every iteration is _____.

$$\frac{V_R}{2^n} = \frac{8V}{2^5} = \frac{V}{4}$$

3.65

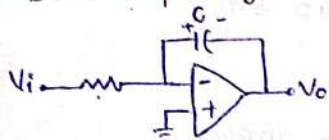
$$\frac{10000}{16/4 = 4} \xrightarrow{0} \frac{01000}{8/4 = 2} \xrightarrow{1} \frac{01100}{3V} \xrightarrow{1} 01110 \xrightarrow{1} 01111$$

5.8V

$$\frac{10000}{4V} \xrightarrow{1} \frac{11000}{6V} \xrightarrow{0} 10100 \xrightarrow{1} 10110 \xrightarrow{1} 10111 \xrightarrow{1} 10111$$

1. It is faster than all converters except flash type.
2. The Converter time is n iterations (n clock cycles) for n bit conversions.
3. The complexity is moderate
4. Most commonly used A/D.

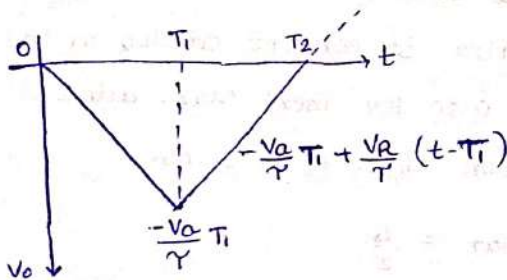
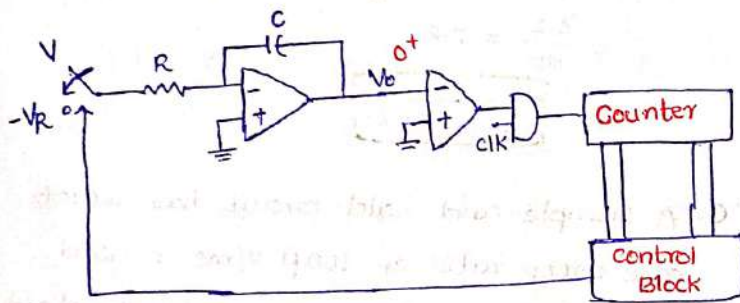
4. Dual Slope type A/D (Integrator type A/D)



$$V_R = \frac{1}{C} \int i_C dt = \frac{1}{C} \int \frac{V_i}{R} dt = \frac{1}{RC} \int V_i dt$$

$$V_O = \frac{1}{\gamma} \int V_i dt = -\frac{V_i}{\gamma} \int 1 \cdot dt = -\frac{V_i}{\gamma} [t]_0^{T_1}$$

$$V_O = -\frac{V_i T_1}{\gamma} \rightarrow \text{Integrator.}$$



$$V_O(t) = -\frac{V_a}{\gamma} T_1 + \frac{V_R}{\gamma} (t - T_1)$$

$$\text{at } t = T_2, V_O(t) = 0$$

$$\rightarrow 0 = -\frac{V_a}{\gamma} T_1 + \frac{V_R}{\gamma} (T_2 - T_1)$$

$$\frac{V_a}{\gamma} T_1 = \frac{V_R}{\gamma} (T_2 - T_1)$$

$$V_a T_1 = V_R (T_2 - T_1)$$

$$V_a t_i = V_R t_d$$

$$V_a 2^n T_{CLK} = V_R N T_{CLK}$$

$$V_a = \frac{V_R}{2^n} \cdot N$$

charging $\rightarrow$ -ve discharging $\rightarrow$ -ve

$$\text{Discharging} \rightarrow 2^n T_{CLK} + (2^n - 1) T_{CLK}$$

$$t_n \rightarrow (2^{n+1} - 1) T_{CLK}$$

1. It is slowest Analog to Digital Converter
2. The Conversion time is depended Analog Sample voltage V_a and Max. Conversion time $2^{n+1} - 1$ clock cycles
3. It is capable of rejecting power fluctuations
4. It is used in digital Voltmeter.

08-08-25

Q. for a dual APC type $3 \frac{1}{2}$ digit DVM, the reference voltage is 100mV and the first integration time is set to 300ms. for some input voltage, the de-integration period is 370.2 ms. The DVM will indicate _____.

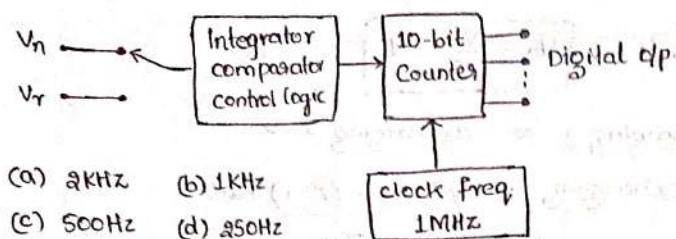
$$V_a t_i = V_R t_d$$

$$V_a = \frac{V_R t_d}{t_i}$$

$$= \frac{100\text{mV} \times 370.2\text{ms}}{300\text{ms}}$$

$$= 123.4\text{mV}$$

Q. The simplified block diagram of a 10-bit A/D converter of dual slope integrator type is shown. The 10-bit counter at the output is clocked by a 1MHz clock. Assume Negligible Timing overhead for the control logic, the max. frequency of the analog signal that can be converted using the A/D converter approximately:



$$f_s(\max) = \frac{1}{t_A} = \frac{1}{(2^{n+1}-1)T_{clk}}$$

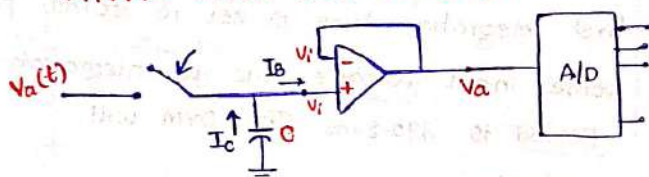
$$= \frac{f_{clk}}{(2^{n+1}-1)} = \frac{1M - 2^{20}}{500} = 488.5$$

$$f_s(\max) = 2 f_m(\max)$$

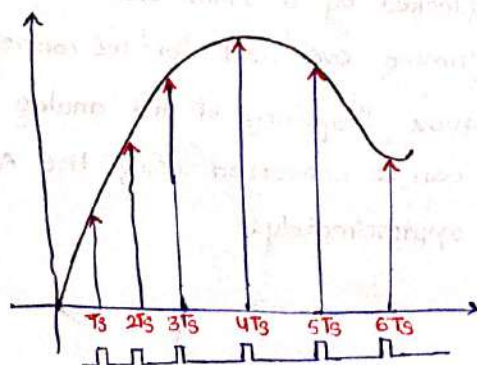
$$f_m(\max) = \frac{f_s(\max)}{2} = \frac{488.5}{2} = 244.25$$

To match the options, $f_m \approx 250\text{Hz}$.

Sample and Hold Circuit



$V_a(t)$ is the analog signal, when switch is ON capacitor charges with time. The amplitude of analog signal is stored in Capacitor. If we off the switch there is no path to discharge. That voltage is transferred to o/p (i.e. virtually short) But due to 'ON' resistance of switch and base current of op-amp makes the leakage current I_c .



If the leakage current changes the capacitor voltage effects the o/p V_o . This effect conversion The max. change allowed is $(dV_c)_{\max} = \frac{\Delta}{2}$

Acquisition time:

The time required to acquire the sample.

Droop rate: Rate of change of voltage across capacitor during hold.

$$\frac{dV_c}{dt} = \text{Droop rate}$$

$$C \cdot \frac{dV_c}{dt} = I_c \text{ (leakage current)}$$

$$\downarrow \frac{dV_c}{dt} = I_c \uparrow \Rightarrow \frac{dV_c}{dt} = 0 \text{ (Ideally)}$$

$$(dV_c)_{\max} = \frac{\Delta}{2}$$

$$\text{Droop rate} = \frac{dV_c}{dt}$$

$$\frac{\Delta/2}{dt} = D \cdot R$$

$$dt = \frac{\Delta}{2 \times D \cdot R}$$

Q. A sample and hold circuit has worst case droop rate of $100 \mu\text{V/ms}$ is used with 10 bit analog to digital converter to convert analog in the range 0 to 10V. Then Max. allowed conversion time is _____ ns.

$$(dV_c)_{\max} = \frac{\Delta}{2}$$

$$\Delta = \frac{10\text{V} - 0\text{V}}{2^{10}} = \frac{10}{2^{10}} \text{V}$$

$$\text{Droop rate} = 100 \mu\text{V/ms}$$

$$= 100 \times \frac{10^{-6}}{10^{-3}} = 10^{-1} = 0.1$$

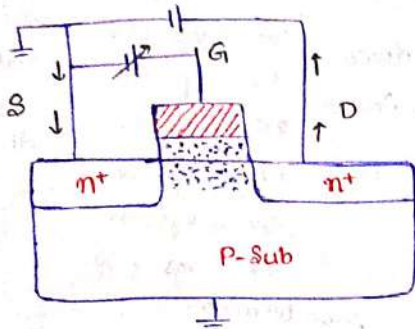
$$dt = \frac{\Delta}{2 \times D \cdot R} = \frac{10(2^{10})}{2(0.1)} = \frac{100}{2^{11}}$$

$$dt = 0.0488$$

$$= 48.8 \approx 49 \text{ms}$$

Complementary Metaloxide Semiconductor (CMOS)

N-channel Mosfet :-



Higher potential - Drain
Lower potential - Source.

→ voltage applied b/w Gate & Source

$$V_{GS} \geq V_T \rightarrow \text{ON}$$

$$V_{GS} < V_T \rightarrow \text{OFF}$$

$$\rightarrow I_{DS} = \mu_n C_{ox} \frac{W}{L} \left\{ (V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right\}$$

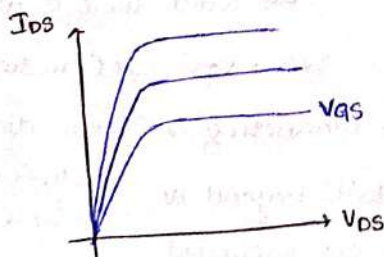
$$V_{GS} > V_T \quad V_{DS} < V_{GS} - V_T$$

$$I_{DS} = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T) V_{DS} \rightarrow \text{ohmic}$$

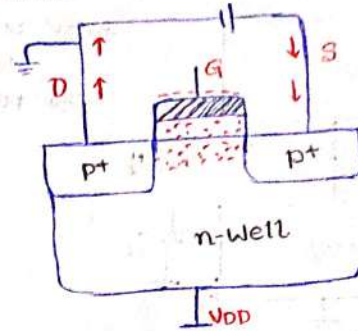
$$\frac{V_{DS}}{I_{DS}} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T)} = R_{on}$$

$$V_{DS} \geq V_{GS} - V_T \rightarrow \text{Saturation.}$$

$$I_{DS} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T)^2$$



P-channel Mosfet :-



→ N-well is connected to V_{DD} , Since V_{DD} is highest, p-junction can't be forward Biased.

→ When $V_{GS} = -Ve$, $V_{TP} = -Ve$

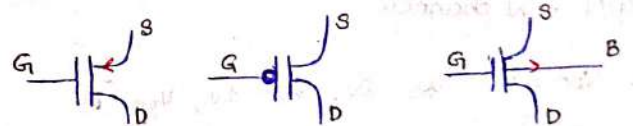
It is rippling existing electrons & attracting holes. It is creating p. at p

→ When voltage applied b/w drain & Source, the -ve terminal is attracting holes. +ve terminal is $\rightarrow$ Drain is rippling holes. $\rightarrow$ Source (Higher potential)

$\therefore V_{GS} \leq V_{TP}$
 $\rightarrow$ Threshold voltage of p channel mosfet.

$$\Rightarrow V_{GS} \leq V_{TP} \rightarrow (\text{ON})$$

$$V_{GS} > V_{TP} \rightarrow (\text{OFF})$$



Current equations,

$$I_{SD} = \mu_p C_{ox} \frac{W}{L} \left\{ (V_{GS} - V_{TP}) V_{DS} \right\} - \frac{V_{DS}^2}{2}$$

$$V_{DS} > V_{GS} - V_{TP}$$

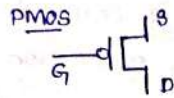
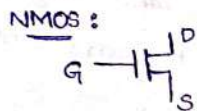
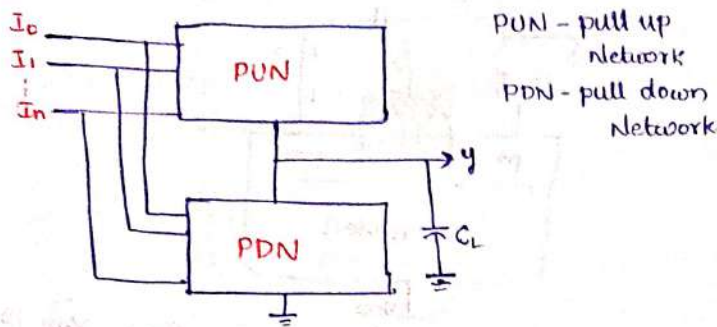
$$I_{SD} = \mu_p C_{ox} \frac{W}{L} (V_{GS} - V_{TP}) V_{DS}$$

$$\frac{V_{DS}}{I_{SD}} = \frac{1}{\mu_p C_{ox} \frac{W}{L} (V_{GS} - V_{TP})} = R_{op}$$

$$V_{DS} \leq V_{GS} - V_{TP}$$

$$I_{SD} = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} (V_{GS} - V_{TP})^2$$

CMOS



→ PUN & PDN are dual in nature & Constructed using CMOS.

→ When i/p's are static any one of Network is one, there is no connection b/w supply and Ground. Hence No static power dissipation (is eliminated)

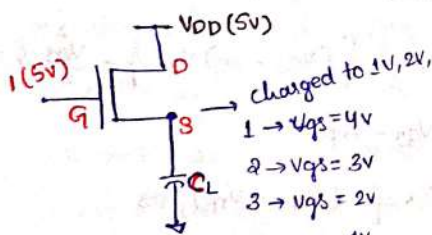
→ But o/p changes (i.e., capacitor charges & discharges.

Ⓘ PUN → p channel } why can't reverse?
PDN → n channel

Let $V_{DD} = 5V$, $V_{SS} = 0V$, $V_{TN} = 1V$, $V_{TP} = -1V$

(i) NMOS why only in PDN.

logic 1 → 5V
0 → 0V



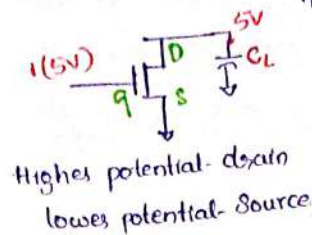
charged to 1V, 2V,
1 → $V_{GS} = 4V$
2 → $V_{GS} = 3V$
3 → $V_{GS} = 2V$
4 → $V_{GS} = 1V$
4.1 → $V_{GS} = 0.9 < V_{TN}$ (OFF)

Beyond 4V it can't charge. But we expecting until V_{DD} .
→ At o/p, V_{DD} is not possible

→ Source is charging, Gain to Source potential is effected.

→ Beyond $V_{DD} - V_{TN}$, It can't charge, bause the difference b/w Gain to Source must be atleast V_{TN} .

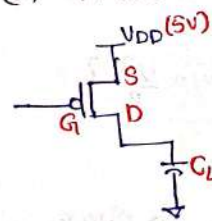
Now, let take NMOS in PDN.



The output at capacitor is already charged to V_{DD} i.e., 5V.
5V → $V_{GS} = 5V$ → Switch on
4V → $V_{GS} = 5V$ then it start
3V → $V_{GS} = 5V$ conducting
2V → $V_{GS} = 5V$ means
1V → $V_{GS} = 5V$ discharging
0.5V → $V_{GS} = 5V$ started.
0V → $V_{GS} = 5V$

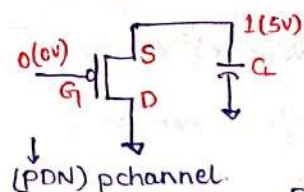
During, Drain terminal discharging the capacitor is not affecting. (V_{GS})
→ It is working effectively charging not working effectively.
Hence, NMOS is PDN placed.

(ii) PMOS



Gate to Source potential = $0 - 5 = -5V$.
 $V_{GS} \gg V_{TP}$ Can switch on P-channel?
→ Bause of this its conducting.
→ The capacitor is charging
Say charging is 1V → $V_{GS} = -5V$

→ Bause, drain is charging. 2V → $V_{GS} = -5V$
At o/p load, we get logic 1 3V → $V_{GS} = -5V$
Strong, effectively it can 5V → $V_{GS} = -5V$
Work during charging in pull up.



let, Say output is 1 which is already charged to 5V.
We want logic 0 at o/p.

5V → $V_{GS} = -5V$ (can switch on)

$V_{GS} \gg V_{TP}$ → Conducting → Starts discharging
It can't conduct beyond 1V.
So, we can't get grounded o/p. i.e., 0V.

4V → $V_{GS} = -4V$
3V → $V_{GS} = -3V$
1V → $V_{GS} = -1V$
0.9 → $V_{GS} = -0.9$ (OFF)

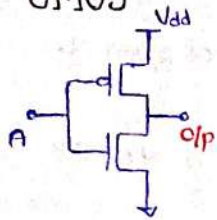
* Conclution:

→ PMOS can pass a strong 1 (charging) Weak 0

→ NMOS can pass a strong 0 (discharging) Weak 1

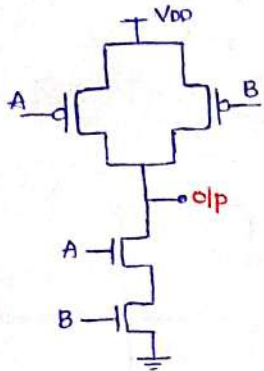
Hence PMOS → PUN
NMOS → PDN

CMOS



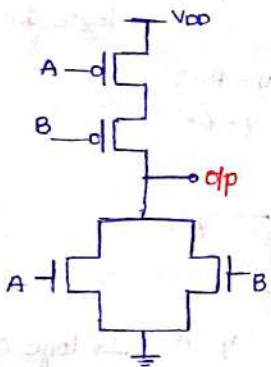
A	Y = $\bar{A}$
0	1
1	0

CMOS NAND



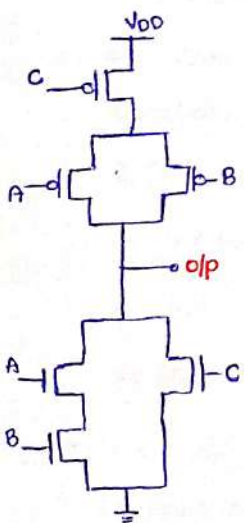
A	B	Y = $\bar{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

CMOS NOR

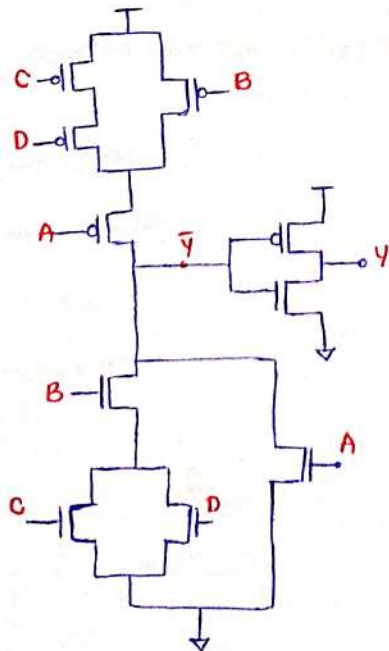


A	B	Y = $\bar{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

(i) $\overline{AB+C}$



(ii) $A+B(C+D)$

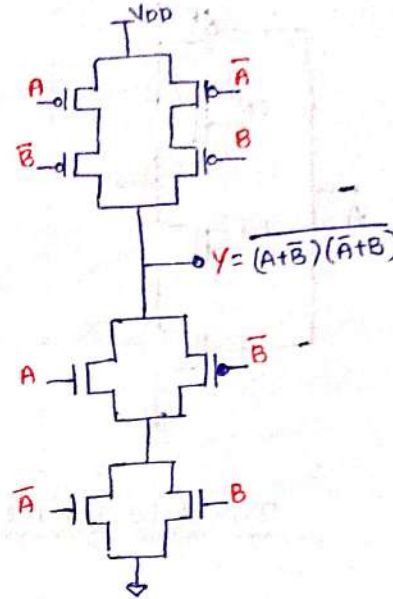
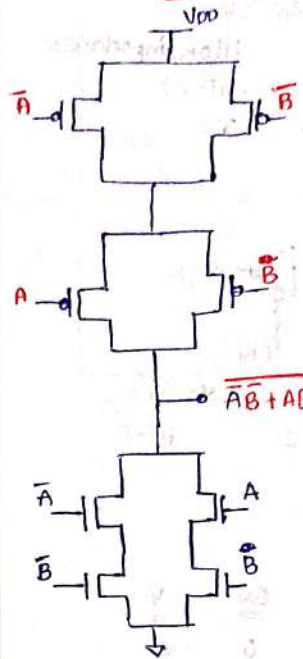


09/08/25

(iii) EXOR

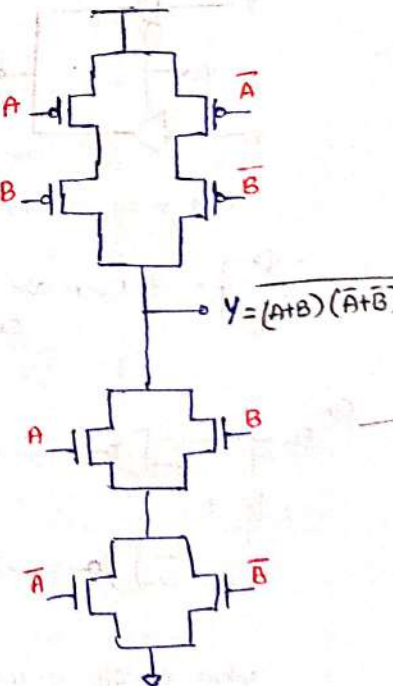
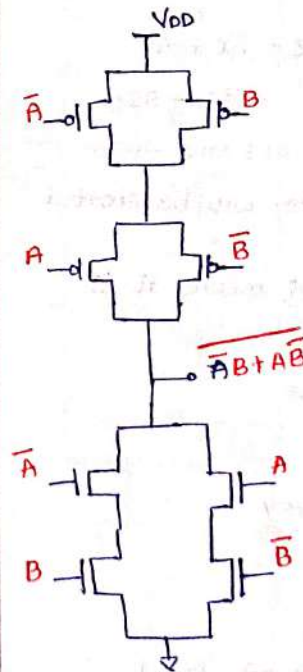
$$\overline{A \odot B} = \overline{AB + \bar{A}\bar{B}} = \overline{AB} \cdot \overline{\bar{A}\bar{B}} = (\bar{A} + \bar{B})(A + B) = \bar{A}B + A\bar{B}$$

$$A \oplus B = \bar{A}B + A\bar{B} = \overline{A \odot B} = \overline{\bar{A}\bar{B} + AB} = \overline{(\bar{A} + \bar{B})(A + B)} =$$

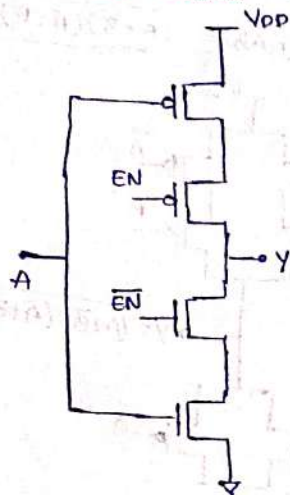


(iv) EXNOR

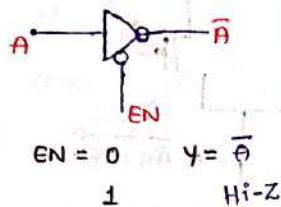
$$A \odot B = \bar{A}\bar{B} + AB = \overline{A \oplus B} = \overline{\bar{A}B + A\bar{B}} = \overline{(A+B)(\bar{A}+\bar{B})} = Y$$



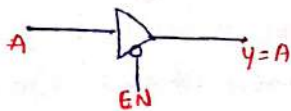
Tri-State Inverter



Enable Y
1 High Impedance (Hi-Z)
0 $\bar{A}$

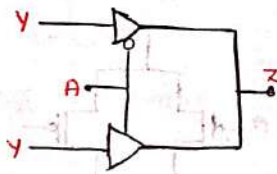


Tri-State Buffer



EN Y
0 A
1 Hi-Z

*

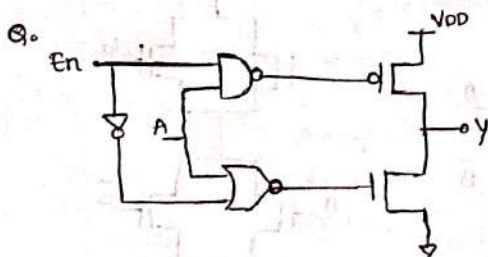
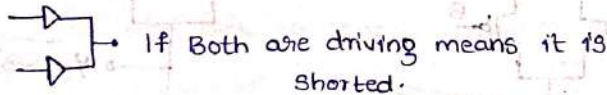


$$Z = \bar{A}B + AY$$

$$= \bar{A}B + AY$$

2:1 Mux.

1 may active & other will be shorted



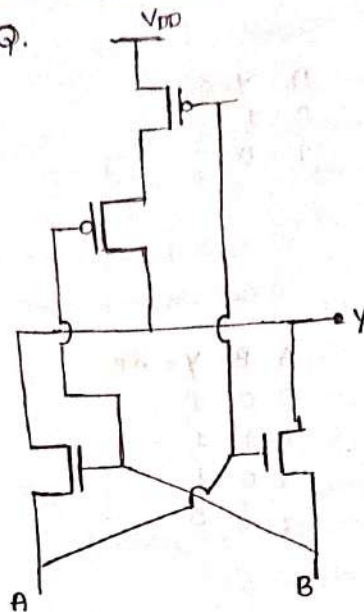
what is o/p y when $EN=0$, $EN=1$

EN	A	Y
0	0	Hi-Z
0	1	Hi-Z
1	0	0
1	1	1

$EN=0$; $Y=Hi-Z$

$EN=1$; $Y=\bar{\bar{A}}=A$

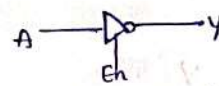
Q.



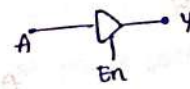
o/p of given y.

A	B	$Y = A \odot B$
0	0	1
0	1	0
1	0	0
1	1	1

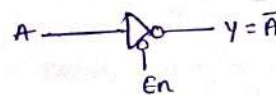
↳ Tri state Inverter



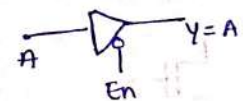
En = 0 $Y = Hi-Z$
En = 1 $Y = \bar{A}$



En = 0 $Y = Hi-Z$
En = 1 $Y = A$

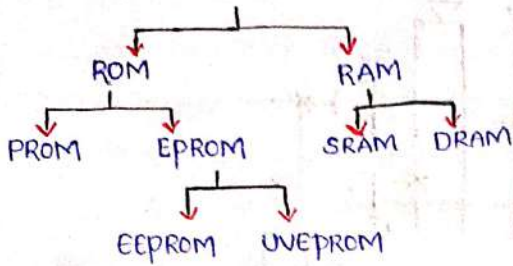


En = 0 $Y = \bar{A}$
En = 1 $Y = Hi-Z$

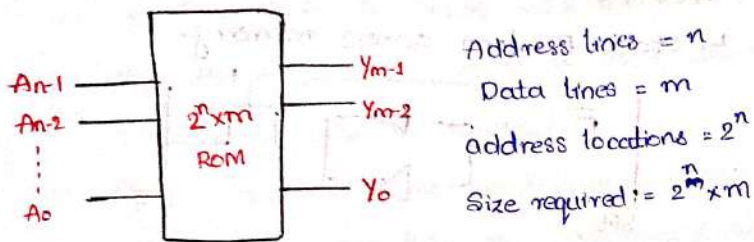


En = 0 $Y = A \rightarrow \text{logic 0}$
En = 1 $Y = Hi-Z$

Semi conductor Memories



ROM (Read only Memory) :-



Q. The ROM size required to store 2 bit Multiplier is

$$\frac{A_1 A_0 \times B_1 B_0}{Y_3 Y_2 Y_1 Y_0} = 2^4 \times 4$$

Q. The ROM size required to store 2 bit Square is

$$\frac{A_1 A_0 \times A_1 A_0}{Y_3 Y_2 Y_1 Y_0} = 2^2 \times 4$$

Q. The ROM size required for the following functions,

$$f_1(A, B, C) = \sum m(1, 2, 3)$$

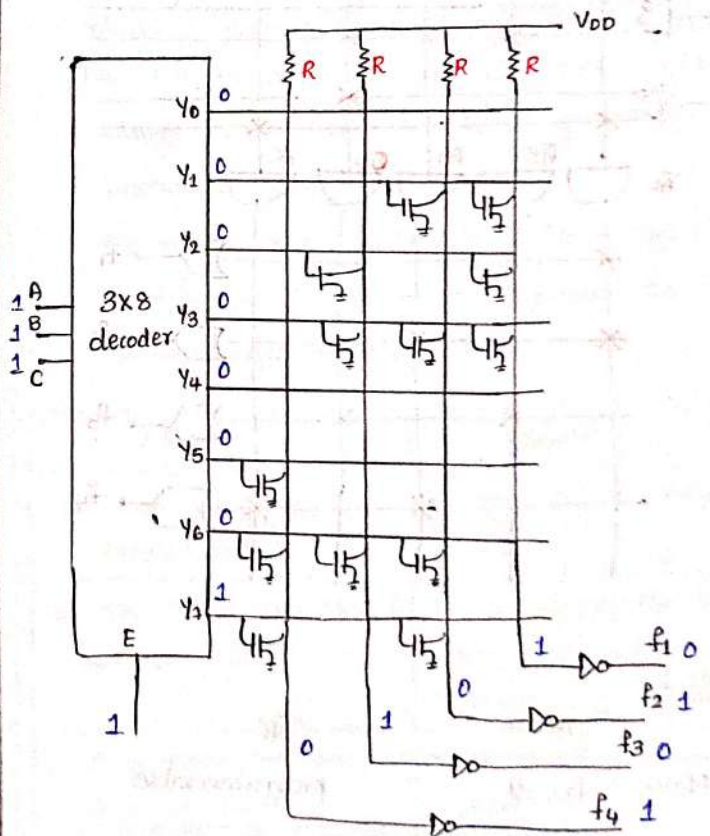
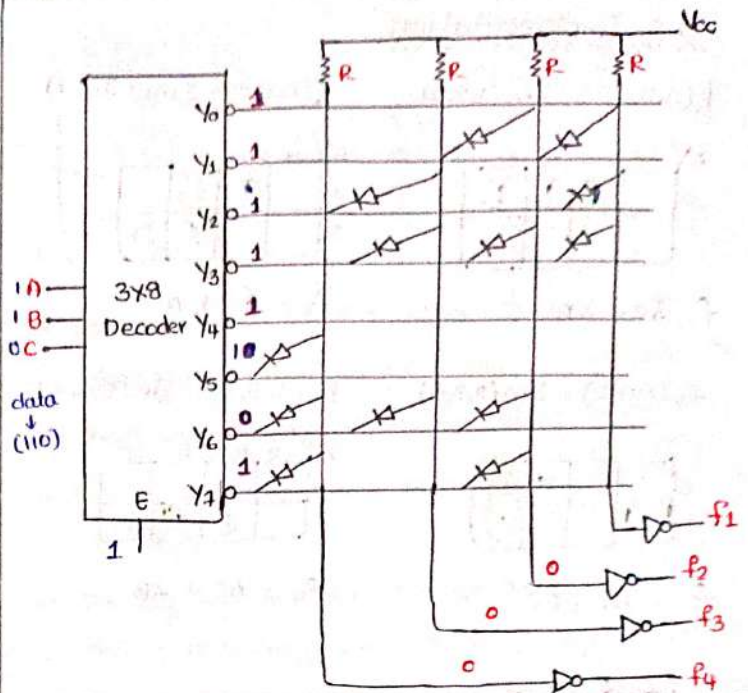
$$f_2(A, B, C) = \sum m(1, 3, 6, 7)$$

$$f_3(A, B, C) = \sum m(2, 3, 6)$$

$$f_4(A, B, C) = \sum m(5, 6, 7)$$

$$m=4 \quad n=3$$

$$\begin{aligned} \text{ROM size required} &= 2^n \times m \\ &= 2^3 \times 2^2 \times 4 \end{aligned}$$



PLA Implementation

$$f_1(A, B, C) = \sum m(1, 2, 3)$$

A	BC	00	01	11	10
0			1	1	
1			1		

$$f_1 = \bar{A}C + \bar{A}B$$

$$f_3(A, B, C) = \sum m(2, 3, 6)$$

A	BC	00	01	11	10
0				1	
1			1		1

$$f_3 = \bar{A}B + B\bar{C}$$

$$f_2(A, B, C) = \sum m(1, 3, 6, 7)$$

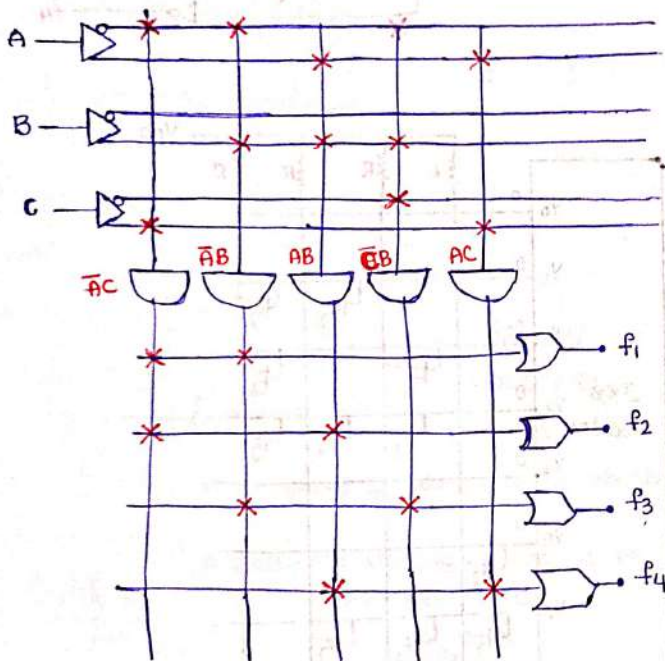
A	BC	00	01	11	10
0			1	1	
1			1	1	

$$f_2 = \bar{A}C + AB$$

$$f_4(A, B, C) = \sum m(5, 6, 7)$$

A	BC	00	01	11	10
0				1	
1			1	1	1

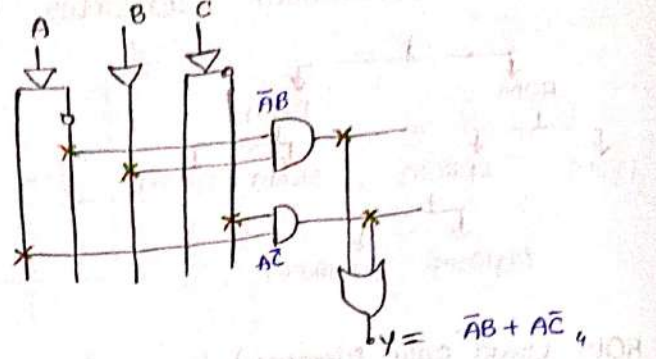
$$f_4 = AC + AB$$



Note :

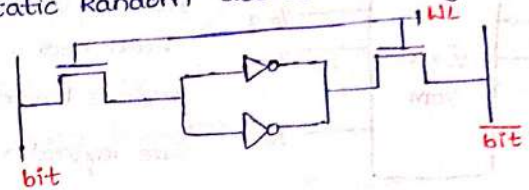
	AND	OR
PROM	fixed	programmable
PLA	programmable	fixed
PAL	programmable	fixed

Q.

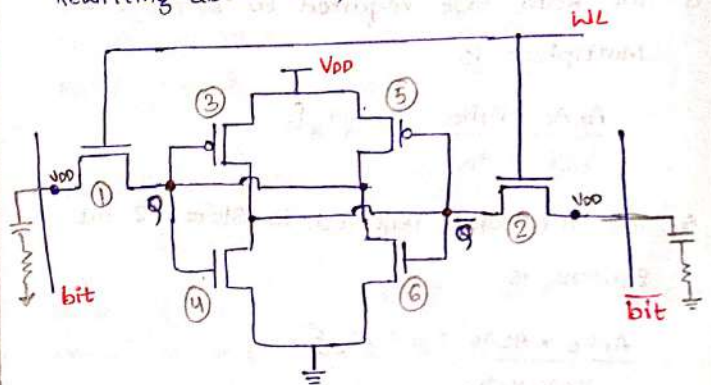


SRAM

→ Static Random access memory.



Rewriting above ckt,



→ A bistable ckt is used here to store 1 bit data. There is no external interference.

→ When $WL = 0$, (Hold state)

$\bar{bit} = 0$ (i.e., off). Transistors (1, 2) are off.

means the ckt is in remembering condition.

→ Assume, $bit = 1$ then $\bar{bit} = 0$. Then V_{cc} say 5v.

Then there is no effect of transistor 1 & 2.

Write Operation :

$WL = 1$, makes the transistor ON (1 & 2)

→ To write 1, set $BL = 1$ (VDD) & $\bar{BL} = 0$ v.

write 0, Set $BL = 0$ (0v) & $\bar{BL} = 1$ (VDD)

→ Suppose 1 is writing at node Q.

$BL = 1$ drives $Q \rightarrow 1$

$\overline{BL} = 0$, drives $\overline{Q} = 0$

Cross Coupled inverters,

If $Q = 1 \rightarrow$ transistor 4 OFF, 3 OFF.

$\overline{Q} = 0 \rightarrow$ 6 ON, 5 ON.

Bit & $\overline{Bit}$ = i/p $\leftarrow$ write

o/p $\leftarrow$ Read

Read operation:

→ Let $Q = 1, \overline{Q} = 0$. To access bit & $\overline{bit}$ $WL = 1$
the those bit lines act as o/p lines which
can read data in memory.

→ These Capacitors having significance in read/
write oper. which they are precharged (V_{DD})

→ Voltage difference - V_{DD} & $Q \rightarrow$ No diff (1 & 1)

V_{DD} & $\overline{Q} \rightarrow$ Yes (1 & 0)
so Capacitor discharge.

Voltage decrease.

→ Bit & $\overline{Bit}$ will send data to Sense Amplifier
will act as comparator

↳ bit $\downarrow \rightarrow$ o/p = 1

Another Case

→ $Q = 0, \overline{Q} = 1$; $WL = 1$

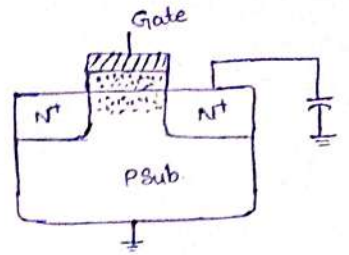
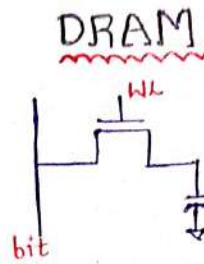
Voltage difference - V_{DD} & Q - No diff
yes (1 & 0)
Voltage $\downarrow$

V_{DD} & $\overline{Q} \rightarrow$ No diff (1 & 1)

→ Bit & $\overline{Bit}$ Sense data to Sense Amplifier.

↳ Comparator

↳ bit $\downarrow$ (V) $\rightarrow$ o/p = 1



→ When bit = 1, $WL = 1$ then Capacitor is
charging. When $WL = 0$, there is no path for
discharging, charge in capacitor remain same.

→ Min 1 transistor is required to store 1
bit data.

→ Ideally we can say, charge remain same.
But practically, we can discharge, can leakage
Because drain and bulk junction is conducting.

↳ At Gate terminal, Say = 0V, The capacitor
which is having some charge, there is leakage
b/w p-n junction. After sometime, charge ($\downarrow$)
voltage across capacitor ($\downarrow$). Logic 1 is
becoming logic 0.

→ To restore logic it has to refresh again,
additional circuitry is required. So it
is dynamic nature.

SRAM	DRAM
1. Minimum no of transistors 6.	1. Min. No of transistors are 1.
2. The logic can restore on its own.	2. To restore the logic we need additional circuitry
3. It is storing in the form of voltage	3. It is storing in the form of charge.
4. It is faster	4. It is slower.
5. Density is less $\downarrow$ No of bits per unit area.	5. Density is more.